Maithripala D. H. S.,
Dept. of Mechanical Engineering,
Faculty of Engineering,
University of Peradeniya,
Sri Lanka.

Lecture Notes on Introduction to Vibration Analysis

Class notes for ME3030

Supplementary .ipynb interactive Notebook:
https://github.com/mugalan/vibration-analysis/blob/main/class-notes/Note_Book_Dynamic_Systems_Vibrations.ipynb

November 2, 2026

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Preface

This is a compilation of notes that originated as class notes for ME301 at the department of Mechanical Engineering, Faculty of Engineering, University of Peradeniya. Subsequently they have served as class material for ME320 and ME327. I am currently using it as class notes for the vibration part of ME3030. They are far from complete and I will be frequently updating them as time permits. I am deeply indebted to all the students who have suffered through them and have tried out the exercises and have provided me with valuable input. I am sure there are many errata and will greatly appreciate if you can please bring them to my notice by sending an e-mail to mugalan at gmail.com.

The GitHub repository: vibration-analysis, provides a collection of interactive IPython notebooks and supplementary material.

Peradeniya, Sri Lanka, *D. H. S. Maithripala*
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Chapter 1
Introduction to Vibration Analysis

1.1 Introduction

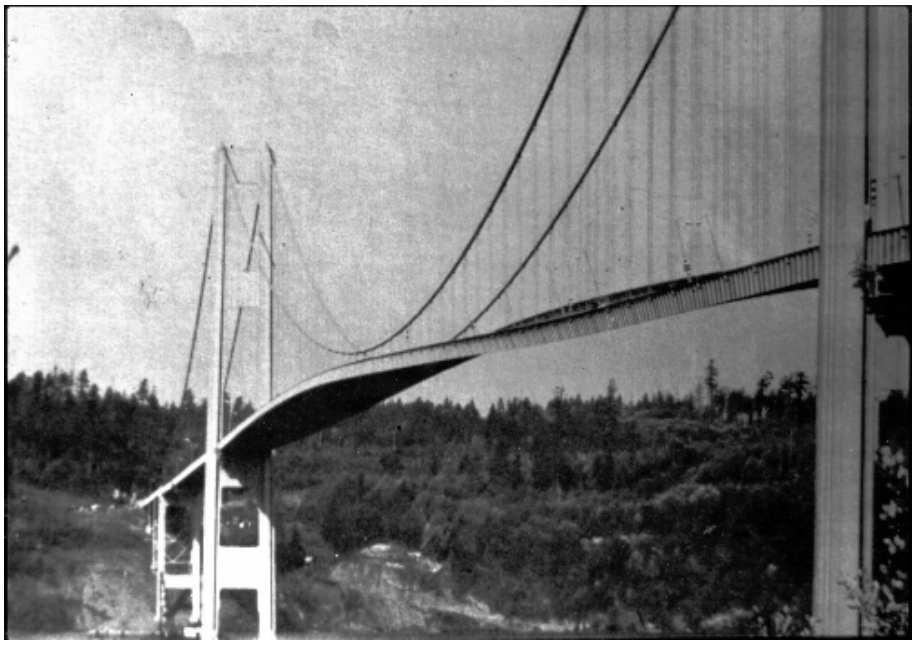


Fig. 1.1 Tacoma Narrows bridge minutes before collapse. Figure curtesy of Wikipedia.

Mechanical vibration is oscillatory motion of a body or structure about a reference configuration, typically an equilibrium configuration. In many engineering applications, this motion has a relatively small amplitude and occurs rapidly compared with the intended overall motion of the system. For example, a rotating shaft may undergo small lateral oscillations while completing its intended rotation, and the body of a moving vehicle may vibrate as it travels along a road. Although small amplitudes and relatively high frequencies are characteristic of many practical vibration problems, they are not essential requirements: large structures may vibrate slowly, and initially small vibrations may grow to large amplitudes.

Vibrations arise when inertia interacts with restoring forces, such as those produced by elastic deformation or gravity. In a simple mechanical system, energy passes repeatedly between kinetic energy associated with motion and potential energy associated with deformation or displacement. Dissipative effects, such as friction and material damping, remove mechanical energy and tend to reduce the motion. External forces or interactions with the surroundings may supply energy and sustain or amplify it. Understanding these competing effects is central to the study of vibration.

1.1.1 Why do vibrations matter?

Rotating machinery provides a common source of vibration. Ideally, the mass of a rotor is distributed so that its centre of mass lies on its axis of rotation. In practice, manufacturing tolerances, wear, damage, or attached components may cause an imbalance. As the rotor turns, this imbalance produces a rotating force that is transmitted through the bearings to the supporting structure. At a constant rotational speed, each transverse component of this force varies periodically at the rotational frequency. Other mechanisms, such as gear engagement and reciprocating motion, may introduce additional excitation frequencies.

The resulting vibration depends on both the excitation and the mechanical properties of the system. A structure has characteristic frequencies, called its *natural frequencies*, at which it can oscillate freely. When a periodic force acts close to one of these frequencies and couples to the corresponding pattern of motion, the response may become much larger than the response to the same force at other frequencies. This phenomenon is called *resonance*. Damping limits the resonant response in many practical systems, but the resulting amplitude may still be unacceptable. In an ideal undamped linear system, forcing at a natural frequency can produce oscillations whose amplitude grows without bound.

Large vibrations can cause excessive deformation, loss of clearance between moving components, loosening of connections, and structural failure. Even when the amplitude remains relatively small, repeated loading may cause fatigue damage over time. Small displacement amplitudes also do not necessarily imply small accelerations. For example, the harmonic displacement $x(t) = A\cos(\omega t)$ has acceleration $\ddot{x}(t) = -A\omega^2\cos(\omega t)$. Thus, at a sufficiently high frequency, a small displacement can be associated with a large acceleration and substantial inertial forces.

A dramatic illustration of the consequences of growing structural oscillations is the collapse of the Tacoma Narrows Bridge on 7 November 1940. Wind interacting with the flexible bridge deck produced large torsional oscillations that ultimately led to structural failure. In this case, the aerodynamic forces depended on the motion of the bridge itself, allowing energy from the wind to amplify the oscillations. This phenomenon, known as *aeroelastic flutter*, illustrates how interaction between a structure and its surroundings can cause initially small vibrations to grow to destructive amplitudes.

These observations lead to an important engineering question: *How can we prevent vibrations from becoming damaging or unstable?* Answering this question requires us to identify

the mechanisms that generate the motion, the conditions under which it becomes large, and the physical changes that can reduce it.

1.1.2 Comfort, noise, and vibration transmission

Structural safety is only one reason for studying vibration. Vibrations may also reduce comfort, interfere with precision equipment, and generate unwanted sound. In a vehicle, irregularities in the road excite the wheels, suspension, and vehicle body. The suspension must limit the motion transmitted to the occupants while maintaining adequate tyre contact with the road and acceptable vehicle handling. The desired response therefore involves several competing requirements.

Similarly, a machine may transmit oscillatory forces through its supports into a foundation and neighbouring structures. The resulting motion may disturb occupants or affect sensitive equipment, even when the machine itself operates satisfactorily. Vibrating surfaces can also radiate sound into the surrounding air. Reducing vibration at its source, limiting its transmission through supports, and reducing the associated sound radiation are therefore closely related engineering tasks.

In each case, the relevant measure of vibration depends on the application. We may need to limit displacement, velocity, acceleration, transmitted force, stress, or sound pressure. A modification that improves one of these quantities over one frequency range may worsen another quantity or increase the response elsewhere. Effective mitigation consequently requires more than simply making a structure stiffer or adding a damper: it requires an understanding of how the complete system responds.

1.1.3 From physical systems to mathematical models

Our first task is to understand and predict the behaviour of a vibrating system. This is the purpose of *vibration analysis*. We approach it by representing selected features of the physical system using mathematical objects, particularly differential equations. The construction of this representation is called *mathematical modelling*.

A model is an approximation developed for a particular purpose. We choose quantities that describe the motion, identify the important forces and interactions, and introduce assumptions that make the problem tractable. For example, a component may be treated as a rigid mass, the flexibility of its supports may be represented by springs, and energy dissipation may be approximated by viscous dampers. Applying the laws of mechanics then gives differential equations governing the chosen quantities. Together with initial conditions, the differential equation determines a predicted motion. Properties of its solutions describe features such as oscillation frequency, decay rate, response amplitude, and dependence on forcing frequency. In this way, questions about physical behaviour become mathematical questions about solutions of differential equations. Whether a disturbance decays, persists, or grows becomes a question about the stability and long-term behaviour of those solutions.

The correspondence between the mathematical solution and the physical motion is limited by the modelling assumptions. In particular, linear spring and damping laws are often local approximations valid for sufficiently small motions. If a model predicts a rapidly growing amplitude, it may indicate that the system is approaching an unsafe operating condition; it does not imply that the physical system will continue to follow the same linear equation at arbitrarily large amplitudes. Predictions must therefore be interpreted in the light of the assumptions and, where possible, checked against observations or experiments.

Some systems can be represented adequately by a finite number of displacement coordinates, leading to *lumped parameter models* governed by ordinary differential equations. Others, such as flexible beams, require displacement to be described as a function of both position and time. These *distributed parameter models* are governed by partial differential equations. Both descriptions follow the same principle: represent the important physical mechanisms mathematically and use the resulting equations to understand the motion.

1.1.4 From vibration analysis to mitigation and control

Once we understand the behaviour, the next task is to determine how it can be modified. This is the purpose of *vibration mitigation and control*. Possible measures include balancing rotating components, changing operating speeds, modifying mass or stiffness, adding damping, introducing vibration absorbers, and isolating equipment from its surroundings. Active control provides another possibility: sensors measure the response, and actuators apply forces designed to modify it.

Each physical intervention has a mathematical counterpart. Changing mass, stiffness, or damping changes the coefficients of the governing equations. Balancing a rotor changes the excitation. Adding an absorber introduces additional coordinates and coupled equations. An active controller introduces a force that depends on measured motion or other available information. We can therefore use a model to investigate how a proposed intervention changes the response before implementing it physically.

The objective is not necessarily to eliminate all motion. It is to achieve an acceptable response over the relevant operating conditions while respecting constraints such as cost, weight, space, reliability, and performance. A successful design must also remain effective when physical parameters and excitation conditions differ from their nominal values.

These notes introduce the basic ideas needed for this progression from physical behaviour to mathematical modelling, from modelling to vibration analysis, and from analysis to mitigation and control. We begin with simple lumped parameter systems to develop an understanding of free vibration, forced vibration, damping, and resonance. We then consider coupled systems, natural modes of vibration, and distributed parameter descriptions of flexible structures. Periodically varying parameters provide a further mechanism through which oscillations may grow or stability may change.

Throughout, the central aim is to connect the mathematics to the mechanics: to understand what the equations predict, why the physical system behaves in that way, and how changes to the system can produce a more desirable response.

1.2 First Approximation of a Simple Vibratory System

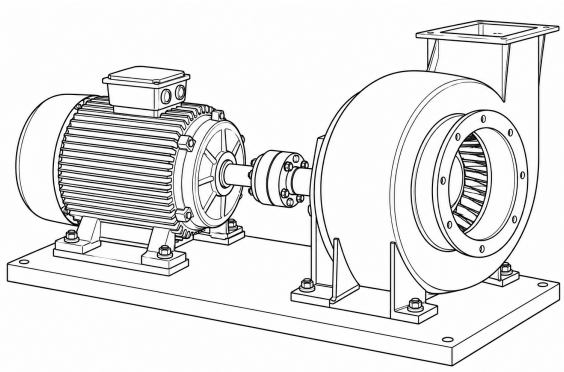


Fig. 1.2 A motor driven blower.

We begin with a simple model that captures three essential features of a vibrating system: inertia, elasticity, and energy dissipation. Using a motor-driven blower as an example, we show how these features lead to a spring–mass–damper model and a differential equation describing its motion. We then examine the responses to initial disturbances and external forcing, introducing natural frequency, damping, and resonance.

Consider the blower shown in Figure. 1.2. As its rotor turns, the machine and its supporting structure may undergo small oscillatory motions. We would like to understand the origin of these vibrations and predict how their amplitude depends on the rotational speed and the mechanical properties of the system.

Although the blower consists of many interconnected components, we need not account for every detail to obtain a useful first approximation of its behaviour. If the dominant motion is a small vertical displacement of the machine, we may represent this motion by a single coordinate $x(t)$. The inertia of the moving assembly, the flexibility of its supports, and the dissipation of mechanical energy can then be represented by a mass, a spring, and a damper, respectively. This gives the idealised system illustrated in Fig. 1.3.

We shall use this example to address the following questions:

- What causes the blower to vibrate, and how is the frequency of the excitation related to the rotational speed of the rotor?
- Under what assumptions can the small-amplitude motion be approximated by a mathematical model?
- How does the system respond to a small initial vertical displacement, such as one caused by an accidental disturbance?
- How does the steady-state vibration amplitude behave at very low and very high rotational speeds?
- How does the steady-state vibration amplitude vary over the full range of rotational speeds, and under what conditions can it become large?

We answer these questions in the sections below.

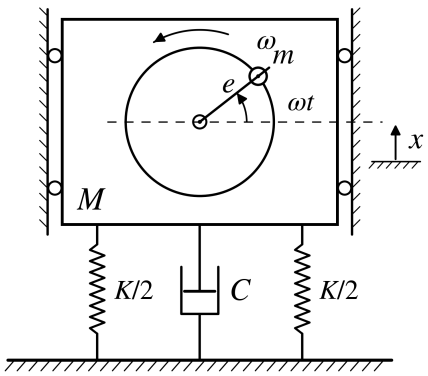


Fig. 1.3 A lumped parameter approximation of the motor driven blower.

1.2.1 The cause of vibration: rotating imbalance

In an ideally balanced rotor, the centre of mass lies on the axis of rotation. In practice, manufacturing tolerances, uneven wear, or material deposited on the rotor may cause its centre of mass to lie a small distance away from this axis. This displacement is called the *eccentricity*, and the resulting mass distribution is said to have a *rotating imbalance*.

Consider the idealisation in Fig. 1.3. Let m denote the mass of the rotor and let e be the distance between its centre of mass and its axis of rotation. Assume that the rotor is rigid and rotates at a prescribed constant angular speed ω.

Due to the eccentricity e of the center of mass of the rotating object a force $me\omega^2$ acts on the center to balance the centrifugal acceleration. We see that the vertical and horizontal components of this force are respectively given by:

$$F_v = me\omega^2 \sin \omega t, \tag{1.1}$$

$$F_h = me\omega^2 \cos \omega t \tag{1.2}$$

All rotating objects have an eccentric center of mass due to practical difficulties of achieving perfect precision. Thus all bodies with rotating objects will inevitably experience sinusoidal forces with frequency equal to the frequency of rotation of the object.

An important feature of rotating imbalance is that changing the rotational speed changes both the frequency and the amplitude of the excitation. Doubling the rotational speed therefore doubles the excitation frequency and increases the force amplitude by a factor of four. However, this does not imply that the displacement amplitude also increases by a factor of four. The displacement depends on the response of the mass, supports, and damping to the applied force, which we must determine from the equations of motion.

Balancing the rotor reduces the product me and hence reduces this source of excitation. Nevertheless, rotating imbalance is only one possible source of vibration; aerodynamic forces,

misalignment, and bearing defects may also contribute. In the following analysis, we isolate the effect of rotating imbalance to develop a first mathematical model of the blower's motion.

Let M denote the total mass of the moving assembly, including the rotating mass m. At static equilibrium, the two springs, each of stiffness $K/2$, undergo an initial compression Δ such that $K\Delta = Mg$. Let $x(t)$ denote the upward displacement of the housing from this equilibrium position. The total upward spring force is then $K(\Delta - x)$, while the damping force is $-C\dot{x}$. Since the vertical displacement of the eccentric mass relative to the shaft centre is $e\sin(\omega t)$, the vertical momentum balance for the complete assembly gives

$$(M-m)\ddot{x} + m\big(\ddot{x} - e\omega^2\sin(\omega t)\big) = K(\Delta - x) - C\dot{x} - Mg.$$

Using $K\Delta = Mg$, this reduces to

$$M\ddot{x} + C\dot{x} + Kx = me\omega^2\sin(\omega t).$$

Thus gravity determines the static spring compression but does not appear explicitly in the equation for motion measured from equilibrium, under the assumed linear spring model.

1.2.2 The motion of a Spring Mass Damper System

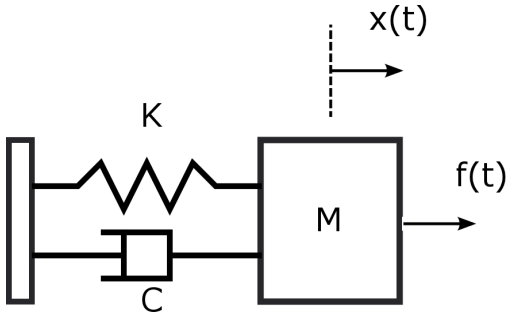


Fig. 1.4 Spring Mass Damper System.

Just like for the blower shown in Figure 1.2 the first approximation of a simple vibrating system would typically result in a spring mass damper system such as the one shown in Figure 1.4. For small deflections and velocities, the viscous damping force exerted by the damper can be approximated by $f_d = -C\dot{x}$ and the spring force can be approximated by $f_s = -Kx$. Considering the free body diagram of the systems shown in figure-1.5 and applying Newton's equations for the mass M yields the following linear ODE.

$$M\ddot{x}(t) + C\dot{x}(t) + Kx(t) = f(t). \tag{1.3}$$

Dividing by M and setting $\omega_n^2 = \frac{K}{M}$ and $2\zeta\omega_n = \frac{C}{M}$ in (1.3) we have

$$\ddot{x}(t) + 2\zeta\omega_n\dot{x}(t) + \omega_n^2 x(t) = \frac{1}{M}f(t). \tag{1.4}$$

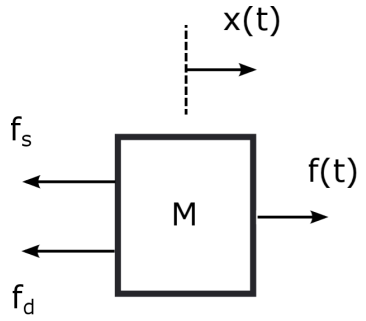


Fig. 1.5 The Free Body Diagram of the Spring Mass Damper System.

The parameter ω_n is called the *undamped natural frequency* of the system and ζ is called the *damping ratio* of the system. The reason for this nomenclature will be apparent when we investigate the solutions of this differential equation.

Being a linear second order ODE one can solve (1.4). Since this differential equation was derived by considering small deflections and velocities of the mass it is interesting to find out if the solutions of the differential equation correspond to the physical behaviour of the SMD system for small deflections.

One way of finding the solutions of this differential equation is the Laplace transform method. For a given $f(t)$ that does not grow faster than an exponential function it can be shown, owing to the linearity of the system, that the solution will also not grow faster than an exponential function. Thus taking Laplace transform of both sides of (1.4) and using the linearity property of the Laplace transform we have

$$\mathscr{L}\left\{\ddot{x}(t)+2\zeta\omega_n\dot{x}(t)+\omega_n^2x(t)\right\}=\frac{1}{M}\mathscr{L}\left\{f(t)\right\}, \tag{1.5}$$

$$(s^2+2\zeta\omega_n s+\omega_n^2)X(s)-\dot{x}(0)-(s+2\zeta\omega_n)x(0)=\frac{1}{M}F(s). \tag{1.6}$$

This yields

$$X(s)=\frac{1}{s^2+2\zeta\omega_n s+\omega_n^2}\dot{x}(0)+\frac{s+2\zeta\omega_n}{s^2+2\zeta\omega_n s+\omega_n^2}x(0)+\frac{1/M}{s^2+2\zeta\omega_n s+\omega_n^2}F(s), \tag{1.7}$$

where $F(s)=\mathscr{L}\left\{f(t)\right\}$. Since the Laplace transform uniquely determines a continuous function of exponential order, the solution is uniquely determined by $x(t)=\mathscr{L}^{-1}\{X(s)\}$. Using the linearity property of the inverse we thus have

$$x(t)=\dot{x}(0)\mathscr{L}^{-1}\left\{\frac{1}{s^2+2\zeta\omega_n s+\omega_n^2}\right\}+x(0)\mathscr{L}^{-1}\left\{\frac{s+2\zeta\omega_n}{s^2+2\zeta\omega_n s+\omega_n^2}\right\}+\mathscr{L}^{-1}\left\{\frac{1/M}{s^2+2\zeta\omega_n s+\omega_n^2}F(s)\right\}. \tag{1.8}$$

Observe that the first two terms depend on the initial conditions and do not depend on the forcing while the last term does depend on the forcing but is in depend of the initial conditions. Therefore we may breakup the solution into two parts such that $x(t)=x_{IC}(t)+x_f(t)$ where

$$x_{IC}(t) = \dot{x}(0)\mathscr{L}^{-1}\left\{\frac{1}{s^2+2\zeta\omega_n s+\omega_n^2}\right\} + x(0)\mathscr{L}^{-1}\left\{\frac{s+2\zeta\omega_n}{s^2+2\zeta\omega_n s+\omega_n^2}\right\}, \tag{1.9}$$

$$x_f(t) = \mathscr{L}^{-1}\left\{\frac{1/M}{s^2+2\zeta\omega_n s+\omega_n^2}F(s)\right\}. \tag{1.10}$$

We will call the first part two parts given by $x_{IC}(t)$ that depends only on the initial conditions the *initial condition response* while we will call the last part that depends only on the forcing the *forced response*.

The inverse Laplace transforms of each of the terms are obtained by expanding the terms in a partial fraction expansion. This depends on the roots of the polynomial

$$\Delta(s) = s^2+2\zeta\omega_n s+\omega_n^2 = (s+\lambda_1)(s+\lambda_2). \tag{1.11}$$

The two roots , $-\lambda_1$ and $-\lambda_2$, of the polynomial are given by

$$-\lambda_1 = \omega_n(-\zeta+\sqrt{\zeta^2-1}),$$
$$-\lambda_2 = \omega_n(-\zeta-\sqrt{\zeta^2-1}).$$

Since the character of the solution is determined by the roots of the above polynomial it is referred to as the *characteristic polynomial* of the system. These roots will be complex and conjugate, real and distinct, or real and repeated depending on if $0 \leq \zeta < 1$, $\zeta > 1$, or $\zeta = 1$ respectively.

Let us consider the case $0 \leq \zeta < 1$. In this case the roots of the characteristic polynomial $\Delta(s) = s^2+2\zeta\omega_n s+\omega_n^2$ of the system are given by $-\lambda_1 = -\zeta\omega_n + i\omega_d$ and $-\lambda_2 = -\zeta\omega_n - i\omega_d$. A typical location of the roots of the characteristic polynomial in this case is shown in figure-1.6. The plot of $x_{IC}(t)$ for several different values of ζ in the range $0 \leq \zeta \leq 1$ is shown in figure-1.7.

Assume that the forcing is such that for distinct $\beta_1, \cdots, \beta_k$ and $\beta_i \neq \lambda_1$ and $\beta_i \neq \lambda_2$ for all i

$$\mathscr{L}\{f(t)\} = F(s) = \frac{N(s)}{(s+\beta_1)(s+\beta_2)\cdots(s+\beta_k)}.$$

When $0 \leq \zeta < 1$ it is a straight forward exercise in partial fraction expansion to show that (1.9) and (1.10) reduce to

$$x_{IC}(t) = e^{-\zeta\omega_n t}\left(\frac{\dot{x}(0)}{\omega_d}\cos(\omega_d t-\pi/2) + \frac{x(0)}{\sqrt{1-\zeta^2}}\sin(\omega_d t+\phi_{IC})\right)1(t), \tag{1.12}$$

$$x_f(t) = \left(2|\alpha|e^{-\zeta\omega_n t}\cos(\omega_d t+\phi) + \left(r_1 e^{-\beta_1 t}+\cdots+r_k e^{-\beta_k t}\right)\right)1(t), \tag{1.13}$$

where $\omega_d = \omega_n\sqrt{1-\zeta^2}$, $\phi = \arg(\alpha)$, $\phi_{IC} = \arcsin\left(\sqrt{1-\zeta^2}\right)$, and the constants α and r_i are determined by,

$$\alpha = (s+\lambda_1)G(s)F(s)\big|_{s=-\lambda_1},$$
$$r_i = (s+\beta_i)G(s)F(s)\big|_{s=-\beta_i}.$$

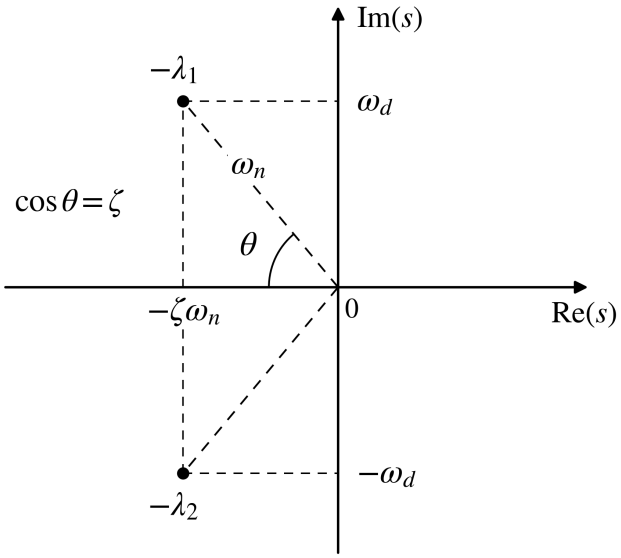


Fig. 1.6 A typical location of the roots, $\lambda = -\lambda_1, \lambda = -\lambda_2$, of the characteristic polynomial $\Delta(s)$ when $0 \leq \zeta < 1$.

When $\zeta = 1$ the system is said to be *critically* damped. When $\zeta = 1$, the characteristic polynomial has a repeated root at $s = -\omega_n$. The initial condition response is

$$x_{IC}(t) = \left[x(0) + \left(\dot{x}(0) + \omega_n x(0)\right)t\right] e^{-\omega_n t}\, 1(t). \tag{1.14}$$

The forced response is

$$x_f(t) = \mathscr{L}^{-1}\left\{\frac{1/M}{(s+\omega_n)^2}F(s)\right\} = \frac{1}{M}\int_0^t (t-\tau)e^{-\omega_n(t-\tau)}f(\tau)\,d\tau. \tag{1.15}$$

For the rational input considered above, assuming that its poles $-\beta_i$ are simple and distinct from $-\omega_n$, this becomes

$$x_f(t) = \frac{1}{M}\left[(A+Bt)e^{-\omega_n t} + \sum_{i=1}^{k} r_i e^{-\beta_i t}\right] 1(t), \tag{1.16}$$

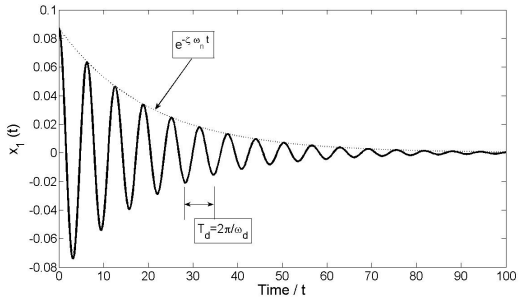
where

$$A = F'(-\omega_n), \qquad B = F(-\omega_n), \qquad r_i = \left.\frac{(s+\beta_i)F(s)}{(s+\omega_n)^2}\right|_{s=-\beta_i}.$$

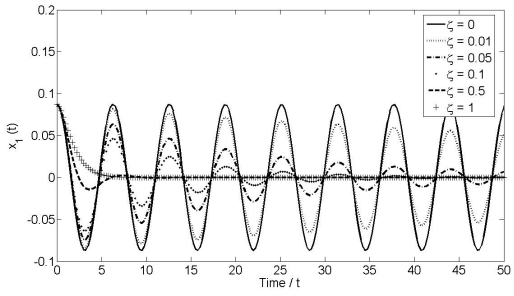
Here $F'(s)$ denotes differentiation with respect to s.

1.2.2.1 Free Vibrations

When the external forcing $f(t)$ is zero, that is $F(s) = 0$, we say that the system is exhibiting *free vibrations* and if not we say that the system is exhibiting a *forced vibration*. In the case



(a) For $\zeta = 0.1$



(b) For different values of ζ in the range $0 \leq \zeta \leq 1$

Fig. 1.7 Initial condition response $x_{IC}(t)$ for several different values of ζ in the range $0 \leq \zeta \leq 1$.

of free vibrations we see that $x_f(t) \equiv 0$ and thus the solution corresponds only to the initial condition part $x_{IC}(t)$. Thus in the case of free vibrations the response of the system is

$$x(t) = x_{IC}(t) = e^{-\zeta\omega_n t}\left(\frac{\dot{x}(0)}{\omega_d}\cos(\omega_d t - \pi/2) + \frac{x(0)}{\sqrt{1-\zeta^2}}\sin(\omega_d t + \phi_{IC})\right)1(t), \tag{1.17}$$

when $0 < \zeta < 1$. Note that this is of the form

$$x(t) = x_{IC}(t) = Ae^{-\zeta\omega_n t}\sin(\omega_d t + \phi)1(t), \tag{1.18}$$

where A and ϕ are constants that depend on ζ, ω_n and the initial conditions $x(0), \dot{x}(0)$.

Figure-1.7 shows the plot of $x_{IC}(t)$ for several different values of ζ. The solution exhibits an amplitude decaying oscillatory behavior, of frequency $\omega_d = \omega_n\sqrt{1-\zeta^2}$ for any initial condition. Thus for $0 < \zeta < 1$ the system said to behave in an *underdamped* manner and ω_d the frequency of oscillation will be referred to as the *damped natural frequency* of the system. Note that the damped free vibration response tends to zero asymptotically as t tends to infinity.

When $\zeta = 0$, that is when there is no damping, the initial condition response is given by

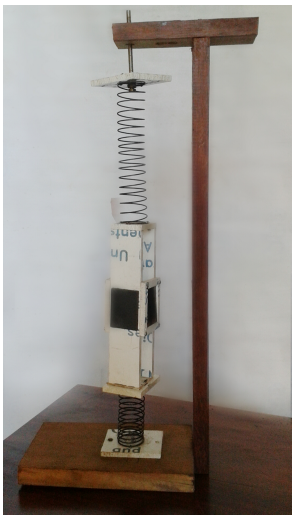
$$x(t) = x_{IC}(t) = \left(\frac{\dot{x}(0)}{\omega_n}\cos(\omega_n t - \pi/2) + x(0)\sin(\omega_n t + \pi/2)\right)1(t), \tag{1.19}$$

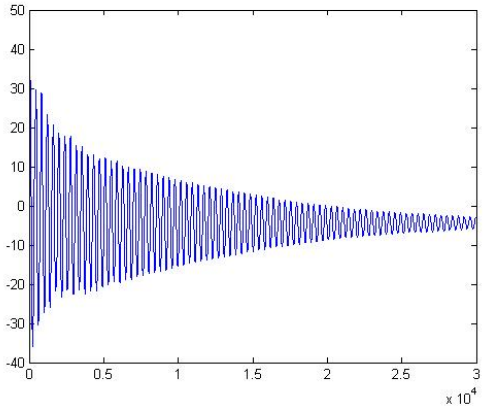
which takes the form $x(t) = x_{IC}(t) = A\cos(\omega_n t + \phi)$ and represents an undamped oscillation of frequency ω_n. This motion is referred to as *simple harmonic motion*.

Twice differentiating (1.18) we see that the acceleration of the mass, when $0 < \zeta < 1$, takes the form

$$\ddot{x}(t) = \ddot{x}_{IC}(t) = A_c e^{-\zeta\omega_n t}\sin(\omega_d t + \phi_c)1(t), \tag{1.20}$$

where A_c and ϕ_c are again constants that depend only on ζ, ω_n and the initial conditions $x(0), \dot{x}(0)$. Comparing this with the experimentally obtained acceleration of the unforced small oscillatory response shown in figure-1.2.2.1 (b) of the spring mass damper system shown in figure-1.2.2.1 (a) we see that the physical behaviour of the system matches very well with the theoretically estimated acceleration of the mass that is given by (1.20).





(a) Experimental setup of a SMD system (b) Experimentally measured acceleration of the SMD system.

1.2.2.2 Forced Vibrations and Resonance of Lightly Damped Systems

In this section we consider the case where $0 < \zeta < 1$. In the case of forced vibrations it suffices to look at only the forced response $x_f(t)$ since the initial condition part of the response dies out when damping is positive. We can show that the forced response also can be broken up into two parts:

$$x_f(t) = x_{tr}(t) + x_{ss}(t),$$

where

$$x_{tr}(t) = \left(2|\alpha|\, e^{-\zeta\omega_n t}\cos\left(\omega_d t + \phi\right)\right) 1(t), \tag{1.21}$$

$$x_{ss}(t) = \left(r_1 e^{-\beta_1 t} + \cdots + r_k e^{-\beta_k t}\right)\ 1(t). \tag{1.22}$$

Once again we see that for positive damping the transient part of the response $x_{tr}(t)$ asymptotically tends to zero as t tends to infinity and hence it suffices to only look at the steady state part of the forced response.

Let us consider a very particular type of forcing

$$f(t) = F_0 \cos\left(\omega t\right) 1(t).$$

The Laplace transform of $f(t)$ is

$$F(s) = F_0\left(\frac{s}{s^2+\omega^2}\right) = F_0\left(\frac{s}{(s+i\omega)(s-i\omega)}\right).$$

From (1.22) it can be shown that the steady state response of the system for sinusoidal forcing is

$$x_{ss}(t) = \chi(\omega)\, F_0 \cos(\omega t + \phi(\omega))\, 1(t), \tag{1.23}$$

where

$$\chi(\omega) = \frac{1/M}{\sqrt{(\omega_n^2 - \omega^2)^2 + 4\zeta^2\omega^2\omega_n^2}} \tag{1.24}$$

and

$$\phi(\omega) = -\operatorname{atan2}\left(2\zeta\omega\omega_n,\, \omega_n^2 - \omega^2\right). \tag{1.25}$$

Thus we see that the steady state response to sinusoidal forcing is also sinusoidal with frequency of oscillation equal to that of the forcing frequency ω. However it differs from the forcing in two distinct ways. One is that the amplitude of the steady state solution is $\chi(\omega)$ times the magnitude of the forcing while the other is that the phase of the steady state solution is shifted by $\phi(\omega)$ from the phase of the forcing. It is also important to note that the amplitude magnification and the phase shift depend on the forcing frequency ω. We will see that this observation has crucial implications for vibration analysis and control system design. In summary:

For positively damped ($\zeta > 0$) systems the solution of the system approaches the steady state response. The steady state response to sinusoidal forcing is also sinusoidal with oscillation frequency equal to that of the forcing frequency ω. However the amplitude of the steady state sinusoidal response is $\chi(\omega)$ times the amplitude of the forcing amplitude and the phase of the steady state sinusoidal response is shifted by $\phi(\omega)$ from that of the forcing. The amplification factor $\chi(\omega)$ and the phase shift $\phi(\omega)$ depend on the forcing frequency ω and are given by (1.24) and (1.25) respectively and are plotted in figure-1.8.

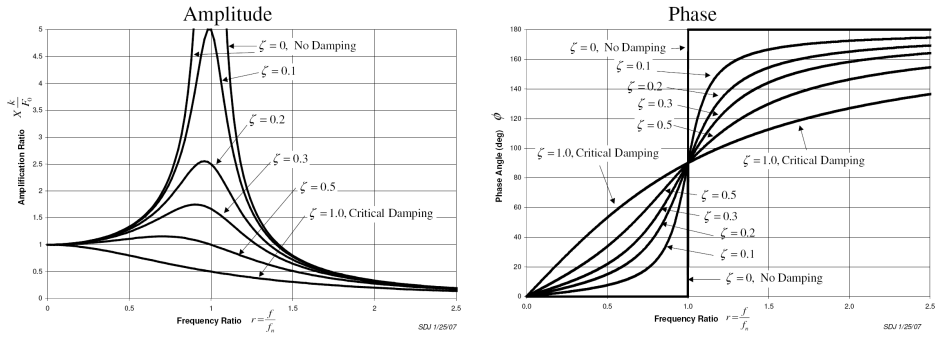


Fig. 1.8 The Frequency response of the Spring Mass Damper system

The important implication of this observation in forced vibration analysis is that the amplitude of the forced response, $\chi(\omega)$, reaches a maximum when the forcing frequency ω is equal to a certain value ω_r. Differentiating (1.24) we find this value to be given by

$$\omega_r = \omega_n\sqrt{1-2\zeta^2}, \qquad 0 < \zeta < \frac{1}{\sqrt{2}}. \tag{1.26}$$

Observe that as the damping becomes negligible the resonance frequency tends to the undamped natural frequency, ω_n. When the forcing frequency reaches the resonance frequency the amplitude of the forced response reaches a maximum value of

$$x_{ssmax} = \frac{F_0/M}{2\zeta\omega_n^2\sqrt{1-\zeta^2}}, \qquad 0 < \zeta < \frac{1}{\sqrt{2}}. \tag{1.27}$$

This phenomena is called *Resonance* and ω_r is called the damped resonance frequency. Observe that when the damping in the system is very small ($\zeta \approx 0$) the amplitude of vibration of the forced system becomes very large and can cause catastrophic damage to the system. Note that these expressions are valid only if F_0 is independent of ω.

We conclude this section by recalling the sometimes useful following results. Observe that $\chi(\omega)$ and $\phi(\omega)$ can be written as

$$\chi(\omega) = |G(i\omega)|$$

and

$$\phi(\omega) = \angle G(i\omega),$$

where

$$G(s) = \frac{1/M}{(s^2 + 2\zeta\omega_n s + \omega_n^2)},$$

is called the *transfer function* of the spring mass damper system. In this context $G(i\omega)$ is called the frequency response of the system. Its magnitude $\chi(\omega)$ is called the *Magnitude Bode* response and its phase $\phi(\omega)$ is called the *Phase Bode* response of the system. These are plotted in figure-1.8 as a function of the ratio $\omega/\omega_n = f/f_n$.

Recall that the convolution theorem gives

$$\mathscr{L}^{-1}\{G(s)U(s)\} = \int_0^t g(t-\tau)u(\tau)\,d\tau, \tag{1.28}$$

where

$$g(t) = \mathscr{L}^{-1}\{G(s)\}, \qquad u(t) = \mathscr{L}^{-1}\{U(s)\}.$$

Hence the forced response can also be equivalently written as

$$x_f(t) = \int_0^t g(t-\tau)u(\tau)\,d\tau. \tag{1.29}$$

In the case where $u(t) = f(t)$ and

$$G(s) = \frac{1/M}{s^2 + 2\zeta\omega_n s + \omega_n^2},$$

we have for $0 < \zeta < 1$

$$x_f(t) = \int_0^t \frac{e^{-\zeta\omega_n(t-\tau)}}{M\omega_d} \sin\big(\omega_d(t-\tau)\big)\, f(\tau)\, d\tau, \qquad \omega_d = \omega_n\sqrt{1-\zeta^2}.$$

The exponential factor causes the contribution of each past excitation to decay with elapsed time when $\zeta > 0$. When $\zeta = 0$, we have $\omega_d = \omega_n$, and the oscillatory contribution of each past excitation persists without decay.

For undamped spring-mass system with harmonic forcing at the undamped natural frequency, we have $\zeta = 0$, $\omega = \omega_n$ and

$$f(t) = F_0 \cos(\omega t).$$

Then the convolution integral gives

$$x_f(t) = \frac{F_0}{2M\omega} t \sin(\omega t),$$

whose amplitude grows linearly with time.

1.3 A 2-DOF Coupled Vibratory System

In this section we will begin to develop the tools necessary to analyze the vibratory motion of coupled systems such as multi-story buildings, rotary machines mounted on elastic foundations, and multi-rotor shafts. Specifically to keep things simple we will address in detail a system that can be approximated as a 2-DOF coupled spring mass damper system with external forcing.

Consider the coupled 2-DOF spring mass damper system shown in figure-1.9. Choosing the equilibrium positions of the masses as the reference points and applying Newton's laws we obtain

$$m_1\ddot{x}_1 + (c_1+c_2)\dot{x}_1 + (k_1+k_2)x_1 - c_2\dot{x}_2 - k_2x_2 = f_1(t), \tag{1.30}$$

$$m_2\ddot{x}_2 + c_2\dot{x}_2 + k_2x_2 - c_2\dot{x}_1 - k_2x_1 = 0, \tag{1.31}$$

where we have set $f_1(t) = p_0\cos(\omega t)$. We can write (1.30)-(1.31) as

$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix}\begin{bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{bmatrix} + \begin{bmatrix} (c_1+c_2) & -c_2 \\ -c_2 & c_2 \end{bmatrix}\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} + \begin{bmatrix} (k_1+k_2) & -k_2 \\ -k_2 & k_2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} f_1(t) \\ 0 \end{bmatrix} \tag{1.32}$$

Let

$$x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix},\ f(t) = \begin{bmatrix} f_1(t) \\ 0 \end{bmatrix},\ M = \begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix},\ C = \begin{bmatrix} (c_1+c_2) & -c_2 \\ -c_2 & c_2 \end{bmatrix}\ K = \begin{bmatrix} (k_1+k_2) & -k_2 \\ -k_2 & k_2 \end{bmatrix}.$$

Then we can write (1.32) as

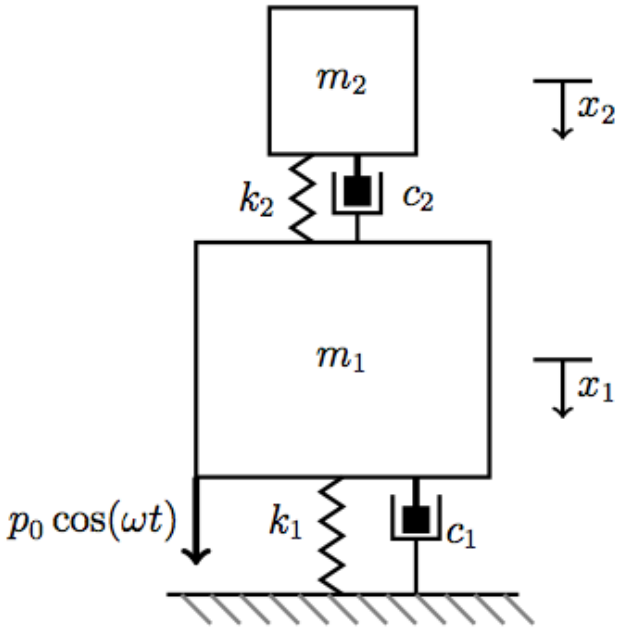


Fig. 1.9 A 2-DOF coupled spring mass damper system

$$M\ddot{x}+C\dot{x}+Kx=f(t). \tag{1.33}$$

Notice the similarity with the 1-DOF spring mass damper system. We are interested in solving this 2nd order matrix ODE in order to understand the vibratory behavior of the system. We do so in two steps. Recalling the crucial role the undamped natural frequency played in the resonance analysis of the 1-DOF spring mass damper system we first investigate undamped free vibrations of the system and then consider the damped forced vibration behavior of the system.

1.3.1 Frequency Response

Similar to the 1-DOF case we begin by taking a close look at the response of the system by taking the Laplace transform of both sides of (1.33)

$$\mathscr{L}\{M\ddot{x}+C\dot{x}+Kx\}=\mathscr{L}\{f(t)\}. \tag{1.34}$$

Then we have

$$(Ms^2+Cs+K)X(s)=(Ms+C)x(0)+M\dot{x}(0)+F(s), \tag{1.35}$$

where

$$X(s)=\mathscr{L}\{x(t)\},\quad F(s)=\mathscr{L}\{f(t)\}.$$

Let

$$\Delta(s) = (Ms^2 + Cs + K) = \begin{bmatrix} \left(m_1 s^2 + (c_1 + c_2)s + (k_1 + k_2)\right) & -(c_2 s + k_2) \\ -(c_2 s + k_2) & (m_2 s^2 + c_2 s + k_2) \end{bmatrix}$$

Then from (1.35) we have that

$$X(s) = \Delta(s)^{-1}\left((Ms + C)x(0) + M\dot{x}(0)\right) + \Delta(s)^{-1}F(s),$$

and that

$$x(t) = \mathscr{L}^{-1}\left\{\Delta(s)^{-1}\left((Ms + C)x(0) + M\dot{x}(0)\right)\right\} + \mathscr{L}^{-1}\left\{\Delta(s)^{-1}F(s)\right\}.$$

The first part of the right hand side of the above equation depends only on the initial conditions while the second part depends only on the forcing. Thus similar to the 1-DOF case we can write

$$x(t) = x_{IC}(t) + x_f(t)$$

where

$$x_{IC}(t) = \mathscr{L}^{-1}\left\{\Delta(s)^{-1}\left((Ms + C)x(0) + M\dot{x}(0)\right)\right\}, \tag{1.36}$$

$$x_f(t) = \mathscr{L}^{-1}\left\{\Delta(s)^{-1}F(s)\right\}. \tag{1.37}$$

Similar to the 1-DOF case it can be shown that when all $k_1, k_2, c_1, c_2 > 0$ the initial condition response $x_{IC}(t)$ tends to zero as t tends to infinity. Thus it suffices to only consider the forced response $x_f(t)$. It can be shown that

$$x_f(t) = \mathscr{L}^{-1}\left\{\begin{bmatrix} G_1(s)F_1(s) \\ G_2(s)F_1(s) \end{bmatrix}\right\},$$

where

$$F_1(s) = \mathscr{L}\{f_1(t)\}, \quad G_1(s) = \frac{m_2 s^2 + c_2 s + k_2}{|\Delta(s)|}, \quad G_2(s) = \frac{c_2 s + k_2}{|\Delta(s)|},$$

and

$$|\Delta(s)| = \det(\Delta(s)) = \left(m_1 s^2 + (c_1 + c_2)s + (k_1 + k_2)\right)(m_2 s^2 + c_2 s + k_2) - (c_2 s + k_2)^2. \tag{1.38}$$

It can be shown that when all $k_1, k_2, c_1, c_2 > 0$ the transient part of the forced response asymptotically tends to zero when t tends to infinity and thus the solution tends to the steady state part of the forced response. Hence if $f_1(t) = p_0 \cos(\omega t)$ we have that

$$x_{ss}(t) = \begin{bmatrix} \chi_1(\omega) p_0 \cos(\omega t + \phi_1(\omega)) \\ \chi_2(\omega) p_0 \cos(\omega t + \phi_2(\omega)) \end{bmatrix},$$

where

$$\chi_1(\omega) = |G_1(i\omega)|, \tag{1.39}$$

$$\chi_2(\omega) = |G_2(i\omega)|, \tag{1.40}$$

and

$$\phi_1(\omega) = \angle G_1(i\omega), \tag{1.41}$$
$$\phi_2(\omega) = \angle G_2(i\omega), \tag{1.42}$$

Introducing the parameters

$$\omega_{n1}^2 = \frac{k_1}{m_1}, \qquad \omega_{n2}^2 = \frac{k_2}{m_2}, \qquad \mu = \frac{m_2}{m_1},$$
$$2\zeta_1\omega_{n1} = \frac{c_1}{m_1}, \qquad 2\zeta_2\omega_{n2} = \frac{c_2}{m_2},$$

the characteristic polynomial (1.38) becomes

$$\det\Delta(s) = m_1m_2\Big[s^4 + 2(\zeta_1\omega_{n1} + (1+\mu)\zeta_2\omega_{n2})s^3 + (\omega_{n1}^2 + (1+\mu)\omega_{n2}^2 + 4\zeta_1\zeta_2\omega_{n1}\omega_{n2})s^2 + 2(\zeta_2\omega_{n2}\omega_{n1}^2 + \zeta_1\omega_{n1}\omega_{n2}^2)s + \omega_{n1}^2\omega_{n2}^2\Big].$$

Here ω_{n1} and ω_{n2} are component frequency parameters; the natural frequencies of the coupled system will be determined below.

Define the real-valued functions

$$A(\omega) = \omega^4 - \big(\omega_{n1}^2 + (1+\mu)\omega_{n2}^2 + 4\zeta_1\zeta_2\omega_{n1}\omega_{n2}\big)\omega^2 + \omega_{n1}^2\omega_{n2}^2,$$
$$B(\omega) = 2\Big[\big(\zeta_2\omega_{n2}\omega_{n1}^2 + \zeta_1\omega_{n1}\omega_{n2}^2\big)\omega - \big(\zeta_1\omega_{n1} + (1+\mu)\zeta_2\omega_{n2}\big)\omega^3\Big].$$

Substituting $s = i\omega$ gives

$$\det\Delta(i\omega) = m_1m_2\big(A(\omega) + iB(\omega)\big).$$

Consequently, the frequency-response magnitudes are

$$\chi_1(\omega) = \frac{1}{m_1}\frac{\sqrt{(\omega_{n2}^2 - \omega^2)^2 + 4\zeta_2^2\omega_{n2}^2\omega^2}}{\sqrt{A(\omega)^2 + B(\omega)^2}},$$
$$\chi_2(\omega) = \frac{1}{m_1}\frac{\sqrt{\omega_{n2}^4 + 4\zeta_2^2\omega_{n2}^2\omega^2}}{\sqrt{A(\omega)^2 + B(\omega)^2}}.$$

Making the further substitution $\omega_{n2} = \alpha\omega_{n1}$, where $\alpha > 0$, define

$$A_\alpha(\omega) = \omega^4 - \big(1 + (1+\mu)\alpha^2 + 4\alpha\zeta_1\zeta_2\big)\omega_{n1}^2\omega^2 + \alpha^2\omega_{n1}^4,$$
$$B_\alpha(\omega) = 2\Big[(\zeta_2\alpha + \zeta_1\alpha^2)\omega_{n1}^3\omega - \big(\zeta_1 + (1+\mu)\zeta_2\alpha\big)\omega_{n1}\omega^3\Big].$$

Then

$$\chi_1(\omega) = \frac{1}{m_1}\frac{\sqrt{(\alpha^2\omega_{n1}^2 - \omega^2)^2 + 4\zeta_2^2\alpha^2\omega_{n1}^2\omega^2}}{\sqrt{A_\alpha(\omega)^2 + B_\alpha(\omega)^2}},$$
$$\chi_2(\omega) = \frac{\alpha\omega_{n1}}{m_1}\frac{\sqrt{\alpha^2\omega_{n1}^2 + 4\zeta_2^2\omega^2}}{\sqrt{A_\alpha(\omega)^2 + B_\alpha(\omega)^2}}.$$

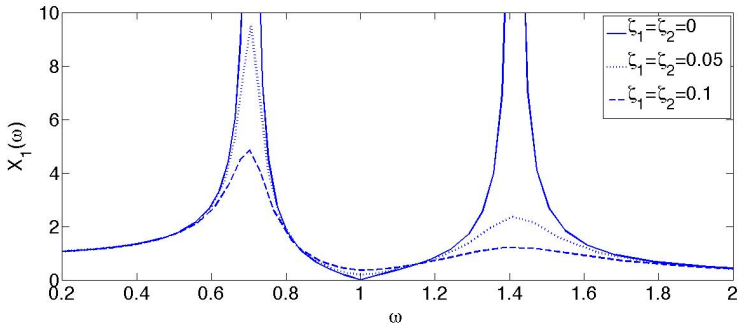


Fig. 1.10 The frequency response of the coupled spring mass damper system when $\omega_{n1} = 1$, $\alpha = 2$ and $\mu = 0.5$.

The frequency response $X_1(\omega)$ of the coupled spring mass damper system when $\alpha = 2$ and $\mu = 0.5$ is plotted in figure-1.10 for several low values of the damping ratios.

To investigate the case of negligible damping, we consider the ideal undamped model obtained by setting $\zeta_1 = \zeta_2 = 0$. Define

$$D(\omega) = \frac{\det \Delta(i\omega)}{m_1 m_2} = \omega^4 - (1 + \alpha^2 + \mu\alpha^2)\omega_{n1}^2\omega^2 + \alpha^2\omega_{n1}^4. \tag{1.43}$$

Thus $D(\omega)$ is the normalised undamped characteristic polynomial evaluated at $s = i\omega$. The frequency-response magnitudes, at frequencies for which $D(\omega) \neq 0$, are

$$\chi_1(\omega) = \frac{|\omega_{n2}^2 - \omega^2|}{m_1|D(\omega)|}, \tag{1.44}$$

$$\chi_2(\omega) = \frac{\omega_{n2}^2}{m_1|D(\omega)|}. \tag{1.45}$$

The four roots of $D(\omega) = 0$ are $\pm\omega_1$ and $\pm\omega_2$, where

$$\omega_1 = \omega_{n1}\left(\frac{1 + \alpha^2 + \mu\alpha^2 - \sqrt{(1 + \alpha^2 + \mu\alpha^2)^2 - 4\alpha^2}}{2}\right)^{1/2}, \tag{1.46}$$

$$\omega_2 = \omega_{n1}\left(\frac{1 + \alpha^2 + \mu\alpha^2 + \sqrt{(1 + \alpha^2 + \mu\alpha^2)^2 - 4\alpha^2}}{2}\right)^{1/2}. \tag{1.47}$$

For positive masses and stiffnesses, $0 < \omega_1 < \omega_2$. These are the two undamped natural frequencies of the coupled system, and

$$D(\omega) = (\omega^2 - \omega_1^2)(\omega^2 - \omega_2^2).$$

The denominators of $\chi_1(\omega)$ and $\chi_2(\omega)$ therefore vanish at $\omega = \omega_1$ and $\omega = \omega_2$. Neither numerator vanishes at these frequencies, so both frequency-response magnitudes become unbounded as either frequency is approached. At either frequency, harmonic forcing of nonzero

amplitude on the first mass produces resonant motion whose amplitude grows with time; there is no bounded harmonic response. Accordingly, ω_1 and ω_2 are also called the *undamped resonance frequencies* of this system.

When damping is small but positive, the response remains bounded and typically exhibits large peaks near these frequencies. The undamped model is therefore useful for identifying potential resonances, but cannot predict their finite peak amplitudes.

From (1.44) we notice that the harmonic particular-response amplitude of the first mass reaches zero when the forcing frequency satisfies $\omega = \omega_{n2}$. This observation will play a crucial role in vibration mitigation in systems to be investigated in the next section.

1.3.2 Tuned Mass Damper Vibration Absorber Design

Let us consider the problem of resonance mitigation in a machine with an unbalanced rotating component. The imbalance produces periodic forcing at the rotational frequency of the component. The operating conditions may require this frequency to vary over a finite range. If the natural frequency of the machine and its mounts lies within this range, large vibrations may occur when the operating frequency approaches the natural frequency. How can we reduce these vibrations over the required range of operating frequencies?

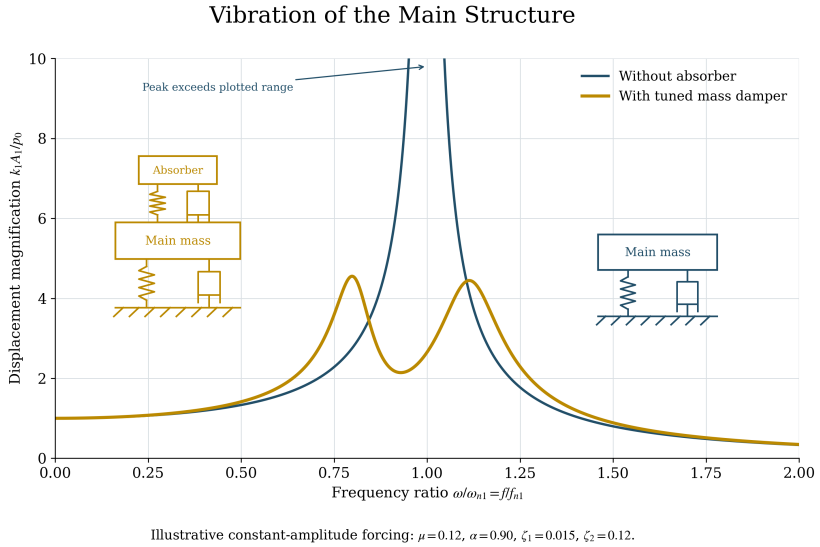


Fig. 1.11 The concept of a vibration absorber.

As a first approximation, the machine and its mounts can be represented by a 1-DOF spring–mass system subjected to a force $f(t)$. We neglect damping initially to understand the basic principle of vibration absorption. An additional mass connected to the machine through a spring produces the coupled 2-DOF system considered in the previous section. Let

$$\omega_{n1} = \sqrt{\frac{k_1}{m_1}}, \qquad \omega_{n2} = \sqrt{\frac{k_2}{m_2}}, \qquad \alpha = \frac{\omega_{n2}}{\omega_{n1}}, \qquad \mu = \frac{m_2}{m_1}.$$

Here ω_{n1} is the natural frequency of the original machine and mounts, while ω_{n2} is the natural frequency of the absorber with its attachment point held fixed.

The additional spring–mass system is called a *dynamic vibration absorber*. As shown below, its motion can generate a spring force that balances the applied harmonic force, making the harmonic response of the main mass zero at one particular excitation frequency. This cancellation is called *antiresonance*. It requires the absorber frequency to equal the excitation frequency, rather than merely being close to the natural frequency of the original machine.

The additional mass also introduces a second mode of vibration. The coupled system therefore has two natural frequencies, ω_1 and ω_2, given by (1.46) and (1.47) respectively. Although the absorber can suppress the response at its tuning frequency, large responses may occur near either of these coupled natural frequencies. The design must therefore consider the response over the entire operating range.

We examine this behaviour using (1.44) and (1.45). For $\omega > 0$, let

$$\gamma = \frac{\omega_{n1}}{\omega}, \qquad Q(\gamma) = 1 - (1 + \alpha^2 + \mu\alpha^2)\gamma^2 + \alpha^2\gamma^4.$$

The frequency-response magnitudes can then be written as

$$\chi_1\left(\frac{\omega_{n1}}{\gamma}\right) = \frac{\gamma^2}{m_1\omega_{n1}^2}\frac{|\alpha^2\gamma^2 - 1|}{|Q(\gamma)|}, \tag{1.48}$$

$$\chi_2\left(\frac{\omega_{n1}}{\gamma}\right) = \frac{\gamma^2}{m_1\omega_{n1}^2}\frac{\alpha^2\gamma^2}{|Q(\gamma)|}. \tag{1.49}$$

These expressions apply away from the coupled resonance frequencies, where $Q(\gamma) \neq 0$.

From (1.48), the harmonic response amplitude of the main mass vanishes when

$$\alpha^2\gamma^2 = 1, \qquad \text{or equivalently} \qquad \omega = \omega_{n2} = \alpha\omega_{n1}.$$

For $\mu > 0$, the denominator is nonzero at this frequency since

$$Q(1/\alpha) = -\mu.$$

Moreover, evaluating the polynomial in (1.43) gives

$$D(\omega_{n2}) = -\mu\omega_{n2}^4 < 0.$$

Since $D(\omega) = (\omega^2 - \omega_1^2)(\omega^2 - \omega_2^2)$, it follows that

$$\omega_1 < \omega_{n2} < \omega_2.$$

Thus the antiresonance frequency lies between the two coupled resonance frequencies.

For harmonic forcing $f_1(t) = p_0\cos(\omega t)$, with $p_0 > 0$, the harmonic response at $\omega = \omega_{n2}$ is

$$x_1(t) = 0, \qquad x_2(t) = -\frac{p_0}{k_2}\cos(\omega_{n2}t).$$

The absorber therefore moves so that the force it exerts on the main mass balances the applied force. These expressions describe the harmonic particular response. Since the present model is undamped, free oscillations associated with the initial conditions generally persist, and the complete response need not approach this particular response.

To examine the separation of the coupled natural frequencies, (1.46) and (1.47) give

$$\omega_2^2 - \omega_1^2 = \omega_{n1}^2\sqrt{(1+\alpha^2+\mu\alpha^2)^2 - 4\alpha^2}. \tag{1.50}$$

They also give

$$\omega_1\omega_2 = \alpha\omega_{n1}^2, \qquad \omega_1^2 + \omega_2^2 = \big(1+(1+\mu)\alpha^2\big)\omega_{n1}^2.$$

Consequently,

$$\omega_2 - \omega_1 = \omega_{n1}\sqrt{(\alpha-1)^2 + \mu\alpha^2}. \tag{1.51}$$

In particular, when the absorber is tuned to the natural frequency of the original machine, $\alpha = 1$, we obtain

$$\omega_2 - \omega_1 = \omega_{n1}\sqrt{\mu}.$$

Thus increasing the absorber mass increases the separation of the resonance frequencies for this tuning.

However, maximising this separation alone does not guarantee small vibration amplitudes over the operating range. The antiresonance frequency must also be appropriately located, and the response must remain acceptable throughout that range. There is no general requirement that the antiresonance lie exactly halfway between the two resonance frequencies. If the additional condition

$$\alpha\omega_{n1} = \frac{\omega_1 + \omega_2}{2}$$

is imposed, it requires

$$(3-\mu)\alpha^2 - 2\alpha - 1 = 0.$$

For a specified mass ratio, this condition constrains the tuning ratio α rather than leaving it free to maximise the frequency separation.

For an excitation with amplitude $p_0(\omega)$, the harmonic displacement amplitudes are

$$A_1(\omega) = p_0(\omega)\chi_1(\omega), \qquad A_2(\omega) = p_0(\omega)\chi_2(\omega).$$

In particular, a rotating imbalance with mass m and eccentricity e produces $p_0(\omega) = me\omega^2$. The frequency dependence of this force amplitude must therefore be included when assessing the machine's response.

The practical design objective is to choose the absorber mass and stiffness so that $A_1(\omega)$ remains acceptably small over the operating range, while respecting constraints on the absorber mass, motion, and spring force. For example, at exact tuning,

$$A_2(\omega_{n2}) = \frac{p_0(\omega_{n2})}{k_2} = \frac{p_0(\omega_{n2})}{m_2\omega_{n2}^2}.$$

For a fixed tuning frequency and excitation amplitude, reducing the absorber mass therefore increases the required absorber motion. This illustrates the compromise between added mass and available travel.

The undamped absorber provides exact harmonic cancellation at a single frequency. Adding damping to the absorber gives a *tuned mass damper*, whose damping can be selected to limit the neighbouring resonance peaks and improve performance over a range of frequencies, at the expense of exact cancellation at the tuning frequency.

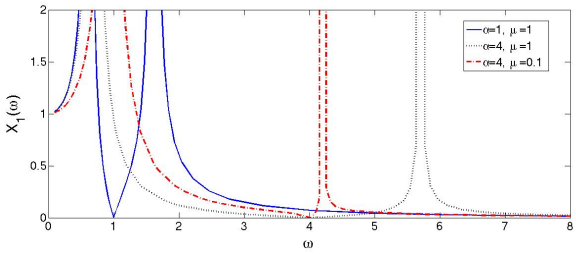


Fig. 1.12 The amplitude magnification $X_1(\omega)$ for several combinations of α and μ.

1.4 Exercises

Exercise 1.1. Consider the torsional vibration system shown in figure-1.13.

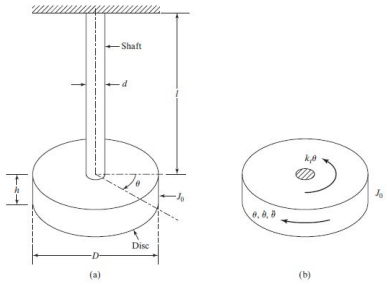


Fig. 1.13 A Tortional Vibration System.

1. Show that for small angular displacements the behavior of θ is given by

$$\ddot{\theta} + 2\zeta\omega_n\dot{\theta} + \omega_n^2\theta = 0,$$

   where $\omega_n^2 = K/J_0$, $2\zeta\omega_n = C/J_0$. Here J_0 is the moment of inertia about the axis of rotation of the system, K is the torsional stiffness of the rod, and C is the damping co-efficient of the system.
2. Show that if the damping is low, the response of the system for small initial displacements and velocities is given by

$$\theta(t) = e^{-\zeta\omega_n t}\left(\frac{\dot{\theta}(0)}{\omega_d}\cos(\omega_d t - \pi/2) + \frac{\theta(0)}{\sqrt{1-\zeta^2}}\sin(\omega_d t + \phi_{IC})\right)1(t).$$

3. Describe the main steps that you would take to experimentally estimate the damping ratio ζ and the frequency of oscillations ω_d of the system.
4. Using the experiment described above estimate the damping ratio and frequency of oscillations of the torsional system in the applied mechanics lab.
5. Using experimental results above estimate the torsional stiffness of the thin rod and calculate the error bounds of your estimate.
6. Validate the accuracy of your results.

Exercise 1.2. Figure 1.14 shows a schematic representation of the Free/Forced vibration apparatus in the applied mechanics lab. The apparatus consists of a beam pivoted at O. A damper is attached to the point A and a spring is attached to the point B. The other end of the spring, P, is constrained to move vertically. When the beam is horizontal and the point P has zero displacement with respect to the reference, that is when $\theta = 0$ and $z = 0$, the spring is at its natural length (unstretched or uncompressed). The spring constant is k the damping constant is C and the moment of inertia of the beam about O is I. Let $OA = L_c$ and $OB = L_k$. Describe a suitable experimental procedure to determine the damping constant of the system. State all assumptions and approximations clearly.

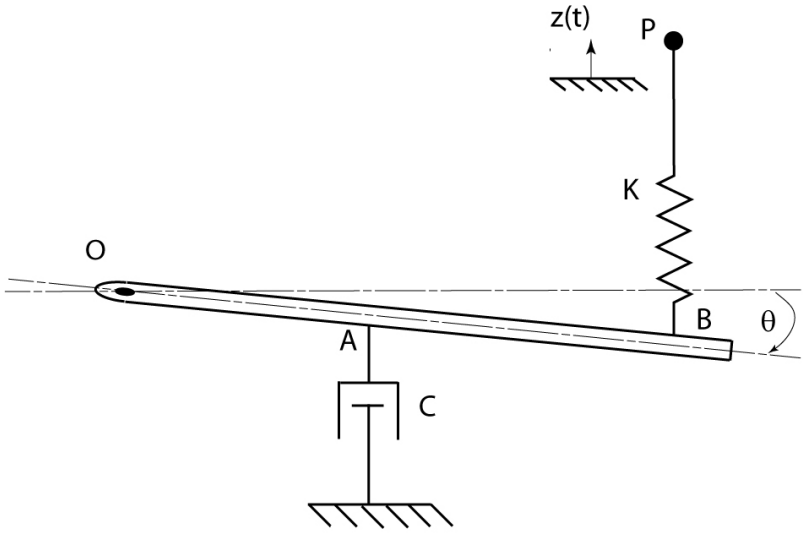


Fig. 1.14 A schematic representation of the Free/Forced Vibration Apparatus in the Applied Mechanics Lab.

Chapter 2
Modal Analysis of Discrete Systems

In the previous chapter, we identified the resonance frequencies of a coupled system through its frequency response. We now ask how the different parts of the system move at these frequencies. Modal analysis answers this question by identifying characteristic patterns of motion and expressing the response as a superposition of these patterns.

In vibration analysis we are often interested in lightly damped systems operating near resonance. For sufficiently small damping, the natural frequencies and mode shapes of the corresponding undamped system generally provide good approximations to the oscillation frequencies and spatial patterns observed physically. The undamped model therefore captures important qualitative features of the motion and provides the foundation for analysing the lightly damped response. However, damping must be retained when estimating decay rates and vibration amplitudes sufficiently close to resonance.

In the following sections we begin this analysis by revisiting the 2-DOF coupled system presented in the previous chapter and then consider the Wilberforce oscillatory system as another example and finally generalize it to a many degrees of freedom system.

2.1 The 2-DOF Example Revisited

In this section we assume that damping is negligible and hence that the system equations (1.33) are approximately

$$M\ddot{x} + Kx = f(t), \tag{2.1}$$

where $f(t) = [f_1(t)\ \ 0]^T$. We assume that $m_1, m_2, k_1, k_2 > 0$. Multiplying by M^{-1} gives

$$\ddot{x} + \Omega x = M^{-1} f(t), \tag{2.2}$$

where $\Omega = M^{-1}K$. The configuration $x(t)$ belongs to $\mathbb{R}^2$, and $\Omega : \mathbb{R}^2 \to \mathbb{R}^2$ is a linear operator.

Recall that, for notational convenience, in the previous section we had set

$$\omega_{n1}^2 = \frac{k_1}{m_1}, \qquad \omega_{n2}^2 = \frac{k_2}{m_2}, \qquad \mu = \frac{m_2}{m_1}, \qquad \alpha = \frac{\omega_{n2}}{\omega_{n1}}.$$

Then

$$\begin{aligned}\Omega &= \begin{bmatrix} 1/m_1 & 0 \\ 0 & 1/m_2 \end{bmatrix} \begin{bmatrix} k_1+k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix} \\ &= \begin{bmatrix} \frac{k_1+k_2}{m_1} & -\frac{k_2}{m_1} \\ -\frac{k_2}{m_2} & \frac{k_2}{m_2} \end{bmatrix} \\ &= \begin{bmatrix} (1+\alpha^2\mu)\omega_{n1}^2 & -\mu\alpha^2\omega_{n1}^2 \\ -\alpha^2\omega_{n1}^2 & \alpha^2\omega_{n1}^2 \end{bmatrix}.\end{aligned}$$

Also recall from the previous section that the undamped natural frequencies correspond to the positive roots of the polynomial (1.43)

$$\begin{aligned}D(\omega) &= \frac{\det\Delta(i\omega)}{m_1 m_2} \\ &= \omega^4 - (1+\alpha^2+\mu\alpha^2)\omega_{n1}^2\omega^2 + \alpha^2\omega_{n1}^4,\end{aligned}$$

and are given by (1.46) and (1.47):

$$\omega_1 = \omega_{n1}\left(\frac{1+\alpha^2+\mu\alpha^2-\sqrt{(1+\alpha^2+\mu\alpha^2)^2-4\alpha^2}}{2}\right)^{1/2},$$

$$\omega_2 = \omega_{n1}\left(\frac{1+\alpha^2+\mu\alpha^2+\sqrt{(1+\alpha^2+\mu\alpha^2)^2-4\alpha^2}}{2}\right)^{1/2}.$$

Notice that $D(\omega) = \det(-\omega^2 I + \Omega)$ and hence that the undamped natural frequencies ω_1 and ω_2 are the square roots of the eigenvalues of Ω.

2.1.1 Modal Decomposition

Let us investigate what the corresponding eigenvectors tell us. They are the nontrivial solutions of

$$(-\omega_i^2 I + \Omega)\bar{x}_i = 0. \tag{2.3}$$

Recall that eigenvectors corresponding to these distinct eigenvalues are unique up to multiplication by a nonzero scalar. Since the eigenvalues are distinct, $\bar{x}_1$ and $\bar{x}_2$ are linearly independent and

$$T = [\bar{x}_1\ \bar{x}_2]$$

is invertible. Thus $\{\bar{x}_1, \bar{x}_2\}$ forms a basis for $\mathbb{R}^2$.

Although $\Omega = M^{-1}K$ is generally not symmetric in the Euclidean inner product, the underlying mass and stiffness matrices are symmetric. The appropriate geometry is therefore provided by the mass inner product introduced below. This choice also has a physical interpretation: the squared norm of the velocity is twice the kinetic energy. Consider the inner product on $\mathbb{R}^2$ is defined by

$$\langle\langle x, y\rangle\rangle = y^T M x.$$

Since $M\Omega = \Omega^T M = K$, we have

$$\omega_1^2 \bar{x}_2^T M \bar{x}_1 = \bar{x}_2^T K \bar{x}_1 = \omega_2^2 \bar{x}_2^T M \bar{x}_1.$$

Since $\omega_1^2 \neq \omega_2^2$, the eigenvectors are orthogonal with respect to this inner product. We choose the normalization $\langle\langle \bar{x}_i, \bar{x}_i\rangle\rangle = 1$, which fixes their magnitude but leaves their sign arbitrary. Thus $\langle\langle \bar{x}_j, \bar{x}_i\rangle\rangle = \delta_{ij}$. Consequently, $T^T M T = I$ and hence $T^{-1} = T^T M$. This normalization allows us to extract the coefficient of each mode using the inner product.

By definition, $\Omega \bar{x}_1 = \omega_1^2 \bar{x}_1$ and $\Omega \bar{x}_2 = \omega_2^2 \bar{x}_2$, and hence $\Omega T = T\Lambda$, where

$$\Lambda = \begin{bmatrix} \omega_1^2 & 0 \\ 0 & \omega_2^2 \end{bmatrix} = T^{-1} \Omega T.$$

Consider the coordinate transformation $x = Tz$, where $z = [z_1 \; z_2]^T$. Substituting into (2.2) and multiplying by T^{-1} gives

$$\begin{bmatrix} \ddot{z}_1 \\ \ddot{z}_2 \end{bmatrix} + \begin{bmatrix} \omega_1^2 & 0 \\ 0 & \omega_2^2 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = T^{-1} \begin{bmatrix} f_1(t)/m_1 \\ 0 \end{bmatrix}.$$

Thus, in the new coordinates z, the coupled pair of ODEs reduces to the uncoupled pair

$$\ddot{z}_1 + \omega_1^2 z_1 = \gamma_1(t), \tag{2.4}$$

$$\ddot{z}_2 + \omega_2^2 z_2 = \gamma_2(t), \tag{2.5}$$

where

$$\gamma_1(t) = \frac{T^{11}}{m_1} f_1(t), \qquad \gamma_2(t) = \frac{T^{21}}{m_1} f_1(t), \qquad T^{-1} = \begin{bmatrix} T^{11} & T^{12} \\ T^{21} & T^{22} \end{bmatrix}.$$

These are the uncoupled modal equations of the system (2.2).

Since $x = Tz$,

$$x(t) = \sum_{i=1}^{2} z_i(t) \bar{x}_i. \tag{2.6}$$

Hence the response is a superposition of the modal contributions $\bar{x}_1 z_1(t)$ and $\bar{x}_2 z_2(t)$.

The coordinate transformation $x = Tz$, where the columns of T are the eigenvectors of Ω, transforms the coupled system (2.2) into two uncoupled 1-DOF spring–mass equations (2.4) and (2.5). The physical response is a superposition of their modal contributions.

Even though the external force acts only on mass m_1, it excites both modal coordinates in this example.

Let

$$\bar{x}_1 = \begin{bmatrix} \bar{x}_{11} \\ \bar{x}_{21} \end{bmatrix}, \qquad \bar{x}_2 = \begin{bmatrix} \bar{x}_{12} \\ \bar{x}_{22} \end{bmatrix}$$

denote the normalized eigenvectors corresponding to ω_1^2 and ω_2^2, respectively. From (2.3), their components satisfy

$$r_1 \triangleq \frac{\bar{x}_{11}}{\bar{x}_{21}} = \frac{\alpha^2 - 1 - \mu\alpha^2 + \sqrt{(1+\alpha^2+\mu\alpha^2)^2 - 4\alpha^2}}{2\alpha^2} > 0, \tag{2.7}$$

$$r_2 \triangleq \frac{\bar{x}_{12}}{\bar{x}_{22}} = \frac{\alpha^2 - 1 - \mu\alpha^2 - \sqrt{(1+\alpha^2+\mu\alpha^2)^2 - 4\alpha^2}}{2\alpha^2} < 0. \tag{2.8}$$

Using $x = Tz$, we obtain

$$x(t) = \bar{x}_1 z_1(t) + \bar{x}_2 z_2(t) = \bar{x}_{21} \begin{bmatrix} r_1 \\ 1 \end{bmatrix} z_1(t) + \bar{x}_{22} \begin{bmatrix} r_2 \\ 1 \end{bmatrix} z_2(t). \tag{2.9}$$

The modal decomposition

$$x(t) = \sum_{i=1}^{2} z_i(t)\bar{x}_i$$

expresses the configuration as a linear combination of the mode shapes. Taking the inner product with the basis element gives

$$z_j(t) = \langle\langle \bar{x}_j, x(t) \rangle\rangle.$$

Thus this inner-product formula is precisely the coordinate transformation $z = T^{-1}x$.

Substituting the modal decomposition into (2.2) and using $\Omega\bar{x}_i = \omega_i^2\bar{x}_i$, we obtain

$$\sum_{i=1}^{2} \ddot{z}_i\bar{x}_i + \sum_{i=1}^{2} \omega_i^2 z_i\bar{x}_i = M^{-1}f(t).$$

Taking the inner product with $\bar{x}_j$ gives

$$\sum_{i=1}^{2} (\ddot{z}_i + \omega_i^2 z_i)\langle\langle \bar{x}_j, \bar{x}_i \rangle\rangle = \langle\langle \bar{x}_j, M^{-1}f(t) \rangle\rangle.$$

Hence

$$\ddot{z}_j + \omega_j^2 z_j = \gamma_j(t), \qquad \gamma_j(t) = \langle\langle \bar{x}_j, M^{-1}f(t) \rangle\rangle.$$

In the following we consider the unforced case, or free vibrations, and the forced case, or forced vibrations.

2.1.1.1 Forced Vibrations and Resonance

For harmonic forcing $f_1(t) = f_0\cos(\omega t)$, the modal equations become

$$\ddot{z}_j + \omega_j^2 z_j = f_0\bar{x}_{1j}\cos(\omega t), \qquad j = 1,2,$$

where $\bar{x}_{1j}$ is the first component of the mass-normalized mode shape $\bar{x}_j$.

For $\omega \neq \omega_1, \omega_2$, the harmonic particular response is

$$x_p(t) = f_0 \left(\frac{\bar{x}_{11}\bar{x}_1}{\omega_1^2 - \omega^2} + \frac{\bar{x}_{12}\bar{x}_2}{\omega_2^2 - \omega^2} \right) \cos(\omega t).$$

Since this model is undamped, free oscillations generally persist. In particular, for a system initially at rest,

$$z_j(t) = \frac{f_0 \bar{x}_{1j}}{\omega_j^2 - \omega^2} \big(\cos(\omega t) - \cos(\omega_j t) \big).$$

At exact resonance, $\omega = \omega_j$, this becomes

$$z_j(t) = \frac{f_0 \bar{x}_{1j}}{2\omega_j} t \sin(\omega_j t),$$

showing the growth of the resonant mode.

When the excitation frequency is sufficiently close to an isolated natural frequency ω_j, and the corresponding modal forcing is nonzero, the harmonic particular response is dominated by the mode shape $\bar{x}_j$. In a lightly damped system, after transients decay, the masses therefore oscillate approximately in this mode shape at the forcing frequency ω.

This explains the modal character of the two resonance peaks in Fig. 1.10. The undamped analysis identifies the relevant frequencies and mode shapes, while damping must be retained to predict finite resonant amplitudes.

2.1.1.2 Free Vibrations

When $f_1(t) \equiv 0$, the modal equations reduce to

$$\ddot{z}_i + \omega_i^2 z_i = 0, \qquad i = 1, 2.$$

Hence the general free response is

$$x(t) = A_1 \bar{x}_1 \cos(\omega_1 t + \phi_1) + A_2 \bar{x}_2 \cos(\omega_2 t + \phi_2),$$

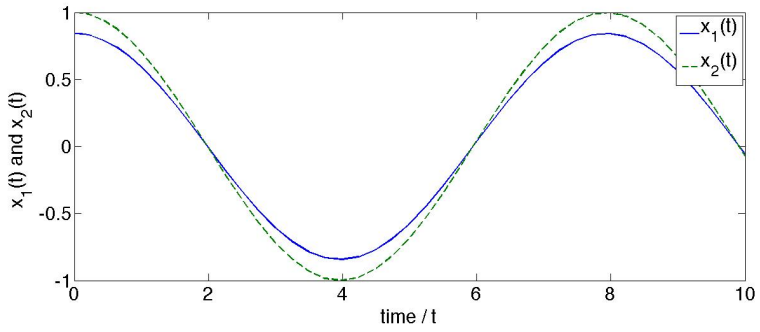
where the amplitudes and phases are chosen to satisfy $z(0) = T^{-1}x(0)$ and $\dot{z}(0) = T^{-1}\dot{x}(0)$.

In particular, choosing $x(0) = \bar{x}_i$ and $\dot{x}(0) = 0$ excites only the ith mode:

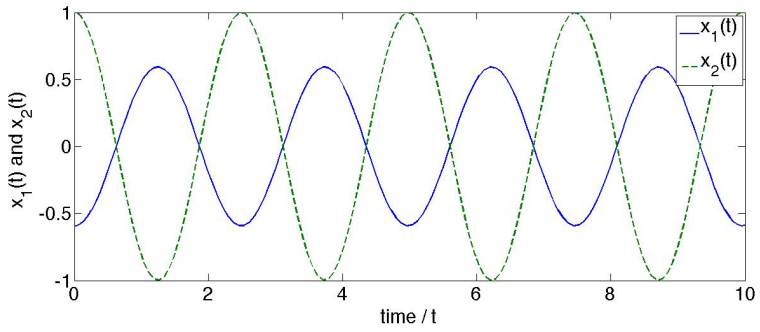
$$x(t) = \bar{x}_i \cos(\omega_i t).$$

Both masses then oscillate at the same frequency with a fixed displacement ratio. In the first mode, $x_1 = r_1 x_2$ with $r_1 > 0$, so the masses move in phase. In the second mode, $x_1 = r_2 x_2$ with $r_2 < 0$, so they move in opposite phase. The ratios are given by (2.7) and (2.8).

These two natural frequencies correspond to the two undamped resonance conditions and explain the two peaks in the lightly damped frequency response shown in Fig. 1.10.



(a) First mode: frequency ω_1.



(b) Second mode: frequency ω_2.

Fig. 2.1 The two modes of free vibration for $\alpha = 2$ and $\mu = 0.5$.

2.2 The Example of the Wilberforce Oscillatory System

The Wilberforce oscillator shown in Figure 2.2 couples the vertical and torsional motions of a suspended mass. As the spring stretches or compresses, it also tends to unwind or wind up. We first consider its forced response and then illustrate beating in free vibration.

For small displacements from static equilibrium and negligible damping, the equations are

$$M\ddot{Y} + KY = F(t),$$

where

$$Y = \begin{bmatrix} x \\ \theta \end{bmatrix}, \qquad M = \begin{bmatrix} m & 0 \\ 0 & \mathbb{I} \end{bmatrix}, \qquad \mathscr{K} = \begin{bmatrix} k_x & \varepsilon \\ \varepsilon & k_\theta \end{bmatrix}, \qquad F(t) = \begin{bmatrix} f(t) \\ 0 \end{bmatrix}.$$

Here x is the vertical displacement, θ is the angular displacement, and I is the moment of inertia about the rotation axis. We assume $m, \mathbb{I}, k_x, k_\theta > 0$ and $\varepsilon^2 < k_x k_\theta$. The sign of θ is chosen so that $\varepsilon > 0$, and weak coupling means $\varepsilon / \sqrt{k_x k_\theta} \ll 1$.

2.2.1 Modal Decomposition and Resonance

Define the mass inner product by

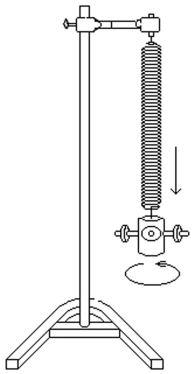


Fig. 2.2 The Wilberforce oscillatory system.

$$\langle\langle X,Y\rangle\rangle_M = Y^T MX.$$

Then $\Omega = M^{-1}K$ is self-adjoint with respect to this inner product. Its eigenvalues are

$$\omega_{1,2}^2 = \frac{\omega_x^2+\omega_\theta^2 \mp \sqrt{(\omega_x^2-\omega_\theta^2)^2+4\varepsilon^2/(m\mathbb{I})}}{2}, \qquad \omega_x^2 = \frac{k_x}{m}, \quad \omega_\theta^2 = \frac{k_\theta}{\mathbb{I}}.$$

To illustrate the behaviour near tuning, consider $\omega_x = \omega_\theta = \omega_n$ and define

$$\varepsilon_n = \frac{\varepsilon}{\sqrt{m\mathbb{I}}}, \qquad 0 < \varepsilon_n \ll \omega_n^2.$$

The natural frequencies and mass-normalized mode shapes are

$$\omega_1 = \sqrt{\omega_n^2-\varepsilon_n}, \qquad \omega_2 = \sqrt{\omega_n^2+\varepsilon_n},$$

$$\bar{Y}_1 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1/\sqrt{m} \\ -1/\sqrt{\mathbb{I}} \end{bmatrix}, \qquad \bar{Y}_2 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1/\sqrt{m} \\ 1/\sqrt{\mathbb{I}} \end{bmatrix}.$$

These satisfy

$$\langle\langle \bar{Y}_i, \bar{Y}_j\rangle\rangle_M = \delta_{ij}.$$

Consequently,

$$Y(t) = Z_1(t)\bar{Y}_1 + Z_2(t)\bar{Y}_2, \qquad Z_j(t) = \langle\langle \bar{Y}_j, Y(t)\rangle\rangle_M.$$

Projecting the equations onto each mode gives

$$\ddot{Z}_j + \omega_j^2 Z_j = \langle\langle \bar{Y}_j, M^{-1}F(t)\rangle\rangle_M = \frac{f(t)}{\sqrt{2m}}, \qquad j = 1,2.$$

Thus a vertical force excites both mass-normalized modes equally.

2.2.2 Free Vibrations and Beating

When $f(t) = 0$, the modal equations give

$$Z_j(t) = A_j \cos(\omega_j t + \phi_j), \qquad j = 1,2.$$

Thus, by the modal decomposition,

$$Y(t) = A_1 \bar{Y}_1 \cos(\omega_1 t + \phi_1) + A_2 \bar{Y}_2 \cos(\omega_2 t + \phi_2),$$

where the constants are determined by the initial conditions.

Releasing the mass from $x(0) = x_0$ with $\theta(0) = \dot{x}(0) = \dot{\theta}(0) = 0$ gives

$$Z_1(t) = x_0 \sqrt{\frac{m}{2}} \cos(\omega_1 t), \qquad Z_2(t) = x_0 \sqrt{\frac{m}{2}} \cos(\omega_2 t).$$

Substituting into the modal decomposition yields

$$\begin{aligned} x(t) &= x_0 \cos(\delta\omega t) \cos(\bar{\omega} t), \\ \theta(t) &= -x_0 \sqrt{\frac{m}{\mathbb{I}}} \sin(\delta\omega t) \sin(\bar{\omega} t), \end{aligned}$$

where

$$\bar{\omega} = \frac{\omega_1 + \omega_2}{2} \approx \omega_n, \qquad \delta\omega = \frac{\omega_2 - \omega_1}{2} \approx \frac{\varepsilon_n}{2\omega_n}$$

for weak coupling. The rapid oscillations have slowly varying amplitudes: vertical motion diminishes as torsional motion grows, and vice versa. This is *beating*, with successive maxima of the vertical amplitude separated by

$$T_{\text{beat}} = \frac{2\pi}{\omega_2 - \omega_1}.$$

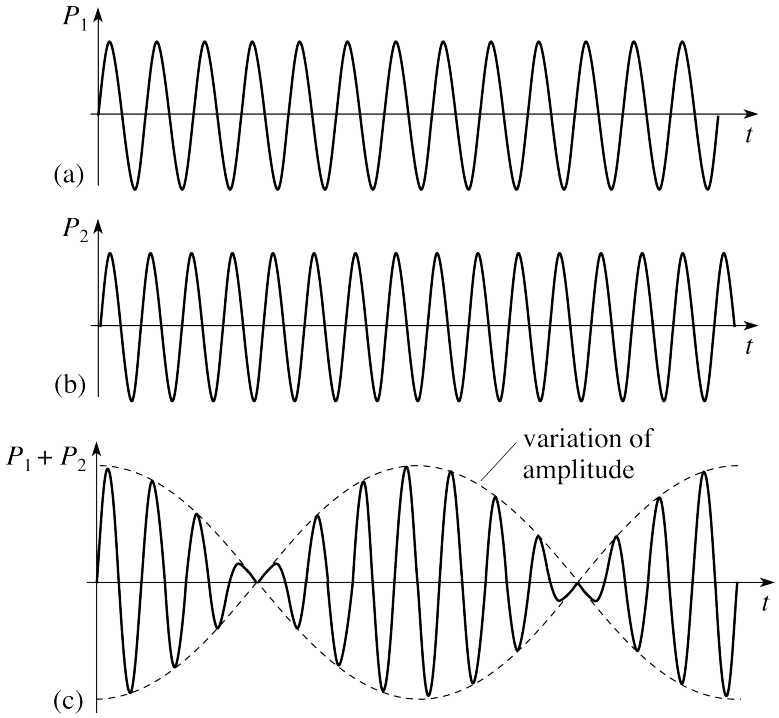


Fig. 2.3 Beating: rapid oscillations modulated by a slowly varying amplitude envelope.

2.3 n-DOF Undamped Vibration Analysis

We now generalize the modal analysis of the two previous 2-DOF examples focusing on the inner-product interpretation.

The linearized equations of an n-DOF undamped system are

$$M\ddot{x} + Kx = f(t), \tag{2.10}$$

where M and K are real symmetric positive definite $n \times n$ matrices and $x(t) \in \mathbb{R}^n$. Multiplying by M^{-1} gives

$$\ddot{x} + \Omega x = M^{-1} f(t), \qquad \Omega = M^{-1} K. \tag{2.11}$$

Define the mass inner product on $\mathbb{R}^n$ by

$$\langle\langle x, y \rangle\rangle_M = y^T M x.$$

Since M is symmetric and positive definite, this is a symmetric, bilinear, positive definite inner product. Furthermore, since K is symmetric,

$$\langle\langle \Omega x, y \rangle\rangle_M = y^T K x = \langle\langle x, \Omega y \rangle\rangle_M.$$

Thus Ω is self-adjoint with respect to the mass inner product. It is also positive definite in the sense that

$$\langle\langle \Omega x, x \rangle\rangle_M = x^T K x > 0 \qquad \text{for } x \neq 0.$$

Consequently, Ω has positive real eigenvalues $\omega_1^2, \ldots, \omega_n^2$, and its eigenvectors can be chosen to form an orthonormal basis with respect to this inner product:

$$\Omega \bar{x}_j = \omega_j^2 \bar{x}_j, \qquad \langle\langle \bar{x}_i, \bar{x}_j \rangle\rangle_M = \delta_{ij}.$$

For repeated eigenvalues, an orthonormal basis is chosen within the corresponding eigenspace. The quantities ω_j are the undamped natural frequencies, and the eigenvectors $\bar{x}_j$ are the corresponding mode shapes.

2.3.1 Modal Decomposition

Since the mode shapes form a basis, the configuration at each instant can be expressed as

$$x(t) = \sum_{j=1}^{n} z_j(t) \bar{x}_j. \tag{2.12}$$

Taking the inner product with $\bar{x}_k$ gives

$$\langle\langle \bar{x}_k, x(t) \rangle\rangle_M = \sum_{j=1}^{n} z_j(t) \langle\langle \bar{x}_k, \bar{x}_j \rangle\rangle_M = z_k(t).$$

Thus

$$z_k(t) = \langle\langle \bar{x}_k, x(t) \rangle\rangle_M$$

is the component of the configuration along the kth mode, measured using the mass inner product.

Substituting (2.12) into (2.11) gives

$$\sum_{j=1}^{n} \left(\ddot{z}_j + \omega_j^2 z_j \right) \bar{x}_j = M^{-1} f(t).$$

Taking the inner product with $\bar{x}_k$ and using orthonormality yields the uncoupled modal equations

$$\ddot{z}_k + \omega_k^2 z_k = \gamma_k(t), \qquad k = 1, \ldots, n, \tag{2.13}$$

where

$$\gamma_k(t) = \langle\langle \bar{x}_k, M^{-1} f(t) \rangle\rangle_M = \bar{x}_k^T f(t).$$

The initial conditions are obtained by the same projection:

$$z_k(0) = \langle\langle \bar{x}_k, x(0) \rangle\rangle_M, \qquad \dot{z}_k(0) = \langle\langle \bar{x}_k, \dot{x}(0) \rangle\rangle_M.$$

Modal analysis expresses the configuration in an orthonormal basis of mode shapes, using the mass inner product. Projecting the equations onto each mode produces an independent scalar oscillator:

$$\ddot{z}_k + \omega_k^2 z_k = \gamma_k(t).$$

The physical response is recovered by superposition:

$$x(t) = \sum_{k=1}^{n} z_k(t)\bar{x}_k.$$

A force excites a mode only through its modal forcing $\gamma_k(t) = \bar{x}_k^T f(t)$.

2.3.1.1 Forced Vibrations and Resonance

Consider harmonic forcing

$$f(t) = \bar{f}\cos(\omega t),$$

where $\bar{f} \in \mathbb{R}^n$ is a constant force-amplitude column. Define

$$b_k = \langle\langle \bar{x}_k, M^{-1}\bar{f}\rangle\rangle_M = \bar{x}_k^T \bar{f}.$$

Then the modal equations are

$$\ddot{z}_k + \omega_k^2 z_k = b_k \cos(\omega t),$$

with initial conditions obtained by projection:

$$z_k(0) = \langle\langle \bar{x}_k, x(0)\rangle\rangle_M, \qquad \dot{z}_k(0) = \langle\langle \bar{x}_k, \dot{x}(0)\rangle\rangle_M.$$

For ω different from every natural frequency, the complete modal response is

$$z_k(t) = z_k(0)\cos(\omega_k t) + \frac{\dot{z}_k(0)}{\omega_k}\sin(\omega_k t) + \frac{b_k}{\omega_k^2 - \omega^2}\left(\cos(\omega t) - \cos(\omega_k t)\right).$$

The physical response is recovered as

$$x(t) = \sum_{k=1}^{n} z_k(t)\bar{x}_k.$$

Since the system is undamped, the natural-frequency oscillations generally persist. For a system initially at rest, the first two terms in the modal response vanish.

At exact resonance, $\omega = \omega_k$, the corresponding modal response is

$$z_k(t) = z_k(0)\cos(\omega_k t) + \frac{\dot{z}_k(0)}{\omega_k}\sin(\omega_k t) + \frac{b_k}{2\omega_k} t\sin(\omega_k t).$$

Thus resonant growth occurs only when $b_k \neq 0$.

When the forcing frequency approaches an isolated natural frequency ω_k with nonzero modal forcing, the harmonic particular response is dominated by the corresponding mode shape $\bar{x}_k$.

For a lightly damped system, this explains why the response near an isolated resonance approximately takes the shape of that mode. Closely spaced or repeated natural frequencies may require several modal contributions. Damping must be retained to determine finite resonant amplitudes and decay rates.

2.3.1.2 Free Vibrations

When $f(t) \equiv 0$, each modal coordinate satisfies

$$\ddot{z}_k + \omega_k^2 z_k = 0.$$

Hence

$$z_k(t) = z_k(0)\cos(\omega_k t) + \frac{\dot{z}_k(0)}{\omega_k}\sin(\omega_k t).$$

The free response

$$x(t) = \sum_{k=1}^{n} z_k(t)\bar{x}_k$$

is therefore a superposition of oscillations at the natural frequencies. If the initial displacement and velocity lie along a single mode shape, only that mode is excited.

2.3.2 Including Damping in the Modal Equations

The undamped modes provide a useful basis for describing the motion even when damping is present. Including damping allows us to account for the decay of free vibrations and finite resonant amplitudes, connecting the modal analysis above with the single-degree-of-freedom results of Chapter 1.

Consider

$$M\ddot{x} + C\dot{x} + Kx = f(t),$$

where M and K are symmetric positive definite and C is symmetric positive semidefinite. Retain the mass-orthonormal undamped mode shapes:

$$\Omega\bar{x}_k = \omega_k^2\bar{x}_k, \qquad \Omega = M^{-1}K, \qquad \langle\langle \bar{x}_j, \bar{x}_k\rangle\rangle_M = \delta_{jk}.$$

Substituting $x(t) = \sum_{k=1}^{n} z_k(t)\bar{x}_k$ and projecting onto $\bar{x}_j$ gives

$$\ddot{z}_j + \sum_{k=1}^{n} c_{jk}\dot{z}_k + \omega_j^2 z_j = \gamma_j(t),$$

where

$$c_{jk} = \langle\langle M^{-1}C\bar{x}_k, \bar{x}_j\rangle\rangle_M = \bar{x}_j^T C\bar{x}_k, \qquad \gamma_j(t) = \bar{x}_j^T f(t).$$

Thus damping generally couples the modal coordinates.

The modal equations remain uncoupled if $M^{-1}C$ and $\Omega = M^{-1}K$ admit a common mass-orthonormal eigenbasis. For example, suppose each spring is accompanied by a parallel viscous damper, with the same ratio $c_e/k_e = b$ for every spring–damper pair. Then $C = bK$, so $M^{-1}C = b\Omega$, and the common-eigenbasis condition holds.

Choosing the mode shapes from this basis gives

$$\Omega\bar{x}_k = \omega_k^2\bar{x}_k, \qquad M^{-1}C\bar{x}_k = 2\zeta_k\omega_k\bar{x}_k, \qquad \zeta_k \geq 0.$$

Consequently,

$$c_{jk} = \langle\langle M^{-1}C\bar{x}_k, \bar{x}_j\rangle\rangle_M = 2\zeta_k\omega_k\delta_{jk},$$

and each modal coordinate satisfies an independent damped oscillator equation:

$$\ddot{z}_j + 2\zeta_j\omega_j\dot{z}_j + \omega_j^2 z_j = \gamma_j(t).$$

Each is a damped single-degree-of-freedom equation, with initial conditions obtained by the same mass-inner-product projections as before. Its initial-condition and forced responses therefore follow directly from Chapter 1.

For example, when $0 \leq \zeta_j < 1$, define $\omega_{dj} = \omega_j\sqrt{1-\zeta_j^2}$. Then

$$z_j(t) = e^{-\zeta_j\omega_j t}\left[z_j(0)\cos(\omega_{dj}t) + \frac{\dot{z}_j(0) + \zeta_j\omega_j z_j(0)}{\omega_{dj}}\sin(\omega_{dj}t)\right] + \int_0^t \frac{e^{-\zeta_j\omega_j(t-\tau)}}{\omega_{dj}}\sin\big(\omega_{dj}(t-\tau)\big)\gamma_j(\tau)\,d\tau.$$

The physical response is recovered through $x(t) = \sum_{j=1}^n z_j(t)\bar{x}_j$. Positive modal damping causes free oscillations to decay and gives a bounded steady-state response to harmonic forcing. If the modal damping matrix is not diagonal, the coupled modal equations must instead be solved together.

2.4 Modes of Atomic Vibrations

In this section we illustrate how atomic vibrations in a solid may be approximated using classical Newtonian mechanics[1]. We consider a one-dimensional polyatomic lattice with N unit cells, periodic boundary conditions, and nearest-neighbour interactions. The cells are indexed by $n = 1,\ldots,N$, and the atoms within each cell by $\alpha = 1,\ldots,s$.

Denote the displacement from equilibrium of atom α in cell n by $q(\alpha,n;t)$. We extend the indices using

[1] This is supplementary reading.

$$q(\alpha,n+N;t)=q(\alpha,n;t),\qquad q(\alpha+rs,n;t)=q(\alpha,n+r;t),\qquad r\in\mathbb{Z}.$$

The first condition identifies the ends of the chain; the second specifies how atom indices continue into neighbouring cells. The masses satisfy $m_{\alpha+s}=m_\alpha>0$.

Let $U_\alpha(r)$ be the interaction potential of the bond joining atoms $(\alpha-1,n)$ and (α,n), expressed in terms of the change in bond length

$$r(\alpha,n;t)=q(\alpha,n;t)-q(\alpha-1,n;t).$$

The bond potentials repeat from cell to cell: $U_{\alpha+s}=U_\alpha$. The equations of motion are

$$m_\alpha\ddot{q}(\alpha,n;t)=-U'_\alpha\big(q(\alpha,n;t)-q(\alpha-1,n;t)\big)+U'_{\alpha+1}\big(q(\alpha+1,n;t)-q(\alpha,n;t)\big). \quad (2.14)$$

Here an overdot denotes differentiation with respect to time and a prime denotes differentiation with respect to bond extension.

For small displacements, we approximate each bond potential by its quadratic expansion about equilibrium:

$$U_\alpha(r)\approx\frac{b_\alpha}{2}r^2,\qquad b_{\alpha+s}=b_\alpha>0,$$

omitting constant terms and equilibrium linear terms that cancel from the equations of motion. Each bond therefore acts like a linear spring of stiffness b_α, giving

$$m_\alpha\ddot{q}(\alpha,n;t)=b_{\alpha+1}q(\alpha+1,n;t)-(b_\alpha+b_{\alpha+1})q(\alpha,n;t)+b_\alpha q(\alpha-1,n;t). \quad (2.15)$$

2.4.1 The Lattice Operator and Inner Product

Collect the Ns atomic displacements into the column

$$Q(t)=\big[q(1,1;t)\cdots\ q(s,1;t)\ q(1,2;t)\cdots\ q(s,N;t)\big]^T.$$

Then (2.15) becomes

$$\ddot{Q}+\Omega Q=0, \quad (2.16)$$

where

$$(\Omega Q)(\alpha,n;t)=\frac{1}{m_\alpha}\big((b_\alpha+b_{\alpha+1})q(\alpha,n;t)-b_\alpha q(\alpha-1,n;t)-b_{\alpha+1}q(\alpha+1,n;t)\big).$$

Thus the lattice is an Ns-DOF system of coupled oscillators, and ΩQ is the restoring force per unit mass with its sign reversed.

As in the preceding examples, define the mass inner product. We allow complex displacements to simplify the description of wave-like modes:

$$\langle\langle U,V\rangle\rangle_M=\sum_{n=1}^{N}\sum_{\alpha=1}^{s}m_\alpha\,\overline{v(\alpha,n)}\,u(\alpha,n).$$

This inner product is linear in its first argument and conjugate-linear in its second. Physical displacements remain real.

Using the periodic boundary conditions gives

$$\begin{aligned}\langle\langle \Omega U, V\rangle\rangle_M &= \sum_{n,\alpha} b_\alpha \overline{\big(v(\alpha,n)-v(\alpha-1,n)\big)}\big(u(\alpha,n)-u(\alpha-1,n)\big)\\ &= \langle\langle U, \Omega V\rangle\rangle_M.\end{aligned}$$

Hence Ω is self-adjoint with respect to the mass inner product. Moreover,

$$\langle\langle \Omega U, U\rangle\rangle_M = \sum_{n,\alpha} b_\alpha |u(\alpha,n)-u(\alpha-1,n)|^2 \geq 0.$$

This expression is twice the elastic potential energy for a real displacement U. It depends only on relative atomic displacements: moving every atom by the same amount does not stretch any bond.

Consequently, Ω has real nonnegative eigenvalues, and its eigenvectors can be chosen to form a mass-orthonormal basis.

2.4.2 Natural Frequencies and Mode Shapes

Let the unit-cell length be sa. Since the lattice repeats from cell to cell, we seek eigenvectors of the form

$$\psi_{\beta k}(\alpha,n) = \frac{1}{\sqrt{m_\alpha N}}\phi_{\beta k}(\alpha)e^{ikasn}.$$

The factor e^{ikasn} specifies the phase change between cells, while $\phi_{\beta k}(\alpha)/\sqrt{m_\alpha}$ specifies the relative amplitudes and phases of the atoms within a cell.

Periodicity over N cells requires

$$e^{ikasN} = 1, \qquad k = \frac{2\pi\ell}{Nsa}, \qquad \ell = 0,\ldots,N-1.$$

Wave numbers are understood modulo $2\pi/(sa)$. We extend the cell components by

$$\phi_{\beta k}(\alpha+s) = e^{ikas}\phi_{\beta k}(\alpha).$$

These conditions ensure that $\psi_{\beta k}$ satisfies the lattice boundary conditions.

Substitution into

$$\Omega\,\psi_{\beta k} = \omega_{\beta k}^2 \psi_{\beta k}$$

reduces the Ns-dimensional eigenvalue problem to an s-dimensional problem for each allowed k:

$$A(k)\phi_{\beta k} = \omega_{\beta k}^2 \phi_{\beta k}, \tag{2.17}$$

where

$$(A(k)\phi)(\alpha) = \frac{b_\alpha + b_{\alpha+1}}{m_\alpha}\phi(\alpha) - \frac{b_\alpha}{\sqrt{m_\alpha m_{\alpha-1}}}\phi(\alpha-1) - \frac{b_{\alpha+1}}{\sqrt{m_\alpha m_{\alpha+1}}}\phi(\alpha+1),$$

using the quasiperiodic extension above.

The matrix $A(k)$ is Hermitian and positive semidefinite. Its eigenvalues are determined by

$$\det\big(\omega^2 I - A(k)\big) = 0. \tag{2.18}$$

For each k, there are s eigenvalues $\omega_{\beta k}^2$, counted with multiplicity, and corresponding frequencies

$$\omega_{\beta k} = \omega_\beta(k) \geq 0, \qquad \beta = 1, \ldots, s.$$

The s relations $\omega = \omega_\beta(k)$ are the *branches of the dispersion relation*. They describe how the vibration frequency depends on the spatial phase pattern. A finite periodic lattice samples these branches at its N allowed wave numbers.

A single harmonic lattice vibration has the form

$$q(\alpha, n; t) = \mathrm{Re}\left\{c_{\beta k}\psi_{\beta k}(\alpha, n)e^{-i\omega_{\beta k}t}\right\}.$$

All atoms in this mode oscillate at the same frequency, with their relative amplitudes and phases fixed by the mode shape.

Choose the cell eigenvectors so that

$$\sum_{\alpha=1}^{s} \overline{\phi_{\beta' k}(\alpha)}\phi_{\beta k}(\alpha) = \delta_{\beta\beta'}.$$

For repeated eigenvalues, choose an orthonormal basis within the corresponding eigenspace. Together with

$$\frac{1}{N}\sum_{n=1}^{N} e^{i(k-k')asn} = \delta_{kk'},$$

this gives

$$\langle\langle \psi_{\beta k}, \psi_{\beta' k'} \rangle\rangle_M = \delta_{\beta\beta'}\delta_{kk'}.$$

Thus the Ns lattice modes form a mass-orthonormal basis.

2.4.3 Modal Decomposition

The displacement column can therefore be expanded as

$$Q(t) = \sum_{\beta=1}^{s}\sum_{k} \eta_{\beta k}(t)\psi_{\beta k}, \qquad \eta_{\beta k}(t) = \langle\langle Q(t), \psi_{\beta k}\rangle\rangle_M. \tag{2.19}$$

Here $\eta_{\beta k}$ is the scalar modal coordinate: it measures the contribution of the mode $\psi_{\beta k}$ to the instantaneous lattice displacement.

Substituting into (2.16) gives

$$\sum_{\beta,k}\left(\ddot{\eta}_{\beta k}+\omega_{\beta k}^2\eta_{\beta k}\right)\psi_{\beta k}=0.$$

Projection onto each mode yields

$$\ddot{\eta}_{\beta k}+\omega_{\beta k}^2\eta_{\beta k}=0. \tag{2.20}$$

The initial conditions are likewise obtained by projection:

$$\eta_{\beta k}(0)=\langle\langle Q(0),\psi_{\beta k}\rangle\rangle_M, \qquad \dot{\eta}_{\beta k}(0)=\langle\langle \dot{Q}(0),\psi_{\beta k}\rangle\rangle_M.$$

For $\omega_{\beta k}>0$,

$$\eta_{\beta k}(t)=\eta_{\beta k}(0)\cos(\omega_{\beta k}t)+\frac{\dot{\eta}_{\beta k}(0)}{\omega_{\beta k}}\sin(\omega_{\beta k}t).$$

Since $A(-k)=\overline{A(k)}$, the eigenvectors may be chosen consistently so that

$$\psi_{\beta,-k}=\overline{\psi_{\beta k}}, \qquad \omega_{\beta,-k}=\omega_{\beta k}.$$

The condition that $Q(t)$ be real then becomes

$$\eta_{\beta,-k}(t)=\overline{\eta_{\beta k}(t)},$$

where $-k$ is understood modulo $2\pi/(sa)$. The paired complex contributions therefore combine to give real atomic displacements.

Atomic vibrations admit the same modal decomposition as the spring–mass systems considered earlier. Each mode specifies a collective pattern of atomic motion, labelled by a wave number k and a branch index β.

Projection using the mass inner product reduces the coupled lattice equations to independent scalar modal equations. The motion of any individual atom is a superposition of these collective vibrations.

2.4.4 Energy and the Translation Mode

The kinetic and potential energies can be written directly using the mass inner product:

$$\mathrm{KE}=\frac{1}{2}\langle\langle\dot{Q},\dot{Q}\rangle\rangle_M=\frac{1}{2}\sum_{\beta,k}|\dot{\eta}_{\beta k}|^2,$$
$$\mathrm{PE}=\frac{1}{2}\langle\langle\Omega Q,Q\rangle\rangle_M=\frac{1}{2}\sum_{\beta,k}\omega_{\beta k}^2|\eta_{\beta k}|^2.$$

Thus the total energy takes the uncoupled form

$$H_0 = \frac{1}{2}\sum_{\beta,k}\left(|\dot{\eta}_{\beta k}|^2 + \omega_{\beta k}^2|\eta_{\beta k}|^2\right). \tag{2.21}$$

The sums include all allowed wave numbers, with the conjugacy condition imposed. Each modal contribution to the energy is constant: in the harmonic approximation, the modes do not exchange energy.

Uniform translation is the zero-frequency mode:

$$\psi_0(\alpha,n) = \frac{1}{\sqrt{N\sum_{\gamma=1}^{s} m_\gamma}}, \qquad \Omega\,\psi_0 = 0.$$

Every atom has the same displacement in this mode, so no bond is stretched. Its modal equation is

$$\ddot{\eta}_0 = 0,$$

describing uniform centre-of-mass motion rather than an oscillation. All remaining modes have positive frequencies.

The branch approaching this uniform translation as $k \to 0$ is called the *acoustic branch*. At long wavelengths, neighbouring cells move almost together. For $s > 1$, the other $s-1$ branches have nonzero frequencies at $k = 0$ and are called *optical branches*; their modes involve relative motion of atoms within each cell.

In the quantum harmonic treatment, the nonzero-frequency modes are treated as quantum harmonic oscillators. Their quantized excitations are called phonons.

2.4.5 The Monatomic Lattice

Setting $s = 1$ and $\phi_k(1) = 1$ recovers the monatomic lattice. The quasiperiodic extension gives

$$\phi_k(0) = e^{-ika}\phi_k(1), \qquad \phi_k(2) = e^{ika}\phi_k(1),$$

and hence

$$\omega^2(k) = \frac{b_1}{m_1}\left(2 - e^{-ika} - e^{ika}\right) = \frac{4b_1}{m_1}\sin^2\left(\frac{ka}{2}\right).$$

There is one dispersion branch. For $|ka| \ll 1$,

$$\omega(k) \approx a\sqrt{\frac{b_1}{m_1}}\,|k|.$$

Thus long-wavelength vibrations behave approximately as waves with speed $a\sqrt{b_1/m_1}$. At $k = 0$ all atoms translate together, while at $k = \pi/a$, when this wave number is allowed, neighbouring atoms move in opposite directions.

2.4.6 Phonons

The modal decomposition provides a natural connection between classical lattice vibrations and their quantum description. Each nonzero-frequency mode behaves as an independent harmonic oscillator. In quantum mechanics, the energy of such an oscillator takes the discrete values

$$E_{\beta k} = \hbar\omega_{\beta k}\left(n_{\beta k} + \frac{1}{2}\right), \qquad n_{\beta k} = 0, 1, 2, \ldots,$$

where $\hbar$ is the reduced Planck constant. Increasing $n_{\beta k}$ by one increases the energy by $\hbar\omega_{\beta k}$. This elementary excitation is called a *phonon*, and $n_{\beta k}$ is the number of phonons occupying that mode.

A phonon is therefore not an individual vibrating atom. It is a quantum of a collective vibration whose spatial pattern is specified by the mode shape $\psi_{\beta k}$. The branch index β identifies the pattern of motion within each cell, while k specifies its phase variation across the lattice.

Classically, a normal mode can have any amplitude and hence any nonnegative energy. Quantum mechanically, the same mode has discrete energy levels separated by $\hbar\omega_{\beta k}$. A phonon represents one quantum of excitation of that collective mode.

Even when $n_{\beta k} = 0$, the mode has the *zero-point energy* $\hbar\omega_{\beta k}/2$. Thus the quantum ground state does not correspond to all atoms remaining motionless at their equilibrium positions. The zero-frequency uniform translation is treated separately as centre-of-mass motion.

The acoustic and optical branches correspond to acoustic and optical phonons, respectively. Long-wavelength acoustic phonons describe collective motion associated with sound, whereas optical phonons involve relative motion of atoms within a unit cell. In crystals with suitable electrical coupling, optical modes can interact with electromagnetic radiation.

The wave number also assigns a phonon a *crystal momentum* $\hbar k$, defined modulo $2\pi\hbar/(sa)$. This describes its behaviour under lattice translations; it should not be identified directly with the total mechanical momentum of the atoms.

At temperature T, the mean occupation of a mode in thermal equilibrium is

$$\overline{n}_{\beta k} = \frac{1}{\exp\left(\hbar\omega_{\beta k}/(k_B T)\right) - 1},$$

where k_B is Boltzmann's constant. When $k_B T \gg \hbar\omega_{\beta k}$, its mean energy approaches $k_B T$, recovering classical equipartition. At low temperatures, high-frequency modes are weakly excited. This helps explain why the heat capacity of a solid decreases as its temperature is lowered.

In the harmonic approximation, the modes are independent. Anharmonic interactions, defects, and boundaries can scatter phonons and redistribute vibrational energy. The phonon description therefore provides a useful framework for understanding heat capacity, thermal transport, and the relaxation of lattice vibrations.

2.5 Exercises

Exercise 2.1. Consider the 2-DOF translational system shown in figure-2.4.

2.1.a Find the natural frequencies of the system.
2.1.b Find the normal modes of vibration and the normalized equations.
2.1.c Analytically find the frequency response of the system.
2.1.d Numerically plot the frequency response of the system using the MATLAB for various combinations of parameters of the system. The MATLAB function *bode* in the controls toolbox may be used for this purpose.

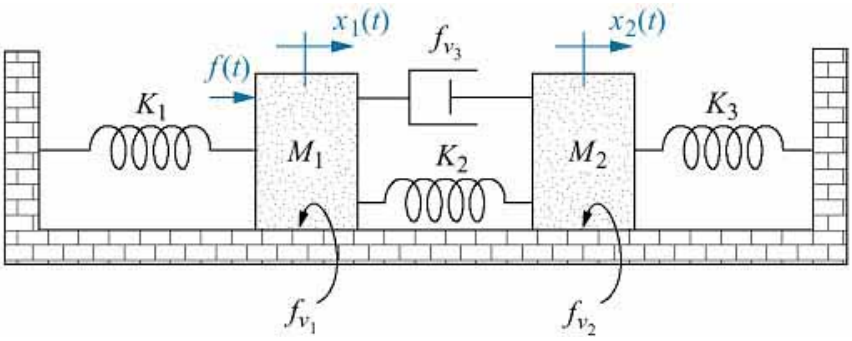


Fig. 2.4 2-DOF Translational Vibratory System

Exercise 2.2. Figure-2.5a shows a schematic of a long rotating shaft mounted on two bearings of damping co-efficient D_1 and D_2 respectively. The shaft is seen to exhibit a torsional vibration. Figure-2.5b shows a simplified approximation of the system. Using this approximation answer the following:

2.2.a Estimate the normal modes of vibration of the shaft.
2.2.b If $T(t) = A\cos(\omega t)$ find the response of θ_2 of the approximated system shown in figure-2.5b as t becomes very large.

Exercise 2.3. An unbalanced small high speed motor is mounted on a heavy platform that is connected to the ground through very stiff springs as shown in figure-2.6. The springs are spaced a distance L away from the center line of the platform and the center of the motor is a distance L_0 away from the center line of the platform. The unbalance distance from the center of rotation is found to be ε and the mass of the rotor of the motor is found to be m. Assuming that the stiffness of the springs in the axial direction is much smaller than the stiffness in the transverse direction. Derive a simplified approximate model for the system and find the normal modes of vibration.

Exercise 2.4. A blower driven by a motor is mounted on the ground using a vibration isolation arrangement similar to that shown in figure-2.7. Due to some modifications in the operation of the system the original blower had to be replaced with a new one with an operating region

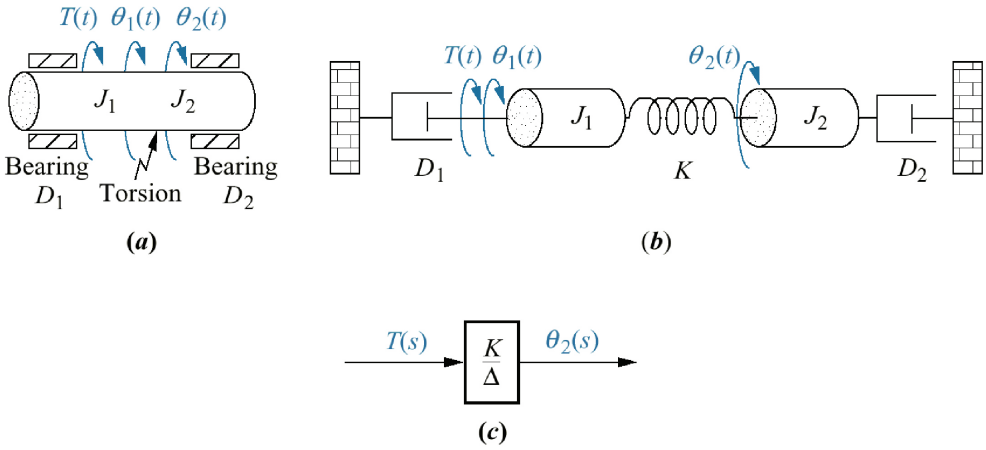


Fig. 2.5 Torsional vibration of a long rotating shaft.

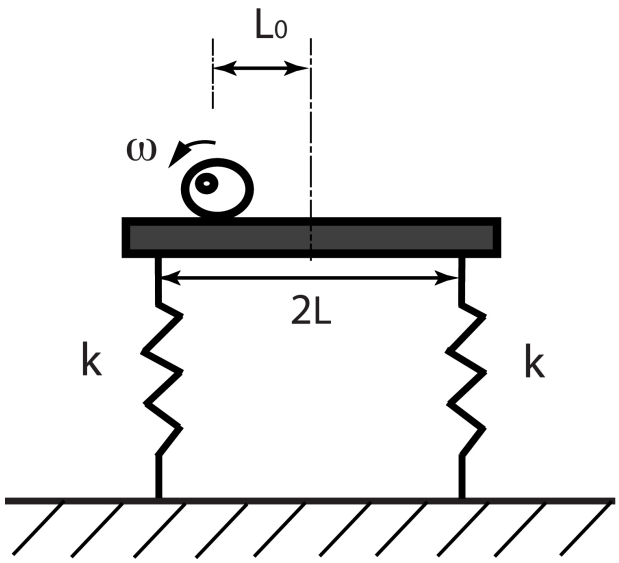


Fig. 2.6 An Unbalance motor on a platform.

between 1000 r.p.m and 2000 r.p.m. The new motor when operated at 990 r. p. m. and 1710 r. p. m. exhibited violent vibrations. You are required to design a vibration absorber to mitigate the vibrations. As a first step one would typically analyzes the approximate behavior of the system and attempt to theoretically predict its behavior.

1. Explain the cause of vibration in the system.
2. Assuming that the motor mass and eccentricities are negligible compared to that of the blower, the system can be assumed to behave similarly to that of the coupled vibration setup in the Applied Mechanics lab. For small deflections and velocities the experimental setup in the lab can be approximately modeled by a lumped parameter system shown in figure-2.8 where the structural damping in the springs is assumed to be negligible. Show

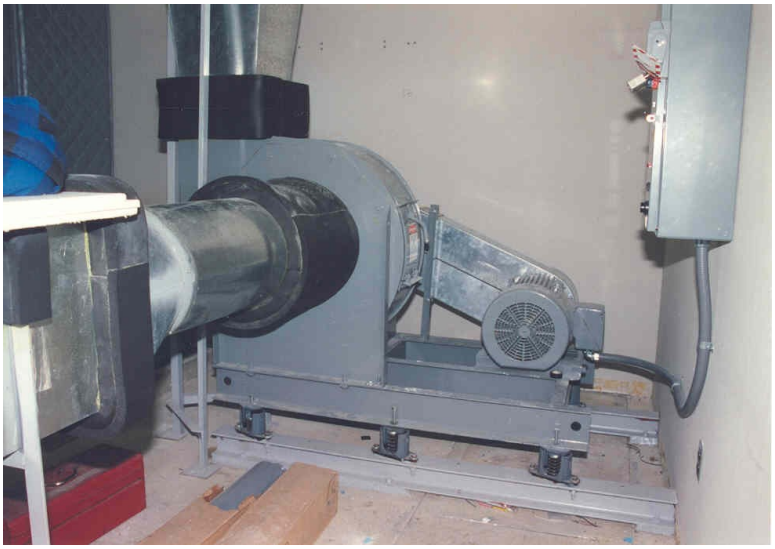


Fig. 2.7 A Vibration Isolation Mounting of Motor Driving a Blower.

that for small displacements, and angles the equations of motion of the system are given by

$$M\ddot{z} + Kz = F(t),$$

where

$$z(t) = \begin{bmatrix} x(t) \\ \theta(t) \end{bmatrix}, \quad F(t) = \begin{bmatrix} f(t) \\ c\,f(t) \end{bmatrix}, \quad M = \begin{bmatrix} m & 0 \\ 0 & I \end{bmatrix}, K = \begin{bmatrix} 2k & -k(l_2 - l_1) \\ -k(l_2 - l_1) & k(l_1^2 + l_2^2) \end{bmatrix}.$$

3. Write down these equations in the form

$$\ddot{z} + \Omega z = M^{-1}F(t),$$

where

$$\Omega = \begin{bmatrix} \omega_{n1}^2 & -(l_1^2 + l_2^2)\mu\,\omega_{n1}^2 \\ -2\mu\,\alpha^2\omega_{n1}^2 & \alpha^2\omega_{n1}^2 \end{bmatrix}.$$

with

$$\omega_{n1}^2 = \frac{2k}{m}, \quad \omega_{n2}^2 = \frac{k(l_1^2 + l_2^2)}{I}, \quad \mu = \frac{(l_2 - l_1)}{2(l_1^2 + l_2^2)}, \quad \alpha = \frac{\omega_{n2}}{\omega_{n1}} = \sqrt{\frac{m(l_1^2 + l_2^2)}{2I}},$$

4. State the conditions for the existence of synchronized motion of the system of the form

$$z_m(t) = \bar{z}_m \cos(\omega t + \phi),$$

for some constant 2×2 matrix $\bar{z}_m = [\bar{x}_m \quad \bar{\theta}_m]^T$.

5. Show that there exists two distinct types of synchronized solutions of the form

$$z_{m1}(t) = \bar{z}_{m1} \cos(\omega_1 t + \phi_1),$$

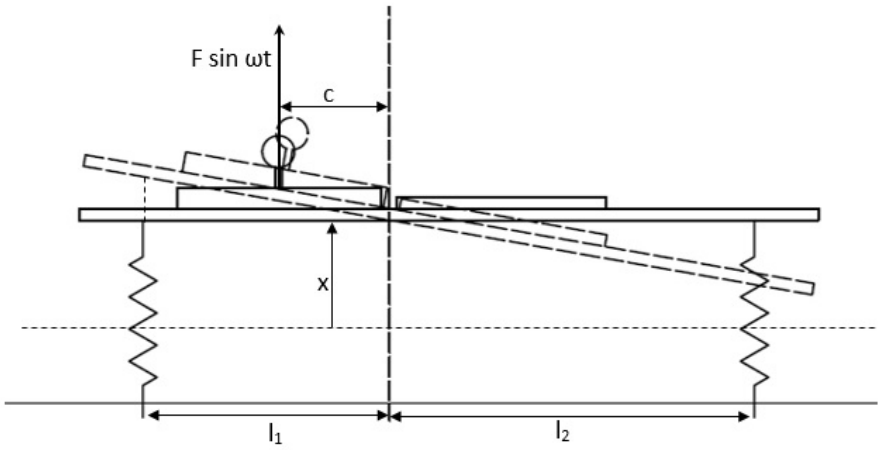


Fig. 2.8 A 2-DOF coupled torsional plus translational system

$$z_{m2}(t) = \bar{z}_{m2}\cos(\omega_2 t + \phi_2),$$

where

$$\omega_1 = \omega_{n1}\left(\frac{(1+\alpha^2) - \sqrt{(1+\alpha^2)^2 - 4\alpha^2(1-2\mu^2(l_1^2+l_2^2))}}{2}\right)^{\frac{1}{2}},$$

$$\omega_2 = \omega_{n1}\left(\frac{(1+\alpha^2) + \sqrt{(1+\alpha^2)^2 - 4\alpha^2(1-2\mu^2(l_1^2+l_2^2))}}{2}\right)^{\frac{1}{2}},$$

and $\bar{z}_{m1} = [\bar{x}_{m1} \quad \bar{\theta}_{m2}]^T$ and $\bar{z}_{m2} = [\bar{x}_{m2} \quad \bar{\theta}_{m2}]^T$ are the eigenvectors corresponding to the eigenvalues ω_1^2 and ω_2^2 respectively while ϕ_1 and ϕ_2 are arbitrary constants.

6. Show that

$$\frac{\bar{x}_{m1}}{\bar{\theta}_{m1}} = \frac{\alpha^2 - 1 + \sqrt{(1+\alpha^2)^2 - 4\alpha^2(1-2\mu^2(l_1^2+l_2^2))}}{4\mu\alpha^2},$$

$$\frac{\bar{x}_{m2}}{\bar{\theta}_{m2}} = \frac{\alpha^2 - 1 - \sqrt{(1+\alpha^2)^2 - 4\alpha^2(1-2\mu^2(l_1^2+l_2^2))}}{4\mu\alpha^2}.$$

or alternatively

$$\frac{\bar{x}_{m1}}{\bar{\theta}_{m1}} = \frac{2(l_1^2+l_2^2)\mu}{1 - \alpha^2 + \sqrt{(1+\alpha^2)^2 - 4\alpha^2(1-2\mu^2(l_1^2+l_2^2))}},$$

$$\frac{\bar{x}_{m2}}{\bar{\theta}_{m2}} = \frac{2(l_1^2+l_2^2)\mu}{1-\alpha^2-\sqrt{(1+\alpha^2)^2-4\alpha^2(1-2\mu^2(l_1^2+l_2^2))}}.$$

7. Describe the qualitative behavior of the two modes of vibration.
8. Discuss the case if the blower is mounted very close to the middle of the platform.
9. The mass of the motor is fairly accurately known. However due to the use over time the spring constants of the mounts are poorly known. Describe a method that you would use to estimate the spring constant of the mount.
10. Describe how you would validate the estimate of the spring constants.
11. Estimate the spring constant of the coupled 2DOF vibratory system in the lab and discuss the validity of your estimate.

Exercise 2.5. A motor that weighs 200 kg is mounted on the floor using a stiff platform of overall stiffness estimated to be 80 kN/m. The unbalances in the rotor induces a periodic forcing on the platform. The operating conditions of the motor are between 60 r.p.m. and 300 r.p.m. Design a tuned mass damper for the system. State clearly all assumptions and approximations made.

Exercise 2.6. Consider the 3-DOF translational vibratory system shown in figure-2.9. If $k_1 = k_2 = k_3 = k_4 = k$ and $C_1 = C_2 = C_3 = C_4 = C$ find the initial condition response of the system when only the first mass is given an initial displacement of x_{10}.

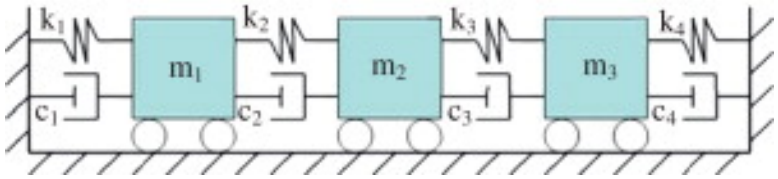


Fig. 2.9 A 3-DOF translational vibratory system

Exercise 2.7. Find the modes of vibration of the 3-DOF translational vibratory system shown in figure-2.10. Comment on the form of the each mode of vibration.

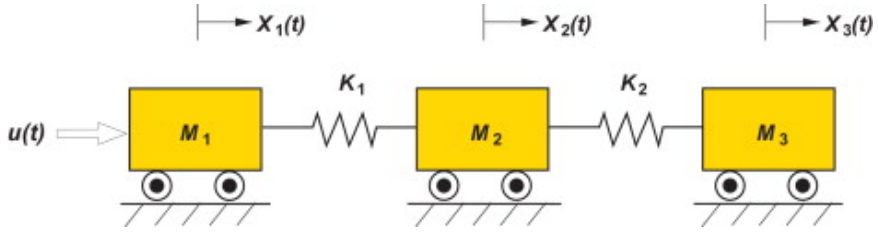


Fig. 2.10 A 3-DOF translational vibratory system

Exercise 2.8. Find the resonance frequencies of the 3-DOF translational vibratory system shown in figure-2.11. Assuming zero initial conditions find the response of the system when $z(t) = 1(t)$ the unit step function.

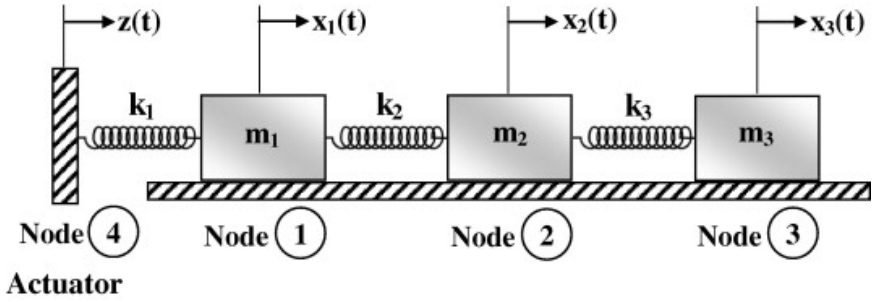


Fig. 2.11 A 3-DOF translational vibratory system

Exercise 2.9. A uniform shaft of 9.52 mm diameter is supported in ball bearings and carries four discs A, B, C, D. The bearings are placed close to the discs, so that the sag of the shaft is reduced to a minimum. The wheel A carries a small spindle which is supported in ball bearings. The spindle carries an out-of balance mass and is belt driven by a variable speed DC motor. A schematic of the setup is shown in figure-2.12. Answer the following with detailed justification:

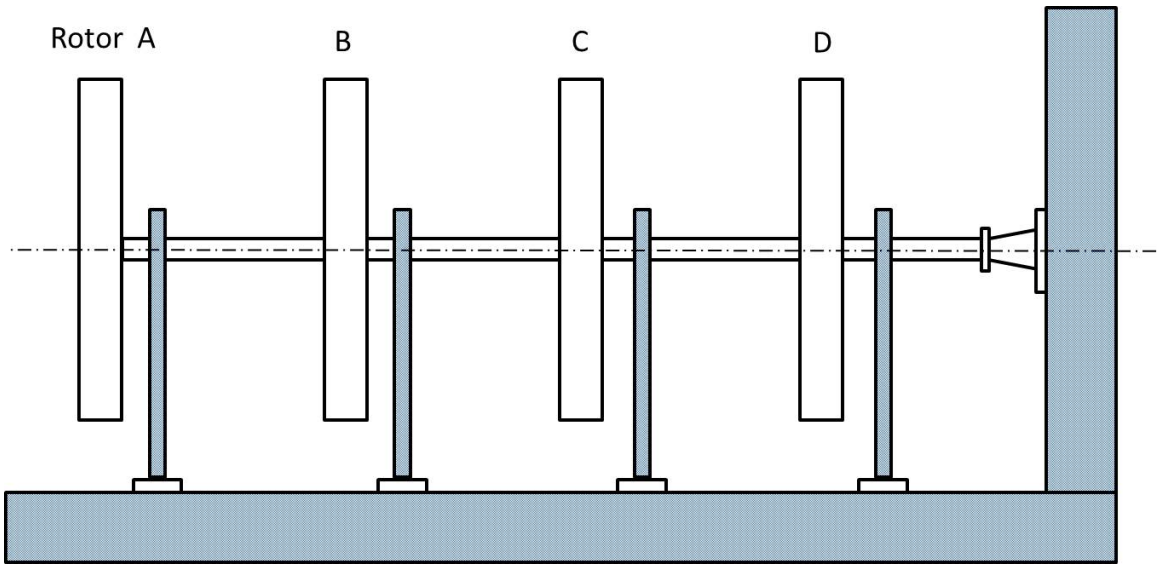


Fig. 2.12 A Multi-Rotor Vibratory System

1. Assuming that the damping in the system is negligible, show that for small angular displacements the equations of motion of the system are given by

$$I\ddot{\theta} + K\theta = T(t),$$

where

$$\theta(t) = \begin{bmatrix} \theta_1(t) \\ \theta_2(t) \\ \theta_3(t) \\ \theta_4(t) \end{bmatrix}, \quad T(t) = \begin{bmatrix} f(t) \\ 0 \\ 0 \\ 0 \end{bmatrix},$$

$$I = \begin{bmatrix} I_1 & 0 & 0 & 0 \\ 0 & I_2 & 0 & 0 \\ 0 & 0 & I_3 & 0 \\ 0 & 0 & 0 & I_4 \end{bmatrix}, K = \begin{bmatrix} k_1 & -k_1 & 0 & 0 \\ -k_1 & (k_1+k_2) & -k_2 & 0 \\ 0 & -k_2 & (k_2+k_3) & -k_3 \\ 0 & 0 & -k_3 & (k_3+k_4) \end{bmatrix}.$$

2. Write down these equations in the form

$$\ddot{\theta} + \Omega\theta = M^{-1}F(t),$$

where

$$\Omega = \begin{bmatrix} \omega_{n1}^2 & -\omega_{n1}^2 & 0 & 0 \\ -\mu_1\omega_{n1}^2 & (\mu_1\omega_{n1}^2 + \omega_{n2}^2) & -\omega_{n2}^2 & 0 \\ 0 & -\mu_2\omega_{n2}^2 & (\mu_2\omega_{n2}^2 + \omega_{n3}^2) & -\omega_{n3}^2 \\ 0 & 0 & -\mu_3\omega_{n3}^2 & (\mu_3\omega_{n3}^2 + \omega_{n4}^2) \end{bmatrix}.$$

with

$$\omega_{n1}^2 = \frac{k_1}{I_1}, \ \omega_{n2}^2 = \frac{k_2}{I_2}, \ \omega_{n3}^2 = \frac{k_3}{I_3}, \ \omega_{n4}^2 = \frac{k_4}{I_4},$$

$$\mu_1 = \frac{I_1}{I_2}, \ \mu_2 = \frac{I_2}{I_3}, \ \mu_3 = \frac{I_3}{I_4},$$

3. For the unforced undamped system, state the conditions for the existence of synchronized motion of the form

$$\theta_m(t) = \bar{z}_m \cos(\omega t + \phi),$$

for some constant 4×1 matrix $\bar{\theta}_m = [\bar{\theta}_{1_m} \quad \bar{\theta}_{2_m} \quad \bar{\theta}_{3_m} \quad \bar{\theta}_{4_m}]^T$.

4. Find the modes of vibration of the system.
5. Qualitatively describe the behavior of each of the modes of vibration.

Chapter 3
Modal Analysis of Continuous Systems

In the preceding chapter, we described vibrating systems using a finite number of displacement coordinates. Such models are called *discrete* or *lumped parameter* systems: their configuration at any instant is represented by a column $x(t) \in \mathbb{R}^n$, and their motion is governed by a finite set of ordinary differential equations. In this chapter, we extend the same ideas to *continuous*, or *distributed parameter*, systems, in which inertia and elasticity are distributed throughout a spatial domain.

For a continuous system, the displacement generally varies from point to point. Its configuration must therefore be described by a field $y(x,t)$, where x denotes position and t denotes time. Examples include the transverse displacement of a beam and the deflection of a stretched membrane. The equations governing these fields involve derivatives with respect to both space and time and are therefore partial differential equations. Boundary conditions describe how the system is supported or constrained, while initial conditions specify its initial displacement and velocity fields.

Despite this change in mathematical description, the central idea of modal analysis remains the same. For a discrete system, the mode shapes are eigenvectors of a matrix, and the configuration is expressed as a linear combination of these eigenvectors. For a continuous system, the mode shapes are eigenfunctions of a spatial differential operator, whose domain incorporates the boundary conditions. Under suitable assumptions, these eigenfunctions form a complete orthonormal basis with respect to an appropriate inner product. The displacement field can then be expressed as

$$y(x,t) = \sum_{k=1}^{\infty} \alpha_k(t)\,\psi_k(x),$$

where $\psi_k(x)$ is a mode shape and $\alpha_k(t)$ is its time-dependent modal coordinate.

Projection onto each mode reduces the governing partial differential equation to an ordinary differential equation for the corresponding modal coordinate. The initial conditions and external forcing are projected in the same way. Thus the physical response is again understood as a superposition of simpler modal responses. The main new feature is that the expansion generally contains infinitely many modes, so completeness, convergence, and the conditions under which the projection is valid require attention. In practical calculations, we retain a finite number of modes and check whether they adequately represent the response of interest.

We begin with orthonormal expansions in finite-dimensional spaces and then introduce Fourier series as a familiar example of the same construction for functions. This prepares the

way for a general formulation of continuous-system modal analysis. We apply the method to an axially loaded beam, a continuous approximation of a circular chain of coupled rotors, and a circular membrane. These examples show how geometry, material properties, and boundary conditions determine the natural frequencies and mode shapes, while the spatial distribution of the forcing determines which modes are excited. Finally, we consider a periodically varying axial load, which introduces parametric excitation and motivates the stability analysis developed in the following chapter.

3.1 The Basic Idea of Fourier Series

Fourier series provide a familiar example of expressing a function in an orthonormal basis. This is the same idea underlying modal decomposition: a spatial pattern is represented as a superposition of simpler patterns, and their coefficients are obtained using an inner product.

3.1.1 Orthonormal Bases in Finite Dimensions

Before considering Fourier series, let us recall the same idea in a finite-dimensional setting. Consider an n-dimensional complex vector space $\mathbb{C}^n$ with the inner product

$$\langle\langle x, y\rangle\rangle = y^* x = \sum_{j=1}^{n} x_j \overline{y_j},$$

where y^* denotes the conjugate transpose of y. This inner product is linear in its first argument. For real columns, it reduces to the Euclidean inner product $y^T x$.

Suppose $\{\psi_1, \ldots, \psi_n\}$ is an orthonormal basis:

$$\langle\langle \psi_k, \psi_l\rangle\rangle = \delta_{kl}.$$

Every $x \in \mathbb{C}^n$ can then be expressed uniquely as

$$x = \sum_{k=1}^{n} \alpha_k \psi_k.$$

To determine the coefficients, take the inner product with ψ_j:

$$\langle\langle x, \psi_j\rangle\rangle = \sum_{k=1}^{n} \alpha_k \langle\langle \psi_k, \psi_j\rangle\rangle = \alpha_j.$$

Hence

$$\boxed{x = \sum_{k=1}^{n} \langle\langle x, \psi_k\rangle\rangle \, \psi_k.}$$

Each coefficient measures the component of x along the corresponding basis element.

For example, the columns

$$\psi_1 = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix}, \qquad \psi_2 = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\end{bmatrix}$$

form an orthonormal basis for $\mathbb{R}^2$. For $x = [x_1\ x_2]^T$,

$$\alpha_1 = \frac{x_1 + x_2}{\sqrt{2}}, \qquad \alpha_2 = \frac{x_1 - x_2}{\sqrt{2}},$$

and

$$x = \alpha_1\psi_1 + \alpha_2\psi_2.$$

The first coefficient describes the common component of the two entries, while the second describes their difference.

If only the first $r < n$ basis elements are retained, define

$$x_r = \sum_{k=1}^{r} \alpha_k \psi_k.$$

The error is orthogonal to every retained basis element:

$$\langle\langle x - x_r, \psi_j\rangle\rangle = 0, \qquad j = 1, \ldots, r.$$

Thus x_r is the orthogonal projection of x onto their span.

Indeed, for any other approximation $v = \sum_{k=1}^{r} c_k\psi_k$, orthogonality gives

$$\|x - v\|^2 = \|x - x_r\|^2 + \|x_r - v\|^2 \geq \|x - x_r\|^2.$$

Therefore projection gives the unique best approximation within the chosen span.

Orthonormality also implies

$$\|x\|^2 = \sum_{k=1}^{n} |\alpha_k|^2, \qquad \|x - x_r\|^2 = \sum_{k=r+1}^{n} |\alpha_k|^2.$$

The approximation error is therefore determined exactly by the omitted coefficients.

An orthonormal basis allows us to resolve a column into independent components using inner products. Retaining only some components gives the best approximation in their span.

Fourier series extend this construction to functions. The finite sum becomes an infinite series, so we must also establish completeness of the basis and specify the sense in which the series converges.

3.1.2 The Space of Square-Integrable Functions

Let

$$\mathscr{F} = L^2(0,L;\mathbb{C})$$

be the space of measurable complex-valued functions satisfying

$$\int_0^L |f(x)|^2\,dx < \infty.$$

Functions that agree almost everywhere are identified as the same element of $\mathscr{F}$. Thus changing a function at finitely many points, or on any set of measure zero, does not change its element in this space.

For $f,g \in \mathscr{F}$, define

$$\langle\langle f,g\rangle\rangle = \frac{1}{L}\int_0^L f(x)\overline{g(x)}\,dx.$$

This inner product is linear in its first argument, conjugate-linear in its second, and satisfies

$$\langle\langle f,g\rangle\rangle = \overline{\langle\langle g,f\rangle\rangle}, \qquad \langle\langle f,f\rangle\rangle \geq 0.$$

Equality in the second expression holds precisely when $f = 0$ almost everywhere. The induced norm is

$$\|f\|_{\mathscr{F}}^2 = \frac{1}{L}\int_0^L |f(x)|^2\,dx.$$

The space is complete in this norm and is therefore a Hilbert space.

3.1.2.1 An Orthonormal Basis

Consider the functions

$$\psi_k(x) = e^{i2\pi kx/L}, \qquad k \in \mathbb{Z}.$$

Their inner products are

$$\langle\langle \psi_k, \psi_l\rangle\rangle = \frac{1}{L}\int_0^L e^{i2\pi(k-l)x/L}\,dx = \delta_{kl}.$$

Thus they form an orthonormal set.

Orthonormality alone does not imply that every function can be represented using these functions. We also require *completeness*: finite linear combinations of the ψ_k must approximate every element of $\mathscr{F}$ arbitrarily closely in the $\mathscr{F}$ norm.

The Fourier completeness theorem states that this is indeed the case. Thus $\{\psi_k\}_{k\in\mathbb{Z}}$ is an orthonormal basis for $\mathscr{F}$. Here "basis" means that infinite expansions converge in the Hilbert-space norm; it does not mean that every function is a finite linear combination.

3.1.2.2 Fourier Coefficients and Convergence

For $f \in \mathscr{F}$, define

$$\alpha_k = \langle\langle f, \psi_k \rangle\rangle = \frac{1}{L}\int_0^L f(x) e^{-i2\pi kx/L}\, dx. \tag{3.1}$$

These integrals exist because square-integrability on a finite interval implies integrability.

The finite Fourier approximation is

$$S_N f(x) = \sum_{k=-N}^{N} \alpha_k \psi_k(x).$$

It is the orthogonal projection of f onto the span of $\psi_{-N}, \ldots, \psi_N$, since

$$\langle\langle f - S_N f, \psi_j \rangle\rangle = 0, \qquad |j| \le N.$$

Consequently, $S_N f$ is the best approximation to f in this finite-dimensional space, measured by $\|\cdot\|_{\mathscr{F}}$.

Completeness gives

$$\lim_{N\to\infty} \frac{1}{L}\int_0^L |f(x) - S_N f(x)|^2\, dx = 0.$$

In this sense,

$$f(x) = \sum_{k=-\infty}^{\infty} \alpha_k e^{i2\pi kx/L}. \tag{3.2}$$

This is the *Fourier series* of f, and its coefficients are unique.

The convergence above is called *mean-square convergence*. It does not, by itself, assert equality at every point. For example, if the periodic extension of f is piecewise continuously differentiable, the Fourier series converges at each point to

$$\frac{f(x^-) + f(x^+)}{2}.$$

It therefore recovers $f(x)$ at continuity points and the average of the two one-sided limits at a jump. At the endpoints, these limits are interpreted using the periodic extension.

No condition $f(0) = f(L)$ is required for an L^2 expansion. Endpoint values do not affect the Fourier coefficients.

Orthonormality and completeness also give Parseval's identity:

$$\frac{1}{L}\int_0^L |f(x)|^2\, dx = \sum_{k=-\infty}^{\infty} |\alpha_k|^2.$$

Thus the squared norm of the function is the sum of the squared magnitudes of its modal coefficients. For a real-valued function,

$$\alpha_{-k} = \overline{\alpha_k},$$

so conjugate pairs combine to give a real Fourier series.

Fourier expansion consists of projecting a function onto a complete orthonormal family:

$$\alpha_k = \langle\langle f, \psi_k \rangle\rangle, \qquad f = \sum_k \alpha_k \psi_k.$$

The equality is understood in the mean-square sense unless stronger convergence conditions are specified. This is the basic structure of modal decomposition.

3.1.2.3 From Fourier Series to the Fourier Transform

Fourier series use a discrete set of frequencies,

$$\omega_k = \frac{2\pi k}{L}, \qquad \Delta\omega = \frac{2\pi}{L}.$$

As the interval length increases, their spacing decreases. This suggests replacing a discrete sum by an integral over a continuous frequency variable.

To see the connection, consider a function on $\mathbb{R}$ and its restriction to $[-L/2, L/2]$. Its Fourier coefficients on this interval are

$$\alpha_k^{(L)} = \frac{1}{L}\int_{-L/2}^{L/2} f(x) e^{-i\omega_k x}\, dx.$$

Writing

$$F_L(\omega) = \int_{-L/2}^{L/2} f(x) e^{-i\omega x}\, dx,$$

the corresponding Fourier series becomes

$$f(x) = \frac{1}{2\pi}\sum_{k=-\infty}^{\infty} F_L(\omega_k) e^{i\omega_k x}\Delta\omega,$$

with equality understood on the interval in the Fourier-series sense.

For sufficiently regular, rapidly decaying functions, the limit $L \to \infty$ leads to the transform pair

$$F(\omega) = \int_{-\infty}^{\infty} f(x) e^{-i\omega x}\, dx, \tag{3.3}$$

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega) e^{i\omega x}\, d\omega. \tag{3.4}$$

These are the *Fourier transform* and the *inverse Fourier transform*, respectively. The sum-to-integral argument motivates the formulas; their validity requires the appropriate convergence theorems.

For $f \in L^1(\mathbb{R})$, the forward integral is absolutely convergent. If additionally $F \in L^1(\mathbb{R})$, the inverse integral recovers f almost everywhere and at its continuity points. For smooth rapidly decaying functions, both formulas hold pointwise.

For a general $f \in L^2(\mathbb{R})$, the forward integral need not converge absolutely. The transform is instead defined by the mean-square limit

$$F = \lim_{R\to\infty} \int_{-R}^{R} f(x) e^{-i\omega x}\, dx \quad \text{in } L^2(\mathbb{R}, d\omega).$$

The inverse is likewise interpreted as

$$f = \lim_{R\to\infty} \frac{1}{2\pi} \int_{-R}^{R} F(\omega) e^{i\omega x}\, d\omega \quad \text{in } L^2(\mathbb{R}, dx).$$

Plancherel's theorem gives

$$\int_{-\infty}^{\infty} |f(x)|^2\, dx = \frac{1}{2\pi} \int_{-\infty}^{\infty} |F(\omega)|^2\, d\omega.$$

This is the continuous-frequency counterpart of Parseval's identity.

If x denotes time, ω is an angular frequency. If x denotes position, it is a spatial angular frequency, or wave number.

3.1.2.4 Functions Defined for Nonnegative Time

For a function defined only on $[0,\infty)$, first extend it by zero:

$$\widetilde{f}(x) = \begin{cases} f(x), & x \geq 0, \\ 0, & x < 0. \end{cases}$$

Its Fourier transform is then

$$F(\omega) = \int_0^{\infty} f(x) e^{-i\omega x}\, dx, \tag{3.5}$$

whenever the integral exists, or in the L^2 sense described above. The inverse transform reconstructs $\widetilde{f}$:

$$\widetilde{f}(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega x}\, d\omega,$$

with the corresponding interpretation of convergence. Thus the one-sided forward integral is the full-line Fourier transform of the zero extension.

3.1.2.5 Connection with Modal Analysis

On a finite interval, Fourier series provide a discrete orthonormal expansion. In vibration problems, the appropriate basis is determined by the spatial operator and its boundary conditions; its eigenfunctions need not be complex exponentials.

On the whole line, the functions $e^{i\omega x}$ are not square-integrable and hence are not elements of $L^2(\mathbb{R})$. The Fourier integral is therefore a continuous spectral representation, rather than a countable orthonormal-basis expansion in that space.

The common idea is to resolve a function into simpler spatial patterns. When those patterns are eigenfunctions of the governing operator, projection also separates the evolution equation into independent modal equations.

3.2 Modal Analysis of Continuous Systems

Consider a linear undamped system described by

$$\frac{\partial^2 y}{\partial t^2} + Hy = q(\mathbf{x},t), \tag{3.6}$$

where $\mathbf{x}$ denotes position in a spatial domain $\mathscr{D}$, $y(\mathbf{x},t)$ is the displacement field, and $q(\mathbf{x},t)$ is the forcing after normalization by the mass density.

At each time, regard $y(\cdot,t)$ as an element of a Hilbert space $\mathscr{F}$. For a scalar displacement, a typical choice is

$$\mathscr{F} = L^2(\mathscr{D}, w\,d\mathbf{x}), \qquad \langle\langle u,v\rangle\rangle = \int_{\mathscr{D}} w(\mathbf{x})u(\mathbf{x})\overline{v(\mathbf{x})}\,d\mathbf{x},$$

where $w(\mathbf{x}) > 0$ is the appropriate mass weight. For real functions, complex conjugation may be omitted. The inner product is linear in its first argument.

The spatial differential operator is written as

$$H : D(H) \subset \mathscr{F} \longrightarrow \mathscr{F}.$$

Its domain $D(H)$ specifies both the required spatial regularity and the homogeneous boundary conditions. The boundary conditions are therefore part of the definition of the operator.

3.2.1 Conditions for the Existence of a Modal Basis

For finite-dimensional systems, the spectral theorem justified our expansion in mode shapes. A corresponding result is needed here: finding some eigenfunctions does not establish that they can represent every admissible displacement. The following assumptions provide sufficient conditions for a complete orthonormal modal basis; their role is to justify the expansion used in the examples that follow.

A sufficient set of assumptions for the modal method is:

- H is densely defined and self-adjoint with respect to the chosen inner product:

  $$H = H^*.$$

  In particular,

  $$\langle\langle Hu,v\rangle\rangle = \langle\langle u,Hv\rangle\rangle, \qquad u,v \in D(H).$$

  For an unbounded differential operator, this identity alone establishes symmetry; self-adjointness also requires equality of the operator and adjoint domains.
- H is nonnegative:

  $$\langle\langle Hu,u\rangle\rangle \geq 0, \qquad u \in D(H).$$
- H has compact resolvent: for some $c > 0$,

  $$(H + cI)^{-1} : \mathscr{F} \longrightarrow \mathscr{F}$$

  is compact.

These assumptions must be checked for the spatial operator, its coefficients, and its boundary conditions. Symmetry of the differential expression alone is insufficient.

Under these assumptions, there is a complete orthonormal basis of eigenfunctions $\{\psi_k\}_{k=1}^{\infty}$ satisfying

$$H\psi_k = \omega_k^2 \psi_k, \qquad \psi_k \in D(H), \qquad \omega_k \in \mathbb{R}, \tag{3.7}$$

and

$$\langle\langle \psi_k, \psi_l \rangle\rangle = \delta_{kl}.$$

The eigenvalues ω_k^2 have finite multiplicity and tend to infinity, counting multiplicities. For repeated eigenvalues, an orthonormal basis is chosen within each eigenspace.

The ω_k are the natural frequencies, and ψ_k are the corresponding mode shapes. A zero eigenvalue represents a displacement pattern with no linear restoring force.

3.2.2 Modal Decomposition

Completeness implies that

$$y(\mathbf{x},t) = \sum_{k=1}^{\infty} \alpha_k(t)\psi_k(\mathbf{x}), \qquad \alpha_k(t) = \langle\langle y(\cdot,t), \psi_k \rangle\rangle. \tag{3.8}$$

The series converges in the norm of $\mathscr{F}$; pointwise convergence requires additional regularity.

For a sufficiently regular solution, project (3.6) onto ψ_k. Self-adjointness gives

$$\langle\langle Hy, \psi_k \rangle\rangle = \langle\langle y, H\psi_k \rangle\rangle = \omega_k^2 \alpha_k.$$

Hence

$$\ddot{\alpha}_k + \omega_k^2 \alpha_k = f_k(t), \qquad f_k(t) = \langle\langle q(\cdot,t), \psi_k \rangle\rangle. \tag{3.9}$$

Thus the distributed forcing excites each mode through its projection onto that mode shape.

For initial conditions

$$y(\cdot,0) = y_0, \qquad \frac{\partial y}{\partial t}(\cdot,0) = v_0,$$

the modal initial conditions are

$$\alpha_k(0) = \langle\langle y_0, \psi_k \rangle\rangle, \qquad \dot{\alpha}_k(0) = \langle\langle v_0, \psi_k \rangle\rangle.$$

A standard sufficient condition for a unique finite-energy solution on $[0,T]$ is

$$y_0 \in D(H^{1/2}), \qquad v_0 \in \mathscr{F}, \qquad q \in L^1(0,T;\mathscr{F}).$$

Here the finite-energy condition means

$$\sum_{k=1}^{\infty} (1+\omega_k^2) |\langle\langle y_0, \psi_k \rangle\rangle|^2 < \infty.$$

The solution satisfies

$$y \in C([0,T];D(H^{1/2})) \cap C^1([0,T];\mathscr{F}),$$

and the field equation is understood weakly unless additional regularity is available. The modal equations remain valid in this setting.

The modal method consists of finding an orthonormal eigenbasis of the spatial operator, projecting the initial conditions and forcing onto that basis, solving the resulting scalar equations, and reconstructing the displacement by superposition.

Completeness of the eigenfunctions is essential: orthogonality alone does not justify the modal expansion.

3.2.3 Free and Forced Responses

For $\omega_k > 0$, the modal solution is

$$\alpha_k(t) = \alpha_k(0)\cos(\omega_k t) + \frac{\dot{\alpha}_k(0)}{\omega_k}\sin(\omega_k t) + \int_0^t \frac{\sin(\omega_k(t-\tau))}{\omega_k} f_k(\tau)\, d\tau.$$

For a zero-frequency mode,

$$\alpha_k(t) = \alpha_k(0) + t\dot{\alpha}_k(0) + \int_0^t (t-\tau) f_k(\tau)\, d\tau.$$

Consider harmonic forcing

$$q(\mathbf{x},t) = q_s(\mathbf{x})\cos(\omega t), \qquad q_s \in \mathscr{F}, \qquad \omega > 0.$$

Then

$$f_k(t) = \beta_k \cos(\omega t), \qquad \beta_k = \langle\langle q_s, \psi_k \rangle\rangle.$$

If ω differs from every natural frequency, the harmonic particular response is

$$\alpha_{k,p}(t) = \frac{\beta_k}{\omega_k^2 - \omega^2}\cos(\omega t),$$

and therefore

$$y_p(\mathbf{x},t) = \sum_{k=1}^{\infty} \frac{\beta_k}{\omega_k^2 - \omega^2}\psi_k(\mathbf{x})\cos(\omega t).$$

The signed coefficient accounts for a phase shift of zero or π relative to the harmonic forcing.

Since the system is undamped, this particular response is generally not the complete response from rest. For zero initial conditions,

$$\alpha_k(t) = \frac{\beta_k}{\omega_k^2 - \omega^2}\left(\cos(\omega t) - \cos(\omega_k t)\right).$$

The natural-frequency contribution persists.

At exact resonance, $\omega = \omega_k > 0$, the corresponding zero-initial-condition response is

$$\alpha_k(t) = \frac{\beta_k}{2\omega_k} t \sin(\omega_k t).$$

Resonant growth therefore requires nonzero modal forcing β_k. For a repeated natural frequency, the forcing must have a nonzero projection onto its eigenspace.

Near an isolated natural frequency with nonzero modal forcing, the harmonic particular response is dominated by the corresponding mode shape. The spatial distribution of the forcing determines which modes are excited.

The undamped model identifies natural frequencies and mode shapes. Damping must be included to predict finite resonant amplitudes and the decay of transients.

In computation, the expansion is truncated to a finite number of modes:

$$y_r(\mathbf{x},t) = \sum_{k=1}^{r} \alpha_k(t)\psi_k(\mathbf{x}).$$

The number retained should be checked by increasing r until the response quantities of interest converge.

3.2.4 Including Damping in the Modal Equations

The undamped analysis identifies the natural frequencies and mode shapes and explains how initial disturbances and external forces excite different patterns of motion. Real systems, however, dissipate mechanical energy through mechanisms such as material deformation, friction, and interaction with the surrounding medium. Consequently, free vibrations generally decay, and sustained harmonic excitation produces finite resonant amplitudes when the relevant modes are damped. Including damping therefore allows us to connect the modal description developed above with these observed features.

For lightly damped systems, the undamped mode shapes often provide a useful basis for describing the motion. We retain this basis and consider the model

$$\frac{\partial^2 y}{\partial t^2} + C\frac{\partial y}{\partial t} + Hy = q(\mathbf{x},t), \tag{3.10}$$

where C is a linear damping operator. We assume that the operators and solution have sufficient regularity for the projections below. The damping is dissipative when

$$\mathrm{Re}\langle\langle C\frac{\partial y}{\partial t}, \frac{\partial y}{\partial t}\rangle\rangle \geq 0$$

for admissible velocity fields $\dfrac{\partial y}{\partial t}$.

Using the modal expansion and projections introduced above,

$$y(\mathbf{x},t)=\sum_{k=1}^{\infty}\alpha_k(t)\psi_k(\mathbf{x}),\qquad f_j(t)=\langle\langle q(\cdot,t),\psi_j\rangle\rangle,$$

gives

$$\ddot{\alpha}_j+\sum_{k=1}^{\infty}c_{jk}\dot{\alpha}_k+\omega_j^2\alpha_j=f_j(t),\qquad c_{jk}=\langle\langle C\psi_k,\psi_j\rangle\rangle. \tag{3.11}$$

Thus damping generally couples the modal coordinates, even though the undamped equations are uncoupled.

A particularly useful case occurs when C acts diagonally on the same mode shapes:

$$C\psi_k=c_k\psi_k,\qquad c_k\geq 0.$$

The modal equations then remain independent:

$$\ddot{\alpha}_k+c_k\dot{\alpha}_k+\omega_k^2\alpha_k=f_k(t). \tag{3.12}$$

For $\omega_k>0$, define the modal damping ratio by

$$c_k=2\zeta_k\omega_k.$$

Each modal equation is therefore precisely the damped single-degree-of-freedom equation studied earlier. The initial modal displacement and velocity are obtained by the same projections as in the undamped case.

For example, uniform damping $C=cI$ gives $c_k=c$ and $\zeta_k=c/(2\omega_k)$. Another common model is *Rayleigh damping*,

$$C=aI+bH,\qquad a,b\geq 0,$$

for which $c_k=a+b\omega_k^2$. Both models preserve modal decoupling.

For an underdamped mode, $0\leq\zeta_k<1$, let

$$\omega_{dk}=\omega_k\sqrt{1-\zeta_k^2}.$$

The single-degree-of-freedom results give

$$\begin{aligned}\alpha_k(t)=e^{-\zeta_k\omega_k t}&\left[\alpha_k(0)\cos(\omega_{dk}t)+\frac{\dot{\alpha}_k(0)+\zeta_k\omega_k\alpha_k(0)}{\omega_{dk}}\sin(\omega_{dk}t)\right]\\&+\int_0^t\frac{e^{-\zeta_k\omega_k(t-\tau)}}{\omega_{dk}}\sin\big(\omega_{dk}(t-\tau)\big)f_k(\tau)\,d\tau.\end{aligned} \tag{3.13}$$

The critically damped and overdamped responses follow from the corresponding scalar equations in the same way.

For real harmonic forcing $q(\mathbf{x},t)=q_s(\mathbf{x})\cos(\omega t)$, choose real mode shapes and set

$$\beta_k=\langle\langle q_s,\psi_k\rangle\rangle.$$

When $\zeta_k > 0$, the modal response approaches

$$\alpha_{k,ss}(t) = \frac{\beta_k \cos\big(\omega t + \phi_k(\omega)\big)}{\sqrt{(\omega_k^2 - \omega^2)^2 + 4\zeta_k^2\omega_k^2\omega^2}}, \tag{3.14}$$

where

$$\phi_k(\omega) = -\operatorname{atan2}\left(2\zeta_k\omega_k\omega, \omega_k^2 - \omega^2\right).$$

The physical response is reconstructed by the same modal superposition as before. At $\omega = \omega_k$, the steady-state modal amplitude is finite:

$$|\alpha_{k,ss}|_{\text{amp}} = \frac{|\beta_k|}{2\zeta_k\omega_k^2}.$$

Zero-frequency modes must be treated separately through $\ddot{\alpha}_k + c_k\dot{\alpha}_k = f_k(t)$; damping can remove their velocity without restoring their displacement to zero.

When damping is diagonal in the undamped modal basis, continuous-system vibration reduces to independent damped scalar oscillators. The mode shapes determine the spatial patterns, while modal damping determines their decay rates and finite harmonic-response amplitudes. For general damping, the projected damping matrix contains off-diagonal terms, and the modal equations must be solved as a coupled system.

3.3 Example: Transversely Forced Axially Loaded Beam

Consider the axially loaded beam in the applied mechanics laboratory, illustrated in Figure 3.1. We first consider periodic transverse forcing with a constant axial load. Periodic variation of the axial load will be considered separately.

We approximate the system by a uniform Euler–Bernoulli beam of length L, flexural rigidity EI, and mass per unit length ρA. The axial load P is positive in compression. Damping is neglected, and the displacement $y(x,t)$ is measured from the static equilibrium configuration. This idealized model neglects the additional inertia of the attached apparatus.

3.3.1 The Spatial Operator

The equation governing small transverse vibrations is

$$\frac{\partial^2 y}{\partial t^2} + \frac{EI}{\rho A}\frac{\partial^4 y}{\partial x^4} + \frac{P}{\rho A}\frac{\partial^2 y}{\partial x^2} = q(x,t), \tag{3.15}$$

where $q(x,t) = p(x,t)/(\rho A)$ and $p(x,t)$ is the transverse force per unit length.

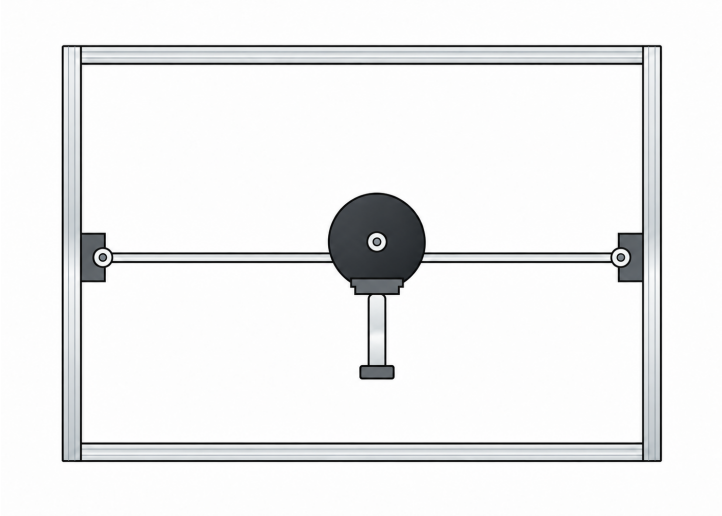


Fig. 3.1 A pin-supported beam carrying a motor-driven eccentric rotor at its centre and a suspended weight. Rotation of the unbalanced rotor produces periodic transverse forcing.

For simply supported ends, the displacement and bending moment vanish:

$$y(0,t) = y(L,t) = 0, \tag{3.16}$$

$$y_{xx}(0,t) = y_{xx}(L,t) = 0. \tag{3.17}$$

Following the general modal framework, take $\mathscr{F} = L^2(0,L)$ with inner product

$$\langle\langle u, v\rangle\rangle = \frac{2}{L}\int_0^L u(x)\overline{v(x)}\,dx.$$

Since the mass per unit length is constant, this differs from the mass inner product only by a constant factor.

Define

$$H = \frac{EI}{\rho A}\frac{d^4}{dx^4} + \frac{P}{\rho A}\frac{d^2}{dx^2} \tag{3.18}$$

on the domain

$$D(H) = \left\{u \in H^4(0,L) : u(0) = u(L) = u''(0) = u''(L) = 0\right\}.$$

Here $H^4(0,L)$ denotes the Sobolev space of functions whose weak derivatives through order four are square-integrable. Thus the boundary conditions belong to the operator domain, rather than to the definition of the ambient space $\mathscr{F}$.

The beam equation becomes

$$y_{tt} + Hy = q. \tag{3.19}$$

Integration by parts gives, for $u, v \in D(H)$,

$$\langle\langle Hu, v\rangle\rangle = \frac{2}{\rho AL}\int_0^L \left(EIu''\overline{v''} - Pu'\overline{v'}\right)dx = \langle\langle u, Hv\rangle\rangle.$$

With the stated domain, H is self-adjoint and has compact resolvent. The negative contribution from compression shows that increasing P reduces the transverse restoring stiffness.

3.3.2 Natural Frequencies and Mode Shapes

The functions

$$\psi_k(x) = \sin\left(\frac{k\pi x}{L}\right), \qquad k = 1,2,\ldots, \tag{3.20}$$

satisfy the boundary conditions and form an orthonormal basis for $\mathscr{F}$:

$$\langle\langle \psi_k, \psi_l \rangle\rangle = \delta_{kl}.$$

Direct substitution gives

$$H\psi_k = \omega_k^2 \psi_k,$$

where

$$\omega_k^2 = \frac{k^2\pi^2}{\rho A L^2}\left(\frac{EIk^2\pi^2}{L^2} - P\right). \tag{3.21}$$

All eigenvalues are positive provided

$$P < P_{\text{cr}}, \qquad P_{\text{cr}} = \frac{\pi^2 EI}{L^2}.$$

Under this condition, the natural frequencies are ω_k We assume $P < P_{\text{cr}}$ in the vibration analysis below.

As the compressive load approaches P_{cr}, the first natural frequency tends to zero. At $P = P_{\text{cr}}$, the first mode loses its linear restoring stiffness; for $P > P_{\text{cr}}$, its eigenvalue is negative and the linearized model predicts exponential growth rather than oscillation. This is the connection between the vibration model and Euler buckling.

3.3.3 Modal Equations

Expand the displacement as

$$y(x,t) = \sum_{k=1}^{\infty} \alpha_k(t)\psi_k(x), \qquad \alpha_k(t) = \frac{2}{L}\int_0^L y(x,t)\psi_k(x)\,dx.$$

Projection of (3.19) onto each mode gives

$$\ddot{\alpha}_k + \omega_k^2 \alpha_k = f_k(t),$$

where

$$f_k(t) = \langle\langle q(\cdot,t), \psi_k \rangle\rangle = \frac{2}{\rho A L}\int_0^L p(x,t)\sin\left(\frac{k\pi x}{L}\right)dx.$$

The initial modal displacement and velocity are obtained by projecting the corresponding initial fields in the same way.

3.3.4 Harmonic Transverse Forcing

Suppose that

$$p(x,t) = p_s(x)\cos(\omega t),$$

where $p_s(x)$ may be supported on only part of the beam. Then

$$f_k(t) = \beta_k \cos(\omega t), \qquad \beta_k = \frac{2}{\rho A L}\int_0^L p_s(x)\sin\left(\frac{k\pi x}{L}\right)dx.$$

For $\omega \neq \omega_k$, including the initial conditions,

$$\alpha_k(t) = \alpha_k(0)\cos(\omega_k t) + \frac{\dot{\alpha}_k(0)}{\omega_k}\sin(\omega_k t) + \frac{\beta_k}{\omega_k^2 - \omega^2}\left(\cos(\omega t) - \cos(\omega_k t)\right),$$

where

$$\alpha_k(0) = \langle\langle y(\cdot,0), \psi_k\rangle\rangle, \qquad \dot{\alpha}_k(0) = \langle\langle y_t(\cdot,0), \psi_k\rangle\rangle.$$

Consequently, when ω differs from every natural frequency,

$$y(x,t) = \sum_{k=1}^{\infty} \alpha_k(t)\sin\left(\frac{k\pi x}{L}\right).$$

The terms involving $\cos(\omega_k t)$ persist because damping has been neglected.

At exact resonance with mode k,

$$\alpha_k(t) = \alpha_k(0)\cos(\omega_k t) + \frac{\dot{\alpha}_k(0)}{\omega_k}\sin(\omega_k t) + \frac{\beta_k}{2\omega_k} t\sin(\omega_k t).$$

Thus resonance requires both $\omega = \omega_k$ and $\beta_k \neq 0$: the force distribution must have a nonzero projection onto the mode.

For example, a force distribution symmetric about the midpoint satisfies $p_s(L-x) = p_s(x)$. Its projection onto every even-indexed mode vanishes:

$$\beta_{2r} = 0, \qquad r = 1, 2, \ldots.$$

Such forcing excites only the odd-indexed modes from rest.

The axial load changes the natural frequencies, while the spatial distribution of the transverse force determines which modes are excited. For this uniform, simply supported model, the sinusoidal mode shapes are independent of the constant axial load.

Near an isolated natural frequency with nonzero modal forcing, the harmonic particular response is dominated by that mode. Damping must be retained to predict a finite resonant amplitude.

3.4 Example: Torsional Vibrations on a Circle

Consider identical rotors mounted at equally spaced positions around a fixed circular support. Each rotor turns about its own fixed axis, and neighbouring rotors are connected by elastic couplings that resist differences in their angular displacements. The last rotor is coupled to the first, forming a closed chain.

This system provides a physical interpretation of Fourier modes on a circle. The angular coordinate θ labels position around the support, while $u(\theta,t)$ denotes the local rotor displacement.

3.4.1 From a Discrete Chain to a Continuous Model

Let there be N rotors, with spacing

$$h = \frac{2\pi}{N}, \qquad \theta_j = jh, \qquad j = 0,\ldots,N-1.$$

If each rotor has moment of inertia J and each coupling has torsional stiffness κ, the equations are

$$J\ddot{u}_j = \kappa(u_{j+1} - 2u_j + u_{j-1}) + T_j(t), \qquad u_{j+N} = u_j.$$

Here $T_j(t)$ is the applied torque on rotor j.

To obtain a continuous model, write

$$J = \mathscr{I}h, \qquad \kappa = \frac{\mathscr{K}}{h}, \qquad T_j(t) = \tau(\theta_j,t)h,$$

where $\mathscr{I}$ is the inertia per unit angular coordinate, $\mathscr{K}$ is the effective coupling stiffness, and τ is the applied torque density. Then

$$\mathscr{I}\ddot{u}_j = \mathscr{K}\frac{u_{j+1} - 2u_j + u_{j-1}}{h^2} + \tau(\theta_j,t).$$

For displacement patterns varying smoothly between neighbouring rotors,

$$\frac{u_{j+1} - 2u_j + u_{j-1}}{h^2} \approx u_{\theta\theta}(\theta_j,t).$$

The continuum equation is therefore

$$u_{tt} - c^2 u_{\theta\theta} = q(\theta,t), \qquad c^2 = \frac{\mathscr{K}}{\mathscr{I}}, \qquad q = \frac{\tau}{\mathscr{I}}. \tag{3.22}$$

The periodic boundary conditions are

$$u(0,t) = u(2\pi,t), \qquad u_\theta(0,t) = u_\theta(2\pi,t).$$

The parameter c is the propagation speed measured in the angular coordinate. If the support has radius R, the corresponding speed along it is Rc.

3.4.2 The Spatial Operator and Fourier Modes

Take $\mathscr{F} = L^2(0,2\pi;\mathbb{C})$ with inner product

$$\langle\langle f,g\rangle\rangle = \frac{1}{2\pi}\int_0^{2\pi} f(\theta)\overline{g(\theta)}\,d\theta.$$

Equation (3.22) becomes

$$u_{tt} + Hu = q, \qquad H = -c^2\frac{d^2}{d\theta^2},$$

with domain

$$D(H) = \left\{v \in H^2(0,2\pi) : v(0) = v(2\pi),\ v'(0) = v'(2\pi)\right\}.$$

Integration by parts, using the periodic boundary conditions, gives

$$\langle\langle Hf,g\rangle\rangle = \frac{c^2}{2\pi}\int_0^{2\pi} f'(\theta)\overline{g'(\theta)}\,d\theta = \langle\langle f,Hg\rangle\rangle.$$

With the stated domain, H is self-adjoint, nonnegative, and has compact resolvent.

Its eigenfunctions are precisely the Fourier modes

$$\psi_k(\theta) = e^{ik\theta}, \qquad k \in \mathbb{Z},$$

since

$$H\psi_k = c^2k^2\psi_k.$$

Thus

$$\omega_k^2 = c^2k^2, \qquad \omega_k = c|k|,$$

and

$$\langle\langle \psi_k, \psi_l\rangle\rangle = \delta_{kl}.$$

The Fourier modes form a complete orthonormal basis for $\mathscr{F}$.

For each $k \geq 1$, the modes ψ_k and ψ_{-k} have the same natural frequency. An equivalent real orthonormal basis is

$$1, \qquad \sqrt{2}\cos(k\theta), \qquad \sqrt{2}\sin(k\theta), \qquad k = 1,2,\ldots.$$

3.4.3 Modal Decomposition

Expand the angular displacement as

$$u(\theta,t) = \sum_{k\in\mathbb{Z}} a_k(t)e^{ik\theta}, \qquad a_k(t) = \frac{1}{2\pi}\int_0^{2\pi} u(\theta,t)e^{-ik\theta}\,d\theta.$$

Projection of (3.22) onto each Fourier mode gives

$$\ddot{a}_k + \omega_k^2 a_k = q_k(t), \qquad q_k(t) = \frac{1}{2\pi}\int_0^{2\pi} q(\theta,t)e^{-ik\theta}\,d\theta.$$

The modal initial conditions are

$$a_k(0) = \langle\langle u_0, \psi_k\rangle\rangle, \qquad \dot{a}_k(0) = \langle\langle v_0, \psi_k\rangle\rangle,$$

where $u_0 = u(\cdot,0)$ and $v_0 = u_t(\cdot,0)$. For a real displacement,

$$a_{-k}(t) = \overline{a_k(t)}.$$

For $k \neq 0$, the modal response is

$$a_k(t) = a_k(0)\cos(\omega_k t) + \frac{\dot{a}_k(0)}{\omega_k}\sin(\omega_k t) + \int_0^t \frac{\sin(\omega_k(t-s))}{\omega_k} q_k(s)\,ds.$$

The mode $k = 0$ requires separate treatment:

$$\ddot{a}_0 = q_0(t), \qquad a_0(t) = \frac{1}{2\pi}\int_0^{2\pi} u(\theta,t)\,d\theta.$$

It represents equal angular displacement of every rotor. Since neighbouring rotors then have identical angles, none of the couplings is strained. Without applied torque, this mode has constant mean angular velocity.

3.4.4 Standing and Travelling Waves

For free vibration, an initial displacement

$$u(\theta,0) = A\cos(k\theta), \qquad u_t(\theta,0) = 0, \qquad k \geq 1,$$

gives

$$u(\theta,t) = A\cos(k\theta)\cos(ckt).$$

This is a standing wave: its nodes remain at fixed positions around the circle.

A travelling wave is obtained by combining the sine and cosine modes with an appropriate temporal phase:

$$u(\theta,t) = A\cos(k\theta - ckt) = A\cos(k\theta)\cos(ckt) + A\sin(k\theta)\sin(ckt).$$

Its pattern travels in the direction of increasing θ at angular speed c. Replacing the minus sign by a plus sign gives propagation in the opposite direction.

Thus the repeated natural frequencies allow both standing and travelling waves. They are different combinations of the same modal solutions.

3.4.5 Spatially Distributed Harmonic Forcing

Consider a torque density

$$\tau(\theta,t) = \tau_0 \cos(m\theta)\cos(\omega t), \qquad m \geq 1.$$

Its only nonzero complex Fourier coefficients occur at $k = \pm m$. Thus, from rest, only that spatial harmonic is excited.

Writing

$$u(\theta,t) = A_m(t)\cos(m\theta),$$

the modal equation is

$$\ddot{A}_m + c^2 m^2 A_m = \frac{\tau_0}{\mathscr{I}}\cos(\omega t).$$

For zero initial conditions and $\omega \neq cm$,

$$A_m(t) = \frac{\tau_0/\mathscr{I}}{c^2m^2 - \omega^2}\left(\cos(\omega t) - \cos(cmt)\right).$$

At exact resonance, $\omega = cm$,

$$A_m(t) = \frac{\tau_0}{2\mathscr{I} cm}\, t\sin(cmt).$$

A spatially uniform torque instead excites only the zero-frequency mode: all rotors respond together.

The Fourier modes describe collective angular motion around a closed chain. Their coefficients are obtained by projection, and each coefficient satisfies an independent scalar equation.

The spatial pattern of the applied torque determines which modes are excited. Equal motion of all rotors produces no restoring torque, while nonuniform motion strains the couplings and produces oscillations.

The continuum approximation is most accurate for wavelengths large compared with the rotor spacing. Indeed, the discrete chain has frequencies

$$\omega_{k,N}^2 = \frac{4\kappa}{J}\sin^2\left(\frac{kh}{2}\right) = \frac{4c^2}{h^2}\sin^2\left(\frac{kh}{2}\right).$$

For $|kh| \ll 1$, this reduces to

$$\omega_{k,N}^2 \approx c^2k^2,$$

recovering the continuum result.

3.5 Example: Vibrations of a Circular Drum

Consider a thin circular membrane of radius R, fixed around its rim and stretched under a uniform tension $\mathscr{T}$ per unit length. Let σ be its mass per unit area and $y(r,\theta,t)$ its small transverse displacement. We neglect bending stiffness, damping, and interaction with the surrounding air.

Under a transverse pressure $p(r,\theta,t)$, the membrane equation is

$$y_{tt} - c^2\left(y_{rr} + \frac{1}{r}y_r + \frac{1}{r^2}y_{\theta\theta}\right) = q(r,\theta,t), \qquad c^2 = \frac{\mathscr{T}}{\sigma}, \qquad q = \frac{p}{\sigma}. \tag{3.23}$$

The fixed rim imposes

$$y(R,\theta,t) = 0.$$

The displacement is 2π-periodic in θ and regular at the centre $r = 0$.

3.5.1 The Spatial Operator

Let $\mathscr{D}$ be the disk of radius R and take $\mathscr{F} = L^2(\mathscr{D})$, with inner product

$$\langle\langle u, v\rangle\rangle = \int_0^{2\pi}\int_0^R u(r,\theta)\overline{v(r,\theta)}\, r\, dr\, d\theta.$$

The factor r comes from the area element in polar coordinates. Since σ is constant, this inner product differs from the mass inner product only by a constant factor.

Equation (3.23) has the form

$$y_{tt} + Hy = q,$$

where

$$H = -c^2\nabla^2 = -c^2\left(\frac{\partial^2}{\partial r^2} + \frac{1}{r}\frac{\partial}{\partial r} + \frac{1}{r^2}\frac{\partial^2}{\partial\theta^2}\right).$$

With domain

$$D(H) = H^2(\mathscr{D}) \cap H_0^1(\mathscr{D}),$$

the operator is self-adjoint, positive, and has compact resolvent. The space $H_0^1(\mathscr{D})$ incorporates the zero-displacement boundary condition at the rim. Thus the general modal framework applies.

3.5.2 Bessel Functions and Mode Shapes

To determine the eigenfunctions, solve

$$H\psi = \omega^2\psi$$

using a separated form

$$\psi(r,\theta) = F(r)\Theta(\theta).$$

Periodicity in θ gives

$$\Theta'' + m^2\Theta = 0, \qquad m = 0,1,2,\ldots.$$

For $m = 0$, the angular factor is constant. For $m \geq 1$, the two independent angular factors are $\cos(m\theta)$ and $\sin(m\theta)$.

Writing $\kappa = \omega/c$, the radial equation becomes

$$r^2F'' + rF' + (\kappa^2 r^2 - m^2)F = 0.$$

This is Bessel's equation of order m. Its solution regular at the centre is proportional to

$$F(r) = J_m(\kappa r),$$

where J_m is the Bessel function of the first kind. The second independent solution is singular at the centre and is excluded.

The fixed-rim condition requires

$$J_m(\kappa R) = 0.$$

Let j_{mn} denote the nth positive zero of J_m, where $n = 1,2,\ldots$. The natural frequencies are therefore

$$\omega_{mn} = \frac{c\, j_{mn}}{R}. \tag{3.24}$$

The corresponding mode shapes are proportional to

$$J_m\left(\frac{j_{mn}r}{R}\right)\cos(m\theta), \qquad J_m\left(\frac{j_{mn}r}{R}\right)\sin(m\theta),$$

with the sine mode omitted when $m = 0$.

For a pure real mode, m gives the number of nodal diameters and $n-1$ gives the number of interior nodal circles. These are locations that remain stationary during that modal vibration. The $m = 0$ modes are axisymmetric.

For $m \geq 1$, the sine and cosine modes have the same natural frequency. Their linear combinations describe the same nodal pattern rotated through an arbitrary angle, reflecting the rotational symmetry of the drum.

3.5.3 Orthonormality and Modal Decomposition

The radial functions satisfy

$$\int_0^R rJ_m\left(\frac{j_{mn}r}{R}\right)J_m\left(\frac{j_{ml}r}{R}\right)dr = \frac{R^2}{2}[J_{m+1}(j_{mn})]^2\delta_{nl}.$$

Combining this with angular orthogonality, define

$$\psi_{0n}(r,\theta) = \frac{J_0(j_{0n}r/R)}{R\sqrt{\pi}\,|J_1(j_{0n})|}$$

and, for $m \geq 1$,

$$\psi^c_{mn}(r,\theta) = \frac{\sqrt{2}J_m(j_{mn}r/R)\cos(m\theta)}{R\sqrt{\pi}\,|J_{m+1}(j_{mn})|},$$
$$\psi^s_{mn}(r,\theta) = \frac{\sqrt{2}J_m(j_{mn}r/R)\sin(m\theta)}{R\sqrt{\pi}\,|J_{m+1}(j_{mn})|}.$$

These functions form an orthonormal basis for $\mathscr{F}$.

The displacement can therefore be expanded as

$$y(r,\theta,t) = \sum_{n=1}^{\infty} a_{0n}(t)\psi_{0n}(r,\theta) + \sum_{m=1}^{\infty}\sum_{n=1}^{\infty}\left(a^c_{mn}(t)\psi^c_{mn}(r,\theta) + a^s_{mn}(t)\psi^s_{mn}(r,\theta)\right).$$

Each coefficient is obtained by projection. For example,

$$a^c_{mn}(t) = \langle\langle y(\cdot,t), \psi^c_{mn}\rangle\rangle.$$

To write all modal equations together, let ν index the axisymmetric, cosine, and sine modes. Then

$$y = \sum_{\nu} a_\nu(t)\psi_\nu, \qquad H\psi_\nu = \omega_\nu^2\psi_\nu.$$

Projecting (3.23) onto ψ_ν gives

$$\ddot{a}_\nu + \omega_\nu^2 a_\nu = f_\nu(t), \tag{3.25}$$

where

$$f_\nu(t) = \langle\langle q(\cdot,t), \psi_\nu\rangle\rangle = \frac{1}{\sigma}\int_0^{2\pi}\int_0^R p(r,\theta,t)\psi_\nu(r,\theta)\,r\,dr\,d\theta.$$

For initial displacement y_0 and velocity v_0,

$$a_\nu(0) = \langle\langle y_0, \psi_\nu\rangle\rangle, \qquad \dot{a}_\nu(0) = \langle\langle v_0, \psi_\nu\rangle\rangle.$$

3.5.4 Free Vibration and Forced Response

The modal solution is

$$a_\nu(t) = a_\nu(0)\cos(\omega_\nu t) + \frac{\dot{a}_\nu(0)}{\omega_\nu}\sin(\omega_\nu t) + \int_0^t \frac{\sin(\omega_\nu(t-\tau))}{\omega_\nu} f_\nu(\tau)\,d\tau.$$

A drum released after an initial deformation, or set in motion by a brief strike, subsequently vibrates as a superposition of its free modal oscillations.

For harmonic pressure

$$p(r,\theta,t) = p_s(r,\theta)\cos(\omega t),$$

write

$$f_\nu(t) = \beta_\nu \cos(\omega t), \qquad \beta_\nu = \frac{1}{\sigma}\langle\langle p_s, \psi_\nu\rangle\rangle.$$

For $\omega \neq \omega_\nu$,

$$a_\nu(t) = a_\nu(0)\cos(\omega_\nu t) + \frac{\dot{a}_\nu(0)}{\omega_\nu}\sin(\omega_\nu t) + \frac{\beta_\nu}{\omega_\nu^2 - \omega^2}\left(\cos(\omega t) - \cos(\omega_\nu t)\right).$$

At exact resonance, the last term becomes

$$\frac{\beta_\nu}{2\omega_\nu} t \sin(\omega_\nu t).$$

Thus a mode grows resonantly only if the applied pressure has a nonzero projection onto it.

The drum's natural frequencies are determined by the zeros of Bessel functions, while its modal amplitudes are determined by projections of the initial conditions and forcing.

Axisymmetric forcing excites only the $m = 0$ modes from rest. A nonsymmetric excitation can also excite modes with nodal diameters. Unlike the ideal vibrating string, the drum's natural frequencies are not generally integer multiples of its fundamental frequency.

3.6 Example: Periodically Varying Axially Loaded Beam

Consider the axially loaded beam (3.15) with no transverse forcing and an axial load

$$P(t) = P_o - f(t), \qquad f(t+T) = f(t),$$

where P_o is constant. The beam equation becomes

$$\frac{\partial^2 y}{\partial t^2} + Hy - \frac{f(t)}{\rho A}\frac{\partial^2 y}{\partial x^2} = 0, \tag{3.26}$$

where

$$H = \frac{EI}{\rho A}\frac{\partial^4}{\partial x^4} + \frac{P_o}{\rho A}\frac{\partial^2}{\partial x^2}.$$

We retain the simply supported boundary conditions and assume $P_o < \pi^2 EI/L^2$, so that the reference natural frequencies are positive.

As shown previously, the eigenfunctions

$$\psi_k(x) = \sin\left(\frac{k\pi x}{L}\right), \qquad k = 1,2,\ldots,$$

form an orthonormal basis for $L^2(0,L)$ with respect to

$$\langle\langle u, v\rangle\rangle = \frac{2}{L}\int_0^L u(x)\overline{v(x)}\,dx.$$

They satisfy

$$H\psi_k = \omega_k^2\psi_k, \qquad \psi_k'' = -\frac{k^2\pi^2}{L^2}\psi_k,$$

where

$$\omega_k^2 = \frac{k^2\pi^2}{\rho AL^2}\left(\frac{EIk^2\pi^2}{L^2} - P_o\right).$$

Thus both spatial operators in (3.26) act diagonally on the same basis.

Use the modal decomposition

$$y(x,t) = \sum_{k=1}^{\infty}\gamma_k(t)\psi_k(x), \qquad \gamma_k(t) = \langle\langle y(\cdot,t), \psi_k\rangle\rangle.$$

Substitution into (3.26) gives, for sufficiently regular solutions,

$$\sum_{k=1}^{\infty}\left[\ddot{\gamma}_k + \left(\omega_k^2 + \frac{k^2\pi^2}{\rho AL^2}f(t)\right)\gamma_k\right]\psi_k = 0.$$

Projecting onto each mode yields

$$\ddot{\gamma}_k + \left(\omega_k^2 + \mu_k f(t)\right)\gamma_k = 0, \qquad \mu_k = \frac{k^2\pi^2}{\rho AL^2}. \tag{3.27}$$

The initial conditions are

$$\gamma_k(0) = \langle\langle y_0, \psi_k\rangle\rangle, \qquad \dot{\gamma}_k(0) = \langle\langle v_0, \psi_k\rangle\rangle,$$

where $y_0(x) = y(x,0)$ and $v_0(x) = y_t(x,0)$.

Unlike transverse forcing, the varying axial load changes the coefficient multiplying each modal displacement. It therefore produces *parametric excitation*. The modes remain uncoupled, but their effective stiffnesses vary periodically. Even if these stiffnesses remain positive at every instant, suitable periodic variation can produce growing oscillations.

Equation (3.27) is a *Hill equation*. When $f(t)$ is sinusoidal, it reduces to the *Mathieu equation* after a rescaling of time. It is therefore of practical interest to determine whether its zero solution is stable.

An unstable modal equation gives an unstable beam motion. For any finite modal truncation, stability is equivalent to stability of all retained modal equations. For the full infinite-dimensional system, stability additionally requires bounds uniform over the modes in the chosen displacement–velocity norm.

For a fixed mode, set $x_1 = \gamma_k$ and $x_2 = \dot{\gamma}_k$. Then (3.27) becomes

$$\frac{d}{dt}\begin{bmatrix}x_1\\x_2\end{bmatrix} = \begin{bmatrix}0 & 1\\ -(\omega_k^2 + \mu_k f(t)) & 0\end{bmatrix}\begin{bmatrix}x_1\\x_2\end{bmatrix}.$$

Thus each modal equation has the form $\dot{x} = A_k(t)x$, where $A_k(t+T) = A_k(t)$. We study such periodically varying linear systems in the next chapter.

3.7 Exercises

Exercise 3.1. Consider the cantilevered beam shown in figure-3.2. The deflection of the beam is modeled by the Euler's beam equaion

$$\frac{\partial^2 y}{\partial t^2} + \frac{EI}{\rho A}\frac{\partial^4 y}{\partial x^4} = q(x,t)$$

Find the modes of vibration of the cantilevered beam and sketch the first three modes of

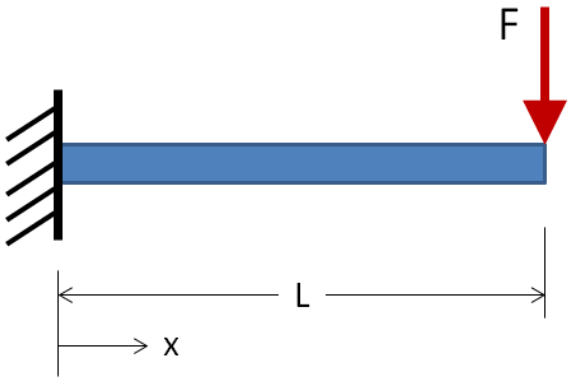


Fig. 3.2 A Cantilevered Beam

vibration.

Exercise 3.2.
The transverse deflection of unforced rotating shafts can be approximately modeled by

$$\frac{\partial^2 y}{\partial t^2} + \frac{EI}{\rho A}\frac{\partial^4 y}{\partial x^4} - \frac{EI}{\kappa GA}\frac{\partial^4 y}{\partial t^2 \partial x^2} = 0.$$

The shaft is simply supported at the ends and hence satisfy the boundary conditions

$$y(0,t) = 0, \quad y(L,t) = 0,$$
$$\frac{\partial^2 y}{\partial x^2}(0,t) = 0, \quad \frac{\partial^2 y}{\partial x^2}(L,t) = 0,$$

Find the modes of vibration and the general solution of the system.

Exercise 3.3. Consider the vibration of a thin circular membrane such as in the case of a drum. The deflection of the membrane of such a rotationally symmetric system is most conveniently described using polar coordinates (r,θ). That is consider the deflection of the membrane to be a function of (t,r,θ). Expressing the wave equation in polar coordinates, one finds that the deflection of the membrane satisfies

$$\frac{\partial^2 y}{\partial t^2} - c^2\left(\frac{\partial^2 y}{\partial r^2} + \frac{1}{r}\frac{\partial y}{\partial r} + \frac{1}{r^2}\frac{\partial^2 y}{\partial \theta^2}\right) = 0,$$

and the boundary conditions $u(t,r_0,\theta) = 0$. Find the modes of vibrations.

Chapter 4
Parametric Excitation and Floquet Theory

The preceding chapter reduced parametric excitation to modal equations with periodically varying coefficients. Unlike the constant-coefficient equations considered earlier, their stability cannot generally be determined from the instantaneous coefficient matrix. Floquet theory instead examines how an initial state evolves over one complete excitation period. Repeated application of this evolution determines whether disturbances remain bounded, decay, or grow.

4.1 Floquet Theory

Consider the time varying linear system

$$\dot{x} = A(t)x, \tag{4.1}$$

where $A : \mathbb{R} \to \mathbb{R}^{n\times n}$ is continuous and $A(t+T) = A(t)$ for all $t \in \mathbb{R}$, with $T > 0$. The study of solutions of this type of system is called Floquet theory.

Let $\Phi(t)$ be the fundamental matrix solution satisfying

$$\dot{\Phi} = A(t)\Phi, \tag{4.2}$$

with initial condition $\Phi(0) = I_{n\times n}$.

Let $\Psi(t) = \Phi(t+T)$. Then

$$\dot{\Psi}(t) = A(t+T)\Phi(t+T) = A(t)\Psi(t).$$

Thus $\Phi(t+T)$ is also a solution of (4.2), with initial condition $\Phi(T)$. Since $\Phi(t)$ is invertible for all t, so is $\Phi(t+T)$. One can show that $\Phi(t)\Phi(T)$ is also a fundamental matrix solution with initial condition $\Phi(T)$. Thus, from uniqueness of solutions,

$$\Phi(t+T) = \Phi(t)C, \tag{4.3}$$

where $C \triangleq \Phi(T)$ is called the monodromy matrix.

Since C is invertible, there exists a possibly complex $n \times n$ matrix ζ such that $C = e^{T\zeta}$. The matrix function $Q(t) = \Phi(t)e^{-t\zeta}$ satisfies

$$Q(t+T) = \Phi(t)e^{T\zeta}e^{-(t+T)\zeta} = \Phi(t)e^{-t\zeta} = Q(t).$$

Thus

$$\Phi(t) = Q(t)e^{t\zeta}, \tag{4.4}$$

where $Q(t)$ is invertible and T-periodic, with $Q(0) = I$. Although $\Phi(t)$ is real, ζ and $Q(t)$ need not be real. Since

$$e^{T(\zeta+2\pi ikI/T)} = e^{T\zeta}, \qquad k \in \mathbb{Z},$$

the matrix ζ is not unique.

Consider the coordinate transformation $x(t) = Q(t)y(t)$. Then

$$\dot{x} = \dot{Q}y + Q\dot{y} = A\Phi e^{-t\zeta}y - \Phi e^{-t\zeta}\zeta y + Q\dot{y} = Ax - Q\zeta y + Q\dot{y}.$$

Since $\dot{x} = Ax$ and Q is invertible, this gives the time invariant system

$$\dot{y} = \zeta y. \tag{4.5}$$

The coordinate transformation $x(t) = Q(t)y(t)$ is known as a Lyapunov transformation. We state these results in the following theorem.

Theorem 4.1 (Floquet Theorem). *Consider the system $\dot{x} = A(t)x$, where $A : \mathbb{R} \to \mathbb{R}^{n\times n}$ is continuous and $A(t+T) = A(t)$ for all $t \in \mathbb{R}$. Let $\Phi(t)$ satisfy $\dot{\Phi} = A(t)\Phi$ with $\Phi(0) = I_{n\times n}$. Then:*

(a) $\Phi(t)$ is defined and invertible for every $t \in \mathbb{R}$.
(b) $\Phi(t+T) = \Phi(t)\Phi(T)$.
(c) There exists a possibly complex $n \times n$ matrix ζ such that $\Phi(T) = e^{T\zeta}$.
(d) There exists an invertible, possibly complex matrix $Q(t)$ such that $Q(t+T) = Q(t)$ and $\Phi(t) = Q(t)e^{t\zeta}$.
(e) The transformation $y = Q(t)^{-1}x$ gives $\dot{y} = \zeta y$.

The eigenvalues of $\Phi(T)$, denoted by $\rho_1, \ldots, \rho_n$, are called the Floquet multipliers (characteristic multipliers). The eigenvalues $\lambda_1, \ldots, \lambda_n$ of ζ are called the characteristic exponents. They are related by

$$\rho_k = e^{\lambda_k T}.$$

The characteristic exponents are determined modulo $2\pi i/T$, while the multipliers are uniquely determined, including their multiplicities.

A solution of (4.1) satisfies

$$x(t) = \Phi(t)x(0) = Q(t)e^{t\zeta}x(0). \tag{4.6}$$

We are interested in whether solutions initially close to the equilibrium $x = 0$ remain near it, converge to it, or become unbounded. Since $Q(t)$ and its inverse are bounded, this behaviour is determined by the eigenvalues and Jordan blocks of ζ.

Let S be an invertible, possibly complex matrix such that $S^{-1}\zeta S = D+N$ is in Jordan form, where D is diagonal, N is nilpotent, and $DN = ND$. With $x(t) = Sz(t)$,

$$z(t) = (S^{-1}Q(t)S)e^{tD}e^{tN}z(0). \tag{4.7}$$

Thus one can conclude the following.

Theorem 4.2 (Stability of Periodic Time Varying Linear Systems).

a. Every solution tends to zero as $t \to \infty$ if and only if all eigenvalues of ζ lie in the strict left half plane; equivalently, $|\rho_k| < 1$ for every k.
b. The origin is stable if and only if $|\rho_k| \leq 1$ for every k and every multiplier on the unit circle is semisimple, meaning that its Jordan blocks all have size one.
c. The system admits a nonzero T-periodic solution if and only if 1 is a Floquet multiplier.
d. The system admits a nonzero solution satisfying $x(t+T) = -x(t)$, and hence $x(t+2T) = x(t)$, if and only if -1 is a Floquet multiplier.
e. The system admits unbounded solutions if a multiplier has magnitude greater than one, or if a multiplier on the unit circle has a Jordan block of size greater than one.

Since the Floquet multipliers determine the stability of the system, the following identity is useful:

$$\rho_1\rho_2\cdots\rho_n = \det\Phi(T) = e^{\text{trace}(T\zeta)} = e^{\int_0^T \text{trace}(A(s))\,ds}. \tag{4.8}$$

The Case where $n = 2$.

For a real system, the monodromy matrix $C = \Phi(T)$ is similar over $\mathbb{R}$ to one of the following forms:

$$\begin{bmatrix}\rho_1 & 0\\ 0 & \rho_2\end{bmatrix}, \qquad \begin{bmatrix}\rho & 0\\ 0 & \rho\end{bmatrix}, \qquad \begin{bmatrix}\rho & 1\\ 0 & \rho\end{bmatrix}, \qquad e^{\sigma T}\begin{bmatrix}\cos(\Omega T) & -\sin(\Omega T)\\ \sin(\Omega T) & \cos(\Omega T)\end{bmatrix}.$$

In the first form, ρ_1 and ρ_2 are distinct real multipliers. In the second and third forms, ρ is a repeated real multiplier. In the fourth form, the multipliers are the nonreal conjugate pair $e^{(\sigma\pm i\Omega)T}$.

The corresponding products of the multipliers are

$$\rho_1\rho_2, \qquad \rho^2, \qquad \rho^2, \qquad e^{2\sigma T},$$

respectively.

From these forms we conclude the following:

For periodic time varying linear systems on $\mathbb{R}^2$ with period T:

(a) The origin is globally asymptotically stable if and only if both multipliers lie strictly inside the unit disk.
(b) If either multiplier lies outside the unit disk, the origin is unstable.

(c) If the multipliers form a nonreal conjugate pair on the unit circle, the origin is stable but not asymptotically stable.
(d) If both multipliers equal $+1$ and C has the second form, all solutions are T-periodic and the origin is stable. If C has the third form, the origin is unstable.
(e) If both multipliers equal -1 and C has the second form, all solutions satisfy $x(t+T) = -x(t)$ and are $2T$-periodic; the origin is stable. If C has the third form, the origin is unstable.

4.2 The Mathieu Equation

We have seen in Section 3.6 how the stability of the modes of a periodically excited axially loaded beam reduces to the stability of the Mathieu equation. The Mathieu equation also arises in electromagnetic waveguides, elliptic membranes, charged-particle traps, and the stabilization of the inverted pendulum. We first apply Floquet theory to study its solutions and then apply these results to the periodically excited axially loaded beam and the inverted pendulum.

Consider the normalized Mathieu equation

$$\ddot{y} + (\alpha + \varepsilon \cos t) y = 0, \tag{4.9}$$

where $\alpha, \varepsilon \in \mathbb{R}$. With $y_1 = y$ and $y_2 = \dot{y}$, this becomes

$$\frac{d}{dt}\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -(\alpha + \varepsilon \cos t) & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}. \tag{4.10}$$

We are interested in determining the values of α and ε for which

a. the origin is globally asymptotically stable;
b. nonzero periodic solutions with period 2π exist;
c. nonzero solutions satisfying $y(t+2\pi) = -y(t)$, and hence having period 4π, exist;
d. the origin is stable.

Writing $x = [y_1 \ y_2]^T$, Floquet theory gives

$$x(t) = P(t, \alpha, \varepsilon) e^{t\zeta(\alpha,\varepsilon)} x(0),$$

where P is invertible and T-periodic, $P(0, \alpha, \varepsilon) = I$, and

$$e^{T\zeta(\alpha,\varepsilon)} = \Phi(T, \alpha, \varepsilon), \qquad T = 2\pi.$$

Here P and ζ may be complex. We have explicitly indicated their dependence on α and ε.

We first consider small ε. When $\varepsilon = 0$ and $\alpha > 0$, the nonzero solutions are periodic with natural period $2\pi/\sqrt{\alpha}$, and the fundamental matrix is

$$\Phi(t, \alpha, 0) = \begin{bmatrix} \cos(\sqrt{\alpha} t) & \frac{1}{\sqrt{\alpha}} \sin(\sqrt{\alpha} t) \\ -\sqrt{\alpha} \sin(\sqrt{\alpha} t) & \cos(\sqrt{\alpha} t) \end{bmatrix}.$$

For comparison with the periodically varying system, we retain the coefficient period $T = 2\pi$, even when $\varepsilon = 0$. The Floquet multipliers are therefore

$$\rho_{1,2}(2\pi, \alpha, 0) = e^{\pm i 2\pi\sqrt{\alpha}}.$$

They lie on the unit circle. They coincide at $+1$ or -1 when $\sqrt{\alpha}$ is an integer or a half-integer, respectively. For every $\alpha > 0$, the unperturbed origin is stable but not asymptotically stable.

As $\alpha \to 0^+$, both multipliers approach 1. At $\alpha = 0$, however,

$$\Phi(t, 0, 0) = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix},$$

so the origin is unstable because solutions with nonzero initial velocity grow linearly. For $\alpha < 0$ and $\varepsilon = 0$, the origin is also unstable, with an exponentially growing solution.

For fixed α and finite t, the fundamental matrix depends analytically on ε, so it admits an expansion

$$\Phi(t, \alpha, \varepsilon) = \Phi(t, \alpha, 0) + \varepsilon \Phi_{\varepsilon}(t, \alpha) + \varepsilon^2 \Phi_{\varepsilon^2}(t, \alpha) + \cdots.$$

The multipliers consequently remain close to their unperturbed values for sufficiently small ε. When the unperturbed multipliers are distinct, each also admits a local power-series expansion. At repeated multipliers, such an expansion cannot be assumed without further analysis.

Since the coefficient matrix in (4.10) has zero trace, (4.8) gives

$$\rho_1(T, \alpha, \varepsilon)\rho_2(T, \alpha, \varepsilon) = 1.$$

Consequently, the origin cannot be asymptotically stable: both multipliers cannot lie strictly inside the unit disk.

The monodromy matrix is real, so its multipliers are either a nonreal conjugate pair or both real. If they are nonreal, their product is their squared magnitude, and hence both lie on the unit circle. If they are real, they are reciprocals. Except when both equal $+1$ or both equal -1, one then has magnitude greater than one, and the origin is unstable.

Thus the stability question reduces to determining where the multipliers form a nonreal conjugate pair on the unit circle, where they are distinct and real, and what happens at the repeated values ± 1. Equivalently, defining

$$d(\alpha, \varepsilon) = \operatorname{trace} \Phi(2\pi, \alpha, \varepsilon),$$

their characteristic equation is

$$\rho^2 - d(\alpha, \varepsilon)\rho + 1 = 0.$$

Hence $|d| < 2$ gives stability, while $|d| > 2$ gives instability. When $d = \pm 2$, stability requires examination of the Jordan form of the monodromy matrix.

The following numerically generated map shows these regions in the (α, ε) plane.

In the interiors of the shaded regions, the Floquet multipliers form a nonreal conjugate pair on the unit circle, and the origin is stable but not asymptotically stable. In the interiors of the unshaded regions, one multiplier lies outside the unit disk, and the origin is unstable.

On the boundaries, both multipliers equal $+1$ or both equal -1. These values guarantee a nonzero 2π-periodic solution or a nonzero solution satisfying $y(t+2\pi) = -y(t)$, respectively. The origin is stable on such a boundary only if the monodromy matrix equals I or $-I$, respectively; otherwise a nontrivial Jordan block produces unbounded solutions.

Figure 4.2(b) shows several typical locations on the stability map, and Figure 4.2(a) shows the corresponding locations of the Floquet multipliers.

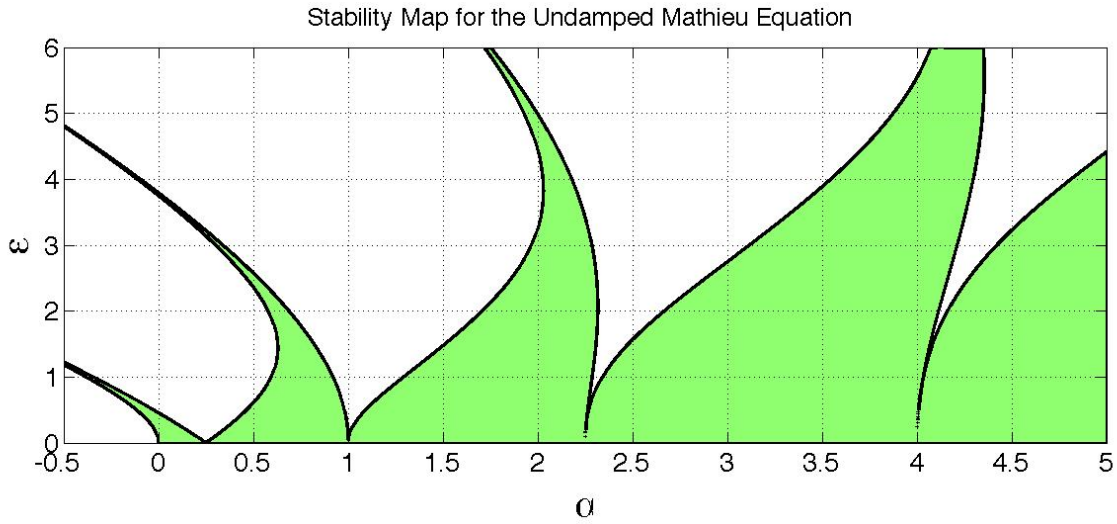


Fig. 4.1 Stability Map for the Mathieu Equation. Figures were created by J. M. Berg using a numerical algorithm in the text 'Nonlinear Ordinary Differential Equations' by Jordan and Smith.

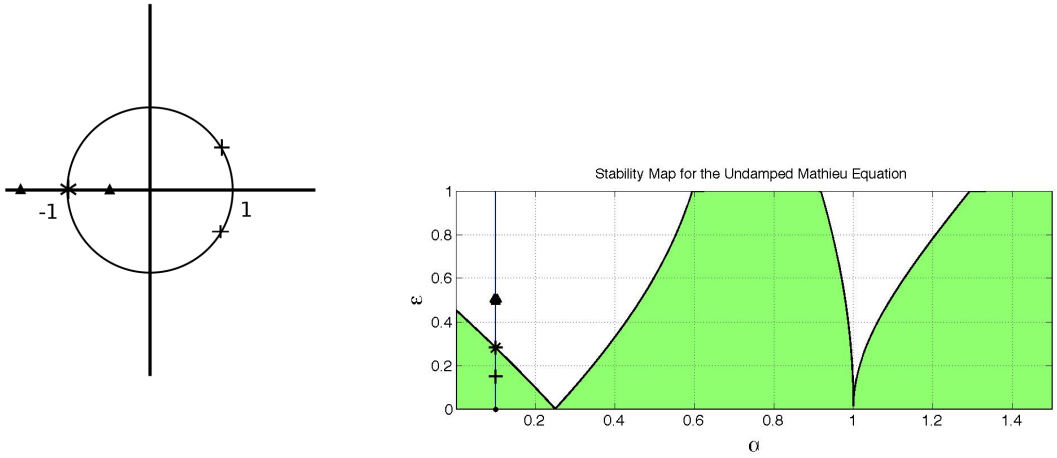
4.2.1 Parametric Resonance of the Axially Loaded Beam

In this section we revisit the periodically varying axially loaded beam considered in Section 3.6. We investigate its stability when the axial load is $P(t) = P_o - a\cos(\omega t)$, where compression is taken as positive, $a > 0$, and $\omega > 0$.

We have seen that the displacement can be written as

$$y(x,t) = \sum_{k=1}^{\infty} \gamma_k(t) \sin\left(\frac{k\pi x}{L}\right),$$

where each modal coordinate satisfies



(a) (b)

Fig. 4.2 Figure (a) shows the Locus of the Floquet Multipliers as ε varies for $\alpha = 0.1$. The corresponding points on the stability map are shown in figure (b)

$$\frac{d^2\gamma_k}{dt^2} + \left(\omega_k^2 + a\mu_k \cos(\omega t)\right)\gamma_k = 0,$$

with

$$\omega_k^2 = \frac{k^2\pi^2}{\rho AL^2}\left(\frac{EIk^2\pi^2}{L^2} - P_o\right), \qquad \mu_k = \frac{k^2\pi^2}{\rho AL^2}.$$

Initially assume that $P_o < P_{cr} = EI\pi^2/L^2$, so that $\omega_k^2 > 0$ for every mode.

Using the time transformation $\tau = \omega t$ and defining

$$\alpha_k = \frac{\omega_k^2}{\omega^2}, \qquad \varepsilon_k = \frac{a\mu_k}{\omega^2},$$

gives the normalized Mathieu equation

$$\frac{d^2\gamma_k}{d\tau^2} + (\alpha_k + \varepsilon_k \cos\tau)\gamma_k = 0.$$

As discussed in Section 4.2, the stability of each mode depends on $(\alpha_k, \varepsilon_k)$. Figure 4.1 shows the corresponding stable and unstable regions.

For a fixed modulation amplitude a, we find that

$$\frac{\varepsilon_k}{\alpha_k} = \frac{a}{EIk^2\pi^2/L^2 - P_o} = m_k.$$

Thus, as ω varies, each mode follows a straight line through the origin in the (α, ε) plane, with slope m_k. Increasing ω moves the point towards the origin. The slope decreases as k increases, and is small when the load modulation is small compared with $EIk^2\pi^2/L^2 - P_o$.

For small nonzero modulation, instability regions emerge near

$$\alpha_k = \frac{n^2}{4}, \qquad n = 1, 2, \ldots,$$

corresponding to excitation frequencies near

$$\omega = \frac{2\omega_k}{n}.$$

The principal instability region occurs near $\alpha_k = 1/4$, or $\omega = 2\omega_k$. The region near $\alpha_k = 1$, or $\omega = \omega_k$, is a higher-order parametric instability region. It is not ordinary forced resonance, since the excitation modulates a coefficient rather than appearing as an additive transverse force.

A mode becomes unstable when its parameter line enters an instability region. In the ideal undamped model, these regions exist for arbitrarily small nonzero modulation, although their widths shrink as the modulation decreases. The higher-order regions are particularly narrow. This phenomenon is called *parametric resonance*.

We also distinguish this instability from static buckling. For a constant axial load, $a = 0$, the first modal frequency vanishes at

$$P_o = P_{cr} = \frac{EI\pi^2}{L^2}.$$

At this load, $\ddot{\gamma}_1 = 0$, so an initial modal velocity produces linear growth. For $P_o > P_{cr}$, $\omega_1^2 < 0$ and the first mode has an exponentially growing solution, indicating instability of the straight configuration. The load P_{cr} is the Euler buckling load.

When $P_o = P_{cr}$ but $a \neq 0$, we instead have $\alpha_1 = 0$. The Mathieu stability map shows that the first mode is stable for sufficiently small nonzero $\varepsilon_1 = a\mu_1/\omega^2$. Thus periodic modulation can stabilize this mode in the linear model. There is no separate "zeroth mode" representing the straight configuration; that configuration corresponds to $\gamma_k = 0$ for every k. Stability of the other modes must also be checked.

Consider a uniform steel rod of length 1.5 m and diameter 3 cm, with $E = 208\,\mathrm{GN/m^2}$ and $\rho = 7780\,\mathrm{kg/m^3}$. If the tensile load has magnitude $(1000 - 100\cos(\omega t))\,\mathrm{N}$, then our compression-positive convention gives

$$P(t) = (-1000 + 100\cos(\omega t))\,\mathrm{N}.$$

The reversal of the modulation sign is equivalent to a shift of its phase and does not change the stability map. The first two natural frequencies are

$$\frac{\omega_1}{2\pi} \approx 27.4\,\mathrm{Hz}, \qquad \frac{\omega_2}{2\pi} \approx 108.7\,\mathrm{Hz}.$$

Here $m_1, m_2 \ll 1$, so the instability regions crossed by the parameter lines are narrow. The principal parametric resonances occur near excitation frequencies 54.9 Hz and 217.3 Hz, respectively. Narrower higher-order instability regions also exist.

Now consider the compressive load

$$P(t) = (1000 - 100\cos(\omega t))\,\mathrm{N}.$$

Its mean is about 3% of the buckling load $P_{cr} \approx 36.3\,\mathrm{kN}$. The first natural frequency is

$$\frac{\omega_1}{2\pi} \approx 26.7\,\mathrm{Hz},$$

so its principal parametric instability occurs near an excitation frequency of 53.4 Hz. Again, the precise unstable interval is determined by the Mathieu stability boundaries.

4.2.2 Stabilization of the Inverted Pendulum with Periodic Vertical Forcing

In the previous section we saw how periodic excitation can destabilize an otherwise stable system. Here we consider the converse: using rapid excitation to stabilize the inverted equilibrium of a pendulum.

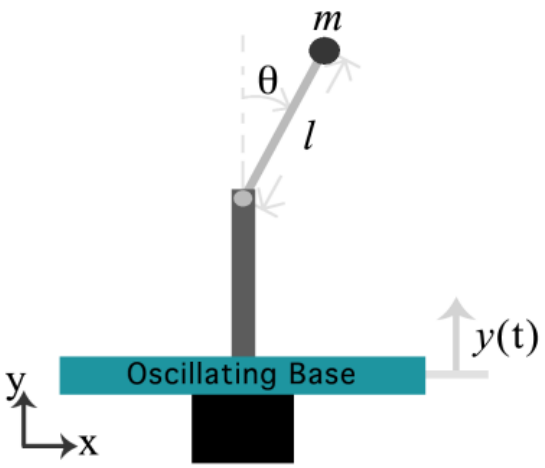


Fig. 4.3 A base-excited inverted pendulum.

Let the upward displacement of the pendulum base be $y(t) = a\cos(\omega t)$, and let θ be measured from the upward vertical. The equation of motion is

$$\ddot{\theta} + \left(-\frac{g}{l} + \frac{a\omega^2}{l}\cos(\omega t)\right)\sin\theta = 0. \tag{4.11}$$

For small displacements from the inverted equilibrium, $\sin\theta \approx \theta$. Introducing $\tau = \omega t$ gives the linearized equation

$$\frac{d^2\theta}{d\tau^2} + (\alpha + \varepsilon\cos\tau)\,\theta = 0, \tag{4.12}$$

where

$$\alpha = -\frac{g}{\omega^2 l}, \qquad \varepsilon = \frac{a}{l}.$$

This is the normalized Mathieu equation. Since $\alpha < 0$, we consider the left-hand portion of its stability map, shown in Figure 4.4.

For fixed a/l, increasing ω moves α towards zero from below. For small ε, the nearby stability boundary is

$$\alpha = -\frac{\varepsilon^2}{2} + O(\varepsilon^4).$$

Consequently, to leading order, the linearized inverted equilibrium is stable when

$$\frac{g}{\omega^2 l} < \frac{a^2}{2l^2}, \qquad \text{or equivalently} \qquad a^2\omega^2 > 2gl.$$

This is an approximate small-amplitude condition; the Mathieu stability map gives the more precise boundary. Since damping is neglected, stability here does not imply convergence to the inverted equilibrium.

For example, if $\varepsilon = a/l = 0.2$, the approximate condition is

$$-\alpha = \frac{g}{\omega^2 l} < 0.02,$$

which gives

$$\omega > 10\sqrt{\frac{g}{2l}}.$$

For a pendulum of length 10 cm and a base excitation amplitude of 2 cm, this gives an approximate threshold

$$\frac{\omega}{2\pi} > 11.1\,\text{Hz}.$$

An excitation frequency of 12 Hz lies above this approximate threshold and within the stable region of the linearized Mathieu model.

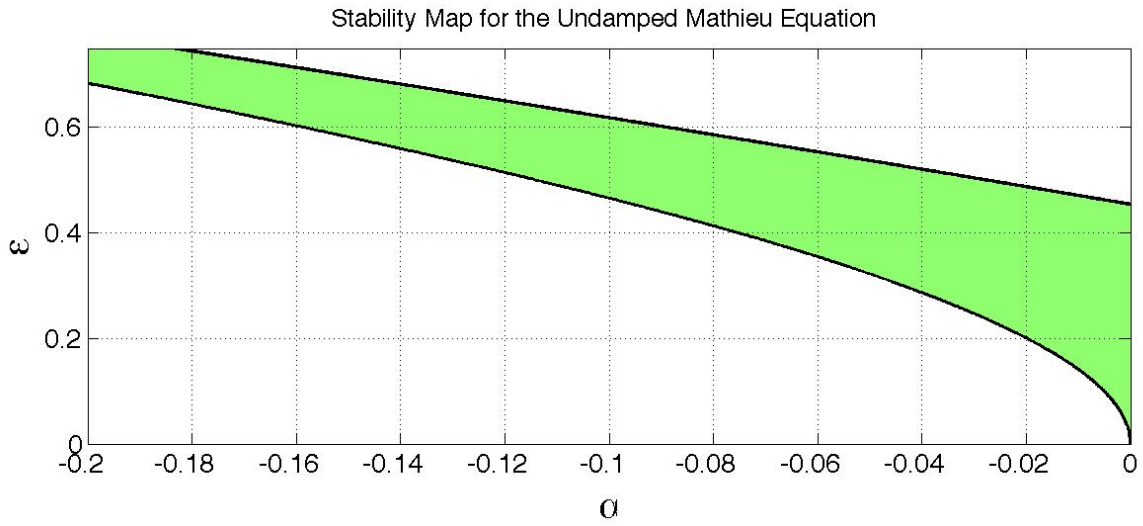


Fig. 4.4 Closeup of the left-hand side of the stability map for the Mathieu equation. Figures were created by J. M. Berg using a numerical algorithm in the text *Nonlinear Ordinary Differential Equations* by Jordan and Smith.